L07 — study notes

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Class 7 — More econometric models · Map of the notes

One claim. Every model of this class is the normal linear regression with one twist, and the twist is handled by **adding something to the Gibbs sampler**: a known Ω is a rotation (GLS); a variance function needs one **Metropolis** step; Student-t errors become normal once each observation carries its own **latent variance**; AR(1) errors are one parameter ρ and **quasi-differencing**. **Hierarchical models** put a prior with estimated mean and variance on unit-level coefficients, so each unit's estimate is a **partial-pooling** average of its own data and the population. **Limited dependent variables** — Tobit, probit, ordered and multinomial probit — are regressions on a **latent** y^* seen only through a cut: draw y^* from a **truncated normal** and it is OLS again, but **coefficients are latent-scale effects**, not effects on what is observed.

Three layers per topic:

Layer	What it means here
U — intuition	why noisy or correlated observations should count less; fat tails as random variances; pooling vs separate fits; corners vs censoring vs truncation; a cut throws away the scale

Layer	What it means here
C — course (examinable)	GLS (5) and the general-Ω full conditionals of note 01; the scale-mixture derivation and the four conditionals of note 02; Ω_ρ, its tridiagonal inverse, Prais–Winsten, the ρ conditional and the truncated-normal draw of note 03; the three panel models, every hierarchical conditional and the partial-pooling weight of note 04; the Tobit likelihood, data augmentation and (17.24)–(17.30) of note 05; the probit identification, Albert–Chib, the 1.6 rule, the ordered-probit cut-point update and the multinomial probit as a SUR of note 06 — **with the slides' and the books' own numbers**
G — graduate	closed-form OLS/GLS variance ratios under AR(1) errors and regressors; the Stein-lemma attenuation $\beta P(y > 0)$; Geweke's moment conditions; slow mixing of cut-points; MNP identification schemes; multivariate probit. **Proved or named, not examinable** unless the slides state them

Reading order

Part I — the general linear model: non-spherical errors

1. L07-01-general-covariance-gls — GLM3 slides 3–15; W §8.4 (pp. 254–262). Settles: $P\Omega P' = I$ makes GLS = OLS on (Py, PX); the full conditionals with Ω^{-1}; WLS = divide every column by $\sqrt{h_i}$; W Example 8.6 **.787 (.063) vs OLS .821 (.104)**; why α in $h = \exp(z'\alpha)$ needs MH; FGLS = the slides' start; OLS loses a factor **4.38** at $\alpha_1 = 1$; the airline posterior ($\alpha_2 = 7.085$). Slip on slide 8 (β).
2. L07-02-student-t-errors — GLM3 slides 16–21; G §4.5, §8.1.2 (pp. 50–52, 114–116). Settles: the t as a **scale mixture**, derived; $\omega_i \sim IG(\frac{\nu+1}{2}, \frac{\nu+\epsilon_i^2/\sigma^2}{2})$ and the weight $(\nu+1)/(\nu+\epsilon_i^2/\sigma^2)$, below 1 iff $|\epsilon_i| > \sigma$; the ν conditional and its MH step; Exp(25) prior puts **0.33** on $\nu \le 10$; Geweke; an outlier downweighted to **0.018**. Slip on slide 6.
3. L07-03-ar1-errors — GLM3 slides 22–30; W §12.3 (pp. 381–386); T §2.9 (pp. 90–93). Settles: ρ^k covariance; tridiagonal $\Omega_\rho^{-1} = P'P$; Prais–Winsten vs Cochrane–Orcutt; barium **2.94 (0.63), $\hat{\rho}$ = .293**; (10)–(11): naive SEs wrong by $(1+\rho\phi)/(1-\rho\phi)$; the "small sin" makes ρ an exact truncated-normal draw. Slip on slide 24.

Part II — hierarchical (multilevel) models

4. L07-04-hierarchical-linear-models — HLM slides 3–23; hierarchicalmodel.html; G §4.6, §9.3; MS §6.4; W §14.2. Settles: pooled / individual-effects / random-coefficient models; every conditional; weight $TV_\alpha/(TV_\alpha + \sigma^2)$; eight schools exactly (τ **median 5.23**, A: 28.39 → **11.64**); pregnant women (pooled **−4.52639 + 0.47633x**, hierarchical $\sigma_\alpha = 3.21$); the RE θ. Slips on slides 9, 13, 15, 16.

Part III — limited dependent variables

5. L07-05-tobit — LDV slides 4–10; G §8.2.1; W §17.2, §17.4. Settles: censoring vs truncation; the likelihood; Chib's augmentation; (17.24)–(17.30) derived; slide 9's **0.500 / 0.249 / 0.104**; OLS slope = $\beta \cdot \mathbf{P(y > 0)}$; Mroz Tobit reproduced (educ **80.65**, APE scale **0.589**). Two slips on slide 8.
6. L07-06-probit-ordered-multinomial — LDV slides 11–15; G §8.2.2–8.2.3, §9.2; W §17.1; MS §6.5. Settles: $\sigma = 1$; effect $\phi(x'\beta)\beta \leq 0.4\beta$ and **logit ≈ 1.6 × probit** (1.61 in Mroz); Albert–Chib; ordered probit cut-points are **uniform** draws (and mix slowly); multinomial probit = latent SUR with a truncated multivariate normal; identification of the scale.

Support

7. L07-07-formulas — the exam specification mapped; every numbered result; every number.
8. L07-08-pitfalls — fourteen confusions with minimal counterexamples and a 30-item True/False self-test.
9. L07-09-python — `L07_python_checks.py`: every sampler of the class in NumPy (no PyMC), with asserts against W Tables 8.1, 12.1, 17.1, 17.3, the site's printout and the slides.

The one result to hold

$$\theta \mid \text{rest} \sim N\Big((V_0^{-1} + \sigma^{-2} X'\Omega^{-1} X)^{-1} (V_0^{-1}\beta_0 + \sigma^{-2} X'\Omega^{-1} y^*),\ (V_0^{-1} + \sigma^{-2} X'\Omega^{-1} X)^{-1} \Big)$$

One conditional, five disguises. Ω known or $h(z, \alpha)$: note 01. $\Omega = \text{diag}(\omega_i)$ with latent ω_i: note 02. $\Omega = \Omega_\rho$: note 03. y^* = data with the hidden part imputed: notes 05–06. And with "β" the unit coefficients and "β_0, V_0" the population mean and variance, it is the partial-pooling update of note 04. What changes between models is only **what else is drawn** around it.

Numbers and cases to have memorized

Case	Number	Meaning
W Ex. 8.6, WLS weights 1/inc	0.787 (0.063) vs OLS 0.821 (0.104)	40% smaller SE if $h = inc$ is right
$h = e^{\alpha_1 x}$, $\alpha_1 = 1$	Var OLS/Var GLS = 4.38	the cost of ignoring Ω
airline variance	$\alpha_2 = 7.085$: +0.1 load factor doubles the variance (×2.03)	Case I in data
$\nu \sim$ Exp(mean 25)	$P(\nu \leq 10) = 0.33$, $P(\nu \geq 40) = 0.20$	slide 17's prior
t weight	$(\nu+1)/(\nu+e^2) = 1$ at $\lvert e\rvert = 1$; 3/38 at e = 6, $\nu = 2$	automatic downweighting
Vella–Verbeek, t_5 vs normal	σ^2 0.105 vs 0.165; $0.105 \times 5/3 = 0.175$; $\log_{10}$ BF 4.83	scale ≠ variance
barium, Prais–Winsten	2.94 (0.63) vs OLS 3.12 (0.48); $\hat{\rho} = 0.293$; t = −1.69	serial correlation inflates SEs

Case	Number	Meaning
$\rho = \varphi = 0.9$	naive OLS variance too small $\times 9.53$	product $\rho\varphi$
eight schools	pooled 7.87 (4.17); τ median 5.23; A 28.39 $\rightarrow$ 10.24 ($\tau = 5.2$), 11.64 (Bayes)	partial pooling
pregnant women	pooled $-4.52639 + 0.47633$x; $\sigma_$y 1.12, σ_α 3.21	heterogeneity $\gg$ noise
slide 9 Tobit	P(y = 0) = 0.500, 0.249, 0.104 at a = 0, 0.7, 1.3	$\Phi(-2a/2.062)$
Mroz Tobit	educ 80.65; APE 0.589 $\times$ 80.65 = 47.5 h vs OLS 28.8	coefficient $\neq$ effect
probit vs logit	$\varphi(0) = 0.399$ vs $1/4 \rightarrow 1.6$; Mroz $0.391/0.243 = 1.61$	scale, not substance

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Figures

Built by `figures/make_figures.py` (run: `python Notas/L07/figures/make_figures.py`; every figure ends in `assert` checks on its plotted numbers).

Figure	Shows	Lives in
L07-gls-efficiency	Var OLS / Var GLS vs α_1 (4.38 at 1); WLS weights	L07-01-general-covariance-gls §4.1
L07-t-scale-mixture	t_ν vs normal tails; Gibbs weight vs residual for four ν	L07-02-student-t-errors §2
L07-ar1-gls	ρ^k; OLS/GLS and true/naive variance ratios vs ρ for $\varphi = 0$, 0.5, 0.9	L07-03-ar1-errors §1
L07-eight-schools-shrinkage	$E(\theta_i \mid y, \tau)$ paths; exact $p(\tau \mid y)$	L07-04-hierarchical-linear-models §5
L07-pregnancy-hierarchical	68 individual fits pulled toward the population; pooled vs woman lines	L07-04-hierarchical-linear-models §6
L07-tobit-moments	βx, $E(y \mid x)$, $E(y \mid y > 0, x)$; slide 9's $P(y = 0)$ and the OLS attenuation	L07-05-tobit §5
L07-probit-logit-ordered	Φ vs $\Lambda(1.6z)$; ordered-probit category probabilities	L07-06-probit-ordered-multinomial §1

Interactive companions

- The Partial-Pooling Dial (`companions/companion-partial-pooling.html`) — move τ, the prior bound A, a data-precision multiplier κ and the highlighted school; read $E(\theta_i \mid y, \tau)$, B_i, $\hat{\mu}$, the fully Bayes mean and $p(\tau \mid y)$ (10.24 for school A at the default $\tau = 5.2$). Linked from note 04.

- The Censoring Dial (`companions/companion-tobit-censoring.html`) — move the covariate mean a, β, σ and an evaluation point x_0; read $P(y = 0)$ (0.249 at slide 9's a = 0.7), the OLS slope $\beta P(y > 0)$ and (17.24)–(17.30) at x_0. Linked from note 05.

Both pages are verified by `companions/check_companions.py` (the pages' own JavaScript under `node` against an independent Python recomputation at both ends of every range, the defaults and the landmarks).

Scope and ceiling

- **Examinable (C):** the derivations and readings of notes 01–06 — the final exam has no code (Aula/COURSE): expect full conditionals, likelihoods, the scale mixture, partial pooling, Tobit moments, and reading printouts like W's tables and the slides' posterior summaries.
- **Proved but above the line (G):** (10)–(11) of note 03; the Stein-lemma attenuation; MNP identification schemes; the multivariate probit's MH step.
- **Named only:** SUR as a topic (Class 9, G §9.1); FE vs RE and the Hausman test (not on the site schedule; W §14.2 cited as the neighbour); the dugong growth curve and WinBUGS (HLM slides 22–33); sample selection (W §17.5); Cowles' cut-point MH.
- **Elsewhere:** Poisson and other GLMs (I §4.6) are in L02 §6.

Course-wide position

Class	Relation to Class 7
1 — Review	OLS = MLE under $\Omega = \mathrm{I}$; AR(1) moments reused for Ω_ρ; conjugate Poisson–gamma updating is the pattern of every conditional here
2 — Statistical learning	the logit is the frequentist twin of the probit; partial effects and cut-offs; heteroskedastic mcycle noise
3+4 — Bayesian learning	Gibbs, MH and conjugate normal / inverse-gamma priors — **the toolkit every note uses**
5+6 — Causal inference	panel data and DiD; the FE side of the hierarchical/RE comparison
7 — More econometric models	**non-spherical errors, hierarchical models, limited dependent variables**
8 — Univariate time series	ARMA errors as models in their own right; Student-t stochastic volatility uses note 02's mixture
9 — Multivariate time series	SUR and VARs: the $I \otimes \Sigma$ GLS of note 06 §4

Rules packs: rules/03 (hierarchical models), rules/06 (probit/Tobit), rules/07 (priors, posteriors, conjugate analysis).

Errata found in the sources

Source, place	Slip	Correct	Note
GLM3 slide 6	Case II "$\omega_i^{-1} \sim IG(\nu/2, \nu/2)$"	$\omega_i \sim IG$, i.e. $\omega_i^{-1} \sim G(\nu/2, \nu/2)$ (as slides 16, 18)	L07-02-student-t-errors §2
GLM3 slide 8	true $\beta = (2.0, 4.0)$	inconsistent with the slide's scatter, OLS start (9.12, 0.62) and chains	L07-01-general-covariance-gls §4
GLM3 slide 24	$\exp\{-0.5\epsilon'\Omega_\rho^{-1}\epsilon\}$	$\exp\{-0.5\sigma^{-2}\epsilon'\Omega_\rho^{-1}\epsilon\}$	L07-03-ar1-errors §4
HLM slide 9	last row of X^* ends in $\tilde{X}_1$	$\tilde{X}_N$	L07-04-hierarchical-linear-models §2
HLM slide 13	μ_α update with $\sigma^{-2}N$, $\sigma^{-2}\sum \alpha_i$	$V_\alpha^{-1}N$, $V_\alpha^{-1}\sum \alpha_i$	L07-04-hierarchical-linear-models §3
HLM slide 15	$V_0\beta_0^{-1}$	$V_0^{-1}\beta_0$	L07-04-hierarchical-linear-models §4
HLM slide 16	$\tilde{V}_\beta^{-1} = V_\beta^{-1} + NV_0$	$+NV_0^{-1}$	L07-04-hierarchical-linear-models §4
LDV slide 8	$V_\beta^{-1} = V_0^{-1} + \sigma^{-2}(X'X)^{-1}$; mean without σ^{-2} on $X'y^*$	$\sigma^{-2}X'X$; $V_\beta(V_0^{-1}\beta_0 + \sigma^{-2}X'y^*)$	L07-05-tobit §4
LDV slides 5, 9–10	censoring called "truncation"	censoring (truncation drops the units)	L07-05-tobit §1
hierarchicalmodel.html §4.4.3	"$p(y_{new} \mid x_{new}, \text{data})$"	posterior of $\alpha_i + \beta_i x$ (no σ_y noise added)	L07-04-hierarchical-linear-models §6

Bibliography actually used in these notes

Book / source	Where
The class decks (`Aula/class-07/materials/`): GLM3.pdf, HLM.pdf, limiteddependentvariablemodels.pdf, hierarchicalmodel.html	the spine of every note
Greenberg (2007) §4.5–4.6, §8.1–8.2, §9.2–9.3	notes 02, 04, 05, 06
Wooldridge 6e §8.4, §12.3, §14.2, §17.1–17.2, §17.4	notes 01, 03, 04, 05, 06 (Tables 8.1, 12.1, 17.1, 17.3 reproduced from the `wooldridge` data)
Tsay 3e §2.9	note 03
Morettin & Singer §6.4–6.5	notes 04, 06

Related project material: readings, practice and Python per class (Map/class-plan); exercise numbers by book (Map/exercises-index). Practice for this class (statements in the books): W 12.C2, 12.C10; G 8.3, 8.5; G 4.6, 9.2, 9.3; W 17.3, 17.C2, 17.C3, 17.C9; G 8.6, 8.7; MS 6.18, 6.21; W 14.C5 — each note's `## Links` gives the pages. The worked solutions are the unit's exercise list (`Listas/`, built separately).

PDF: L07-notes.pdf — all ten notes with the figures, 70 pages (rebuild after any edit: `python -I ~/.claude/commands/templates/notes_pdf.py Notas/L07`).

GLM3 slides 3–15 · Known Ω is OLS after a rotation; unknown Ω costs one Metropolis step

> **Claim.** With $y = X\beta + \epsilon$, $\epsilon \sim N(0, \sigma^2\Omega)$, any matrix P with $P\Omega P' = I_n$ turns the model into a standard regression of Py on PX. So **GLS is OLS on rotated data**, and the Bayesian full conditionals are the familiar normal / inverse-gamma pair with $X'X$ replaced by $X'\Omega^{-1}X$. A diagonal Ω is the heteroskedastic case: GLS = **weighted least squares** with weights $1/\omega_i$. When $\omega_i = h(z_i, \alpha)$ is unknown, α has no conjugate conditional and is drawn by **Metropolis–Hastings**; the frequentist twin is **feasible GLS**, whose first step (regress $\log \hat{e}_i^2$ on z_i) is exactly the slides' starting value. Prev: — · Next: L07-02-student-t-errors · Map: L07-00-map

Notation.

- y $(n \times 1)$, X $(n \times k$, full column rank), β $(k \times 1)$; $\epsilon = y - X\beta$.
- $\sigma^2 > 0$: scale; Ω $(n \times n$, symmetric positive definite): the error correlation/variance pattern, ω_{ij} its entries.
- P: any $n \times n$ matrix with $P\Omega P' = I_n$ (e.g. $P = L^{-1}$, $\Omega = LL'$ Cholesky).
- Prior: $\beta \sim N(\beta_0, V_0)$, $\sigma^2 \sim IG(\nu_0/2, \nu_0 s_0^2/2)$ (slide 4).
- $h(z_i, \alpha) > 0$: variance function, z_i $(p \times 1)$ its covariates, α its parameters (slide 7).
- W's notation: $\mathrm{Var}(u \mid x) = \sigma^2 h(x)$, $h_i = h(x_i)$ (8.21).

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1 · U — why the error covariance matters

OLS stays unbiased when $\mathrm{Var}(\epsilon) = \sigma^2\Omega$ (unbiasedness never used the variance), but two things break: the usual variance formula $\sigma^2(X'X)^{-1}$ is wrong, and OLS is no longer the most precise linear unbiased estimator. Observations with large variance (the noisy ones) carry less information about β and should count less; correlated observations repeat part of each other's information and should not be counted twice. GLS does both, by first transforming the data into something that has spherical errors.

2 · C — the known-Ω case (slides 3–5)

Model (slide 3).

$$y = X\beta + \epsilon, \qquad \epsilon \sim N(0, \sigma^2\Omega). \tag{1}$$

The rotation (slide 4). Take P with $P\Omega P' = I_n$ and multiply (1) by P:

$$Py = PX\beta + P\epsilon. \tag{2}$$

The new error has mean 0 and variance

$$\mathrm{Var}(P\epsilon) = P\,\mathrm{Var}(\epsilon)\,P' = P(\sigma^2\Omega)P' = \sigma^2(P\Omega P') = \sigma^2 I_n. \tag{3}$$

Writing $y^* = Py$, $X^* = PX$, $\epsilon^* = P\epsilon$, (2)–(3) say $y^* = X^*\beta + \epsilon^*$, $\epsilon^* \sim N(0, \sigma^2 I_n)$: **given Ω, a standard linear regression** (slide 4).

GLS is OLS on (2). OLS on the starred data is $(X^{*\prime}X^*)^{-1}X^{*\prime}y^*$. Substitute $X^* = PX$, $y^* = Py$:

$$\hat{\beta}_{GLS} = (X'P'PX)^{-1}X'P'Py. \tag{4}$$

From $P\Omega P' = I$: $\Omega = P^{-1}(P')^{-1}$, so $\Omega^{-1} = P'P$. Substitute into (4):

$$\hat{\beta}_{GLS} = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}y, \qquad \mathrm{Var}(\hat{\beta}_{GLS}) = \sigma^2(X'\Omega^{-1}X)^{-1}. \tag{5}$$

P itself never matters, only $P'P = \Omega^{-1}$. (Checked numerically: `L07_python_checks.py` §1, OLS on (Py, PX) equals (5) to machine precision.)

Full conditionals (slide 5). The likelihood of (1) is

$$p(y \mid \beta, \sigma^2) \propto (\sigma^2)^{-n/2} \exp\Big\{ -\frac{1}{2\sigma^2}(y - X\beta)'\Omega^{-1}(y - X\beta)\Big\}. \tag{6}$$

$\boldsymbol{\sigma^2}$ *given* $\boldsymbol{\beta}$. Multiply (6) by the $IG(\nu_0/2, \nu_0 s_0^2/2)$ kernel $(\sigma^2)^{-\nu_0/2-1}\exp\{-\nu_0 s_0^2/(2\sigma^2)\}$ and collect powers and exponents:

$$p(\sigma^2 \mid \beta, y) \propto (\sigma^2)^{-(\nu_0+n)/2-1} \exp\Big\{ -\frac{\nu_0 s_0^2 + (y - X\beta)'\Omega^{-1}(y - X\beta)}{2\sigma^2}\Big\},$$

$$\sigma^2 \mid \beta, \Omega, y \sim IG(\nu_1/2, \nu_1 s_1^2/2), \quad \nu_1 = \nu_0 + n, \quad \nu_1 s_1^2 = \nu_0 s_0^2 + (y - X\beta)'\Omega^{-1}(y - X\beta). \tag{7}$$

$\boldsymbol{\beta}$ *given* $\boldsymbol{\sigma^2}$. Keep the terms of (6) $\times$ prior that involve β:

$$-\tfrac{1}{2}[\sigma^{-2}(y - X\beta)'\Omega^{-1}(y - X\beta) + (\beta - \beta_0)'V_0^{-1}(\beta - \beta_0)].$$

Expand and keep the quadratic and linear terms in β:

$$-\tfrac{1}{2}[\beta'(V_0^{-1} + \sigma^{-2}X'\Omega^{-1}X)\beta - 2\beta'(V_0^{-1}\beta_0 + \sigma^{-2}X'\Omega^{-1}y)] + \text{const.}$$

Completing the square gives

$$\beta \mid \sigma^2, \Omega, y \sim N(\beta_1, V_1), \quad V_1^{-1} = V_0^{-1} + \sigma^{-2}X'\Omega^{-1}X, \quad V_1^{-1}\beta_1 = V_0^{-1}\beta_0 + \sigma^{-2}X'\Omega^{-1}y. \tag{8}$$

(8) is a precision-weighted average of the prior mean and the GLS estimate (5). With $V_0^{-1} \to 0$, $\beta_1 = \hat{\beta}_{GLS}$ and $V_1 = \sigma^2(X'\Omega^{-1}X)^{-1}$; with $\Omega = I$ it is G's (8.2) (p. 111). Alternating (7) and (8) is G's Algorithm 8.1 (p. 112) with Ω in place.

3 · C — Ω diagonal: weighted least squares (slide 6; W §8.4a)

With $\Omega = \mathrm{diag}(\omega_1, \ldots, \omega_n)$ (slide 6), $P = \mathrm{diag}(\omega_i^{-1/2})$ and (2) divides row i by $\sqrt{\omega_i}$ — W's transformed equation (8.25)–(8.26), p. 255, with $h_i = \omega_i$:

$$\frac{y_i}{\sqrt{h_i}} = \beta_0\frac{1}{\sqrt{h_i}} + \beta_1\frac{x_{i1}}{\sqrt{h_i}} + \cdots + \beta_k\frac{x_{ik}}{\sqrt{h_i}} + \frac{u_i}{\sqrt{h_i}}. \tag{9}$$

The **constant is divided too** — (9) has no intercept column of ones. OLS on (9) minimises

$$\sum_{i=1}^{n} \frac{(y_i - x_i'\beta)^2}{h_i} \qquad \text{(W 8.27, p. 256)}, \tag{10}$$

the residuals weighted by $1/h_i$, the variables by $1/\sqrt{h_i}$. (10) is (5) written out, since $X'\Omega^{-1}X = \sum_i x_i x_i'/h_i$.

Worked: W Example 8.6 (Table 8.1, p. 257). Net total financial wealth (`nettfa`, \$1000s) on income (`inc`, \$1000s), single people ($n = 2{,}017$), WLS under $\mathrm{Var}(u \mid inc) = \sigma^2 inc$ (so $h_i = inc_i$, weight $1/inc_i$):

	intercept	inc	R^2
OLS (robust SE)	−10.57 (2.53)	0.821 (0.104)	0.0827
WLS, weights $1/inc$	−9.58 (1.65)	0.787 (0.063)	0.0709

All eight numbers are reproduced from the `401KSUBS` data in `L07_python_checks.py` §1. The WLS standard error of the income slope is $0.063/0.104 = 0.61$ of the OLS one — "almost 40% smaller" (W p. 257), **provided** the variance model is right. The R^2 of WLS is lower and **not comparable**: it measures fit of the weighted problem (W p. 256).

Case I vs Case II (slide 6). The slides split diagonal Ω into $\omega_i = h(z_i, \alpha)$ (Case I, a deterministic function of covariates — this note) and ω_i random with an inverse-gamma law (Case II — the Student-t model, L07-02-student-t-errors).

4 · C — Case I: ω_i = h(z_i, α), and why α needs Metropolis–Hastings (slides 7–8)

Variance functions (slide 7). $h(z_i, \alpha) = (\alpha_1 z_{i1} + \cdots + \alpha_p z_{ip})^2$ or, more commonly, $h(z_i, \alpha) = \exp\{\alpha_1 z_{i1} + \cdots + \alpha_p z_{ip}\}$ — W's (8.30), p. 259, is the same exponential form.

The conditional of α. Given β and σ^2, with $e_i = y_i - x_i'\beta$, (6) becomes

$$p(\alpha \mid y, \beta, \sigma^2) \propto p(\alpha) \prod_{i=1}^{n} h(z_i, \alpha)^{-1/2} \exp\Big\{ - \frac{e_i^2}{2\sigma^2 h(z_i, \alpha)} \Big\}. \tag{11}$$

For $h = \exp(z_i'\alpha)$, substitute and take logs:

$$\log p(\alpha \mid \cdot) = \log p(\alpha) - \tfrac{1}{2} \sum_i z_i'\alpha - \frac{1}{2\sigma^2} \sum_i e_i^2 e^{-z_i'\alpha} + \text{const.} \tag{12}$$

α appears linearly and inside an exponential: no normal prior makes (12) a known family, so the sampler alternates (7), (8) with $\Omega = \mathrm{diag}\{h(z_i, \alpha)\}$, and a **random-walk MH step** for α: propose $\alpha^\dagger = \alpha + \delta$, $\delta \sim N(0, c^2 I)$, accept with probability $\min\{1, \exp[\log p(\alpha^\dagger \mid \cdot) - \log p(\alpha \mid \cdot)]\}$ (the RW proposal is symmetric, so it cancels).

Identification (slide 8). With an intercept in z, $\sigma^2 h(z, \alpha) = \sigma^2 e^{\alpha_0} e^{\alpha_1 x}$: σ^2 and e^{α_0} enter only through their product, so the slide **fixes** $\sigma^2 = 1$.

The slide-8 design. $n = 100$, $h = \exp\{\alpha_0 + \alpha_1 x_i\}$, $x_i \sim N(0, 1)$, $\alpha = (1, 1)$, priors $\beta \sim N(0, 100 I_2)$, $\alpha \sim N(0, 100 I_2)$; start at $\hat\beta_{ols}$ and at α from the regression of $\log(y_i - x_i'\hat\beta_{ols})^2$ on x_i; burn-in 1000, keep every 20th draw to $M = 1000$.

Slip on slide 8. It prints $\beta = (2.0, 4.0)$, but its own scatter (slide 9: y between 4 and 16, slope well under 1), its OLS start $\hat{\beta}_{ols} = (9.12, 0.62)$ and its chains (centred near 9 and 0.7, slide 10) are not consistent with that β. The true β of the run cannot be recovered from the deck; the α part ($\alpha = (1, 1)$, chains near 1) is consistent.

Re-run here with the printed $\beta = (2, 4)$, $\alpha = (1, 1)$ (`L07_python_checks.py` §1, seeded, 12,000 iterations, 2,000 burn-in, RW step 0.15): MH acceptance 0.55; posterior means $\beta = (2.20, 3.97)$, $\alpha = (0.93, 1.12)$.

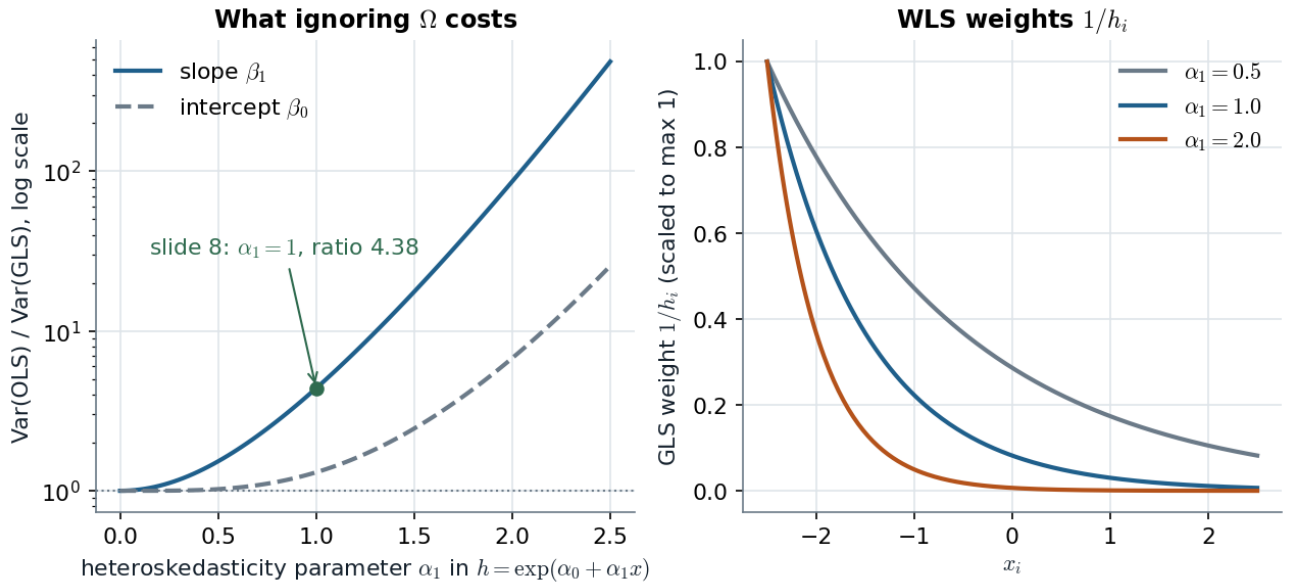
4.1 · Feasible GLS is the slides' starting value (W §8.4b, pp. 259–260)

W's FGLS: (i) OLS residuals $\hat{u}_i$; (ii) regress $\log \hat{u}_i^2$ on the x's (8.32); (iii) $\hat{h}_i = \exp(\hat{g}_i)$ from the fitted values (8.33); (iv) WLS with weights $1/\hat{h}_i$. Step (ii) is footnote 1 of slide 8. Why the log regression works: under (8.30)

$$u^2 = \sigma^2 \exp(\delta_0 + \delta_1 x_1 + \dots)\, v, \quad E(v \mid x) = 1 \;\Rightarrow\; \log u^2 = \alpha_0 + \delta_1 x_1 + \dots + e \qquad (8.31),$$

so the **slopes** are estimated consistently; the intercept absorbs $\log \sigma^2 + E \log v$ (for normal errors $v \sim \chi_1^2$ and $E \log \chi_1^2 = -1.27$), which does not matter because a constant factor in h cancels from WLS.

What OLS loses. For a fixed design and $h = \exp(\alpha_1 x)$: $\mathrm{Var}(\hat{\beta}_{OLS}) = \sigma^2 (X'X)^{-1} X'\Omega X (X'X)^{-1}$ versus (5).



Exact variance ratio, 100 normal-quantile design points standing in for $x_i \sim N(0,1)$ (slide 8); (5) vs the OLS sandwich.

At $\alpha_1 = 0$ the ratio is 1 (OLS is GLS); at the slide's $\alpha_1 = 1$ the OLS slope variance is **4.38 times** the GLS variance, and the curve climbs steeply after that because $e^{\alpha_1 x}$ makes a few large-x points carry almost all the noise. The right panel shows why: GLS gives those points weights near zero, OLS gives them weight one.

Robust standard errors fix the inference, not the efficiency. OLS with HC standard errors (W §8.2) is valid whatever Ω is; GLS/WLS is *better* only if the variance model is right (W §8.4c, p. 261). Example 8.6's WLS SE is 40% smaller *under* $h = inc$.

5 · C — the US airline cost example (slides 11–15)

Greene's data: 6 airlines × 15 years (1970–1984), 90 observations. Mean: log cost on log output, log fuel price and load factor; variance: $h = \exp(\alpha_1 + \alpha_2\, LF)$. Starting values (slide 12): OLS $-10.7112 + 1.1113 \log Q - 0.5049 \log PF + 2.0197\, LF$ and $\log \hat{e}^2 = -6.3723 + 1.7032\, LF$. Priors $N(0, 10^4 I)$ for β and α; RW-MH with variances (0.5, 0.5); $(M_0, \text{lag}, M) = (1000, 100, 1000)$ (slide 13). Posterior (slide 14):

	$E(\theta \mid y)$	sd	2.5%	50%	97.5%
β_1 intercept	−10.733	0.211	−11.146	−10.730	−10.317
β_2 log output	1.117	0.016	1.086	1.117	1.148
β_3 log fuel price	−0.504	0.023	−0.549	−0.503	−0.460
β_4 load factor	1.895	0.374	1.178	1.887	2.626
α_1	−3.968	0.157	−4.256	−3.964	−3.667
α_2	7.085	3.293	0.572	7.077	13.731

Reading. $e^{\alpha_1} = e^{-3.968} = 0.0189$ is the variance scale at $LF = 0$ (slide 15's histogram of e^{α_1} is centred near 0.02). Load factor ranges over about 0.4–0.7 (slide 12); a 0.1 rise multiplies the error variance by $e^{0.1 \times 7.085} = e^{0.7085} = 2.03$. The 95% interval of α_2 excludes 0, so heteroskedasticity in LF is supported, but it is wide (0.57 to 13.7): the variance function is far less precisely learned than the mean. The β's barely move from OLS, as expected — heteroskedasticity changes precision, not location.

Links

- Unit: L07-00-map · next L07-02-student-t-errors (Case II) · L07-03-ar1-errors (non-diagonal Ω) · L07-07-formulas · L07-08-pitfalls · L07-09-python §1.
- Class 1: L01 §6 (OLS = MLE under spherical errors — the special case $\Omega = \text{I}$).
- Class 2: L02 §3 (the mcycle noise is heteroskedastic: the pairs-bootstrap band vs the formula band).
- Later: SUR (GLM3 slides 31–32) is GLS with $\Omega = I_n \otimes \Sigma$; it returns in Class 9 and is used here only for the multinomial probit (L07-06-probit-ordered-multinomial §6).
- Rules pack: rules/03 (the unit's panel/hierarchical topic).
- **Practice** (statements in the books): W 12.C2 (p. 397) and 12.C10 (p. 399) are the AR(1) analogues (L07-03-ar1-errors); G 8.5 (p. 132) is the Case II sampler of the next note. Case I has no assigned exercise: re-derive (11)–(12) for $h = (z_i'\alpha)^2$ as a self-check.

GLM3 slides 16–21 · The Student-t is a normal with a random variance, and that makes it Gibbs-friendly

> **Claim.** If $\epsilon_i \mid \omega_i \sim N(0, \sigma^2\omega_i)$ and $\omega_i \sim IG(\nu/2, \nu/2)$, then marginally $\epsilon_i \sim t_\nu(0, \sigma^2)$ — the t is a **scale mixture of normals**. Keeping the ω_i in the sampler turns a non-conjugate t regression into the heteroskedastic regression of L07-01-general-covariance-gls with **conjugate** inverse-gamma ω_i: each observation gets the weight $E(\omega_i^{-1} \mid \cdot) = (\nu+1)/(\nu + \epsilon_i^2/\sigma^2)$, which is below 1 exactly when $|\epsilon_i| > \sigma$ — **outliers are downweighted automatically**. The degrees of freedom ν get an exponential prior and a Metropolis step. Prev: L07-01-general-covariance-gls · Next: L07-03-ar1-errors · Map: L07-00-map

Notation.

- $\omega_i > 0$: observation-specific variance multiplier (slides 16–18); $\lambda_i = \omega_i^{-1}$ its precision (G writes λ_i, eq. 8.4).
- $\nu > 0$: degrees of freedom; ν^0: prior mean of ν (slide 17); σ^2: scale (not the variance).
- $IG(a, b)$: density $\frac{b^a}{\Gamma(a)} w^{-a-1} e^{-b/w}$; $G(a, b)$: density $\frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda}$ (rate b). $\omega \sim IG(a, b) \iff \omega^{-1} \sim G(a, b)$.
- $t_\nu(0, \sigma^2)$: density $\propto (1 + e^2/(\nu\sigma^2))^{-(\nu+1)/2}$; variance $\sigma^2\nu/(\nu-2)$ for $\nu > 2$.

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1 · U — fat tails as "some observations are noisier"

A Gaussian likelihood treats a residual of 5σ as nearly impossible ($\phi(5)/\phi(0) = e^{-12.5}$), so OLS bends the line toward it. A t likelihood says such residuals happen. The mixture reading gives the mechanism: every observation draws its own variance $\sigma^2\omega_i$; a few draw large ones. After seeing the data, an observation with a huge residual is inferred to have had a large ω_i, and counts little in the β update. G §4.5 (p. 51) motivates the prior on ω_i (his λ_i) by **exchangeability**: each λ_i belongs to observation i, but nothing tells us in advance which observations are the noisy ones, so they are drawn iid from one distribution.

2 · C — the scale-mixture identity (slide 16; G p. 52)

Joint density of (ϵ, ω) given σ^2, from $N(0, \sigma^2\omega)$ times $IG(\nu/2, \nu/2)$:

$$f(\epsilon, \omega \mid \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2\omega}} e^{-\epsilon^2/(2\sigma^2\omega)} \cdot \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)} \omega^{-\nu/2-1} e^{-\nu/(2\omega)}. \tag{1}$$

Collect the powers of ω and the exponents:

$$f(\epsilon, \omega \mid \sigma^2) \propto \omega^{-(\nu+1)/2-1} \exp\Big\{ -\frac{1}{\omega} \cdot \frac{\nu + \epsilon^2/\sigma^2}{2} \Big\}. \tag{2}$$

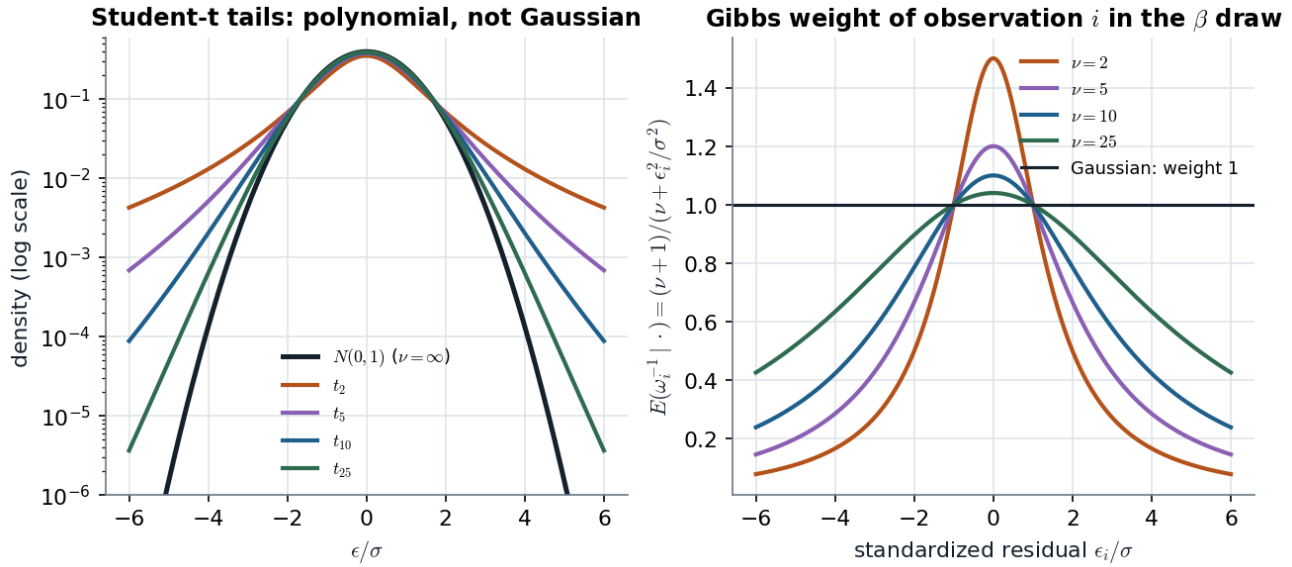
As a function of ω, (2) is an $IG(\frac{\nu+1}{2}, \frac{\nu+\epsilon^2/\sigma^2}{2})$ kernel. Its integral over ω is $\Gamma(a)/b^a$ with $a = (\nu+1)/2$, $b = (\nu + \epsilon^2/\sigma^2)/2$:

$$f(\epsilon \mid \sigma^2) = \int_0^\infty f(\epsilon, \omega \mid \sigma^2)\, d\omega \propto \Big(\frac{\nu + \epsilon^2/\sigma^2}{2}\Big)^{-(\nu+1)/2} \propto \Big(1 + \frac{\epsilon^2}{\nu\sigma^2}\Big)^{-(\nu+1)/2}, \tag{3}$$

the $t_\nu(0, \sigma^2)$ kernel. ■ (Checked by quadrature at 20 (ν, ϵ) pairs, `L07_python_checks.py` §2.)

Conditionally heteroskedastic, marginally homoskedastic (G p. 52): $\mathrm{Var}(\epsilon_i \mid \omega_i) = \sigma^2\omega_i$ differs across i, but every ϵ_i has the same marginal $t_\nu(0, \sigma^2)$.

Slip on slide 6. It writes Case II as "$\omega_i^{-1} \sim IG(\nu/2, \nu/2)$"; slides 16 and 18 (and G's $\lambda_i \sim G(\nu/2, \nu/2)$) use $\omega_i \sim IG$, i.e. $\omega_i^{-1} \sim G(\nu/2, \nu/2)$ — a **gamma**, not an inverse gamma. With $\omega_i^{-1} \sim IG$ the marginal would not be a t.



Left: (3) against the normal, log scale. Right: the conditional mean (5) of ω_i^{-1} — the weight observation i carries in the next β *draw.*

On the log scale the normal is a parabola and the t's are much flatter: at $|e| = 6$ the t_5 density is 6.9×10^{-4} against the normal's 6.1×10^{-9}, about 110,000 times larger. On the right, every curve passes through 1 at $|e/\sigma| = 1$ ($(\nu+1)/(\nu+1)$), exceeds 1 inside, and decays like ν/e^2 outside: a 6σ residual gets weight $3/38 = 0.079$ at $\nu = 2$ and $26/61 = 0.43$ at $\nu = 25$. Larger ν flattens the curve toward the Gaussian constant weight.

3 · C — full conditionals (slides 18; G p. 115)

Model: $y_i = x_i'\beta + \epsilon_i$, $\epsilon_i \mid \omega_i \sim N(0, \sigma^2\omega_i)$, $\omega_i \sim IG(\nu/2, \nu/2)$; priors as in L07-01-general-covariance-gls (β normal, σ^2 inverse gamma), plus a prior on ν.

ω_i. Given everything else, only factor i of the likelihood and the prior of ω_i involve it; with $\epsilon_i = y_i - x_i'\beta$ this is (2):

$$\omega_i \mid y, \beta, \sigma^2, \nu \sim IG\Big(\frac{\nu+1}{2}, \frac{\nu+\epsilon_i^2/\sigma^2}{2}\Big), \quad \text{equivalently} \quad \lambda_i \sim G\Big(\frac{\nu+1}{2}, \frac{\nu+\epsilon_i^2/\sigma^2}{2}\Big). \tag{4}$$

The gamma mean (shape/rate) gives the weight:

$$E(\omega_i^{-1} \mid \cdot) = \frac{(\nu+1)/2}{(\nu+\epsilon_i^2/\sigma^2)/2} = \frac{\nu+1}{\nu+\epsilon_i^2/\sigma^2}. \tag{5}$$

β and σ². Given $\Omega = \mathrm{diag}(\omega_i)$ the model is Case I's: (7)–(8) of note 01 with that Ω, i.e. $V_1^{-1} = V_0^{-1} + \sigma^{-2}\sum_i x_i x_i'/\omega_i$, $V_1^{-1}\beta_1 = V_0^{-1}\beta_0 + \sigma^{-2}\sum_i x_i y_i/\omega_i$, and $\nu_1 s_1^2 = \nu_0 s_0^2 + \sum_i \epsilon_i^2/\omega_i$ (G's $\delta_1 = \delta_0 + (y - X\beta)'\Lambda(y - X\beta)$, p. 115).

ν. Only the ω_i priors and $p(\nu)$ involve ν. Multiply the n IG densities:

$$\prod_{i=1}^{n} \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)} \omega_i^{-\nu/2-1} e^{-\nu/(2\omega_i)} = \Big[\frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)}\Big]^n \exp\Big\{-\frac{\nu}{2}\sum_i \big(\ln\omega_i + \omega_i^{-1}\big)\Big\} \prod_i \omega_i^{-1}.$$

With the exponential prior $p(\nu) \propto e^{-\nu/\nu^0}$ (§4), dropping $\prod_i \omega_i^{-1}$ (free of ν):

$$p(\nu \mid \omega) \propto \Big[\frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)}\Big]^n \exp\Big\{-\Big[\frac{1}{\nu^0} + \frac{1}{2}\sum_{i=1}^{n}\big(\ln\omega_i + \omega_i^{-1}\big)\Big]\nu\Big\}, \quad \nu > 0, \tag{6}$$

exactly slide 18. (6) is no standard family (ν sits inside a Gamma function): **ν needs an MH step**. G (§8.1.2) avoids it by fixing ν (= 5 in his example); G §4.6 (p. 53) is the hierarchical version, a prior on ν.

4 · C — the prior of ν, and Geweke's warning (slide 17)

$\nu \sim G(1, 1/\nu^0) \equiv$ Exp with mean ν^0, so $E(\nu) = \sqrt{V(\nu)} = \nu^0$. With $\nu^0 = 25$:

$$P(\nu \le 10) = 1 - e^{-10/25} = 1 - 0.6703 = 0.3297, \qquad P(\nu \ge 40) = e^{-40/25} = 0.2019.$$

One third of the prior mass on fat tails, one fifth on "practically normal" (the t is close to the normal in its centre once $\nu \gtrsim 30$, G p. 53): this is what "substantial prior on both" means on slide 17.

Geweke (1993), slide 17: with $p(\beta) \propto 1$, $E(\beta \mid y)$ does not exist unless $P\{\nu \in (0,2]\} = 0$, and $V(\beta \mid y)$ unless $P\{\nu \in (0,4]\} = 0$. The t's own moments exist only for $\nu > 1$ (mean) and $\nu > 2$ (variance); under a flat β prior the posterior inherits that tail problem. **Medians and quantiles still exist** — report them, or use a proper prior on β.

5 · C — the simulated example (slides 19–21) and a re-run

Design (slide 19). $n = 100$, $\nu = 10$, $\sigma^2 = 1$, $\beta = (0.5, 1.0, 2.0)$; priors $\beta_0 = 0$, $V_0 = 10I_3$, $\nu_0 = 5$, $s_0^2 = 1$, $\nu^0 = 25$; start at OLS, $\nu = 10$, $\omega_i = 1$; RW proposal $q(\nu \mid \nu^{(j-1)}) = N(\nu^{(j-1)}, 0.25^2)$; burn-in 10,000, every 100th draw to $M = 1000$.

What the slides show (20–21). β and σ² chains mix well and cover the true values; **the ν chain does not** — its ACF is still about 0.5 at lag 30 even after thinning by 100, and its histogram spreads over 5–60 around the true 10. ν is weakly identified by 100 observations: only the few large residuals speak about the tails.

Re-run (`L07_python_checks.py` §2; same design, one observation shifted by +15 to create an outlier; 3,000 burn-in + 20,000 draws, no thinning):

	intercept	β_1	β_2
true	0.5	1.0	2.0
OLS	0.54	1.07	2.03
t-posterior mean	0.43	0.87	1.93

Posterior mean of ν: **3.8** (the outlier drags the tails fatter than the true 10); MH acceptance 0.83 (a 0.25 step is small for ν); posterior mean weight of the outlier **0.018** against a median weight of 1.06. Dropping the outlier moves the OLS intercept by more than twice what it moves the t posterior (asserted). That is the robustness: the t fit **ignores the outlier by itself**, without anyone deleting it.

6 · C — Greenberg's Vella–Verbeek comparison (G §8.1.1–8.1.2, pp. 112–116)

Union-membership coefficient β_U in a log-wage regression:

errors	$E(\beta_U \mid y)$	sd	95% interval	$E(\sigma^2 \mid y)$	$\log_{10}$ marginal likelihood
Gaussian	0.129	0.036	(0.059, 0.198)	0.165	−152.78
Student-t, $\nu = 5$	0.128	0.033	(0.062, 0.192)	0.105	−147.95

The coefficient hardly moves; **σ^2 does, and should**: under t_5 the error variance is $\sigma^2 \cdot 5/3 = 0.105 \times 1.667 = 0.175$, close to the Gaussian 0.165 — the two models describe about the same spread with different scale parameters. The $\log_{10}$ Bayes factor is $-147.95 - (-152.78) = 4.83$ in favour of t errors ("strong evidence", G p. 116). The t sampler's inefficiency factor is higher (1.274 vs 1.074) because the λ_i add autocorrelation.

Links

- Unit: L07-00-map · L07-01-general-covariance-gls (Case I, the conditionals reused in §3) · L07-05-tobit and L07-06-probit-ordered-multinomial (t versions of the latent-data models: G 8.6, 8.7) · L07-07-formulas · L07-08-pitfalls · L07-09-python §2.
- Class 1: L01 §4 (conjugate updating, the gamma–Poisson pattern that (4) repeats).
- Later: the same mixture gives Student-t stochastic volatility in Class 8 (L08).
- **Practice** (statements in the books): G 8.3 (p. 132, t errors with five degrees of freedom and a marginal-likelihood comparison); G 8.5 (p. 132, the sampler of §3); G 4.6 (p. 59, iid ⇒ exchangeable, the assumption behind §1); G 9.3 (p. 151, t errors in the panel model of L07-04-hierarchical-linear-models).

GLM3 slides 22–30 · One parameter ρ fills the whole n × n Ω, and GLS is quasi-differencing

> **Claim.** AR(1) errors $\epsilon_t = \rho\epsilon_{t-1} + u_t$ give $\mathrm{Corr}(\epsilon_t, \epsilon_{t+k}) = \rho^k$: a full $n \times n$ covariance pinned down by **one** parameter. Ω_ρ^{-1} is **tridiagonal**, so GLS is OLS on **quasi-differenced** data $y_t - \rho y_{t-1}$ (Prais–Winsten keeps the first row, rescaled; Cochrane–Orcutt drops it). In the Gibbs sampler β and σ^2 keep the note-01 conditionals with Ω_ρ; ρ is drawn either by MH on its exact conditional or — after "a small sin", conditioning on the first error — **exactly** from a truncated normal, by the inverse-CDF formula of slide 26. How much OLS loses, and how wrong its standard errors are, depends on ρ **times the persistence of the regressor**. Prev: L07-02-student-t-errors · Next: L07-04-hierarchical-linear-models · Map: L07-00-map

Notation.

- $y_t = x_t'\beta + \epsilon_t$, $t = 1, \ldots, n$; $\epsilon_t = \rho\epsilon_{t-1} + u_t$, $u_t \sim N(0, \sigma^2)$ iid, $|\rho| < 1$ (slide 22). W writes $u_t = \rho u_{t-1} + e_t$ (12.26), σ_e^2 for our σ^2.
- Ω_ρ: the $n \times n$ matrix with entries $\rho^{|i-j|}/(1-\rho^2)$, so $\mathrm{Var}(\epsilon) = \sigma^2\Omega_\rho$ (slide 23).
- $\tilde{y}_t = y_t - \rho y_{t-1}$, $\tilde{x}_t = x_t - \rho x_{t-1}$: quasi-differences (W 12.29).
- ρ_0, V_ρ: prior mean and variance of ρ, truncated to $[-1, 1]$ (slide 24).
- §3 only: x_t a stationary AR(1) with coefficient φ and variance σ_x^2.

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1 · C — the covariance of AR(1) errors (slide 22)

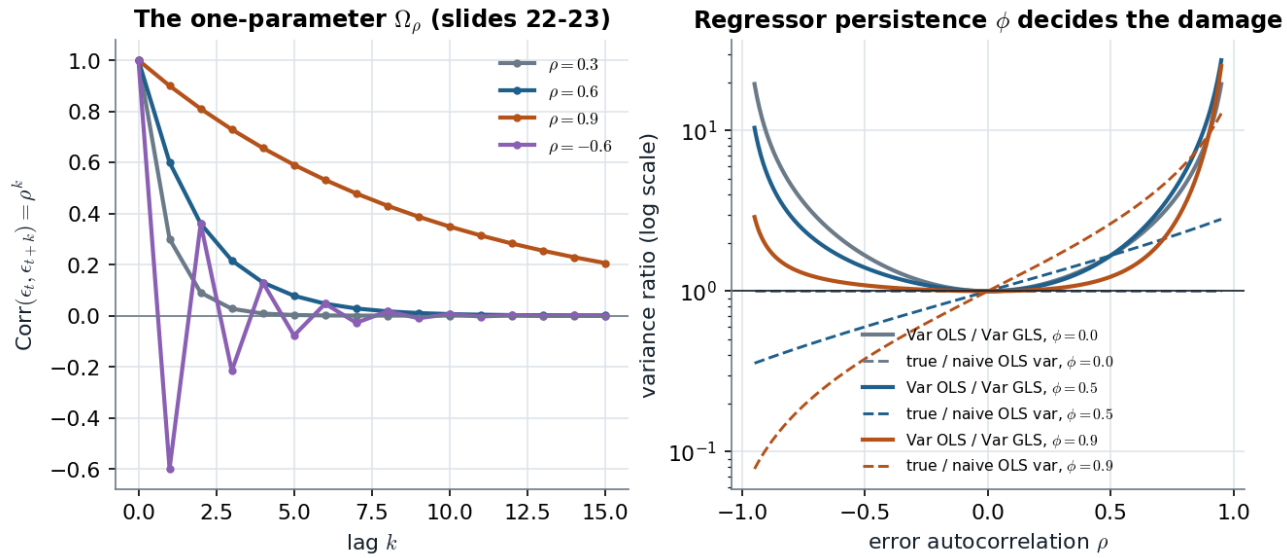
Take variances in $\epsilon_t = \rho\epsilon_{t-1} + u_t$, with u_t independent of ϵ_{t-1} and stationarity $\mathrm{Var}(\epsilon_t) = \mathrm{Var}(\epsilon_{t-1}) = \gamma_0$:

$$\gamma_0 = \rho^2\gamma_0 + \sigma^2 \;\Rightarrow\; \gamma_0 = \frac{\sigma^2}{1-\rho^2} \qquad \text{(W 12.27)}. \tag{1}$$

Substitute the recursion k times: $\epsilon_{t+k} = \rho^k\epsilon_t + \sum_{j=0}^{k-1}\rho^j u_{t+k-j}$; the sum is independent of ϵ_t, so

$$\mathrm{Cov}(\epsilon_t, \epsilon_{t+k}) = \rho^k\gamma_0 = \rho^k\frac{\sigma^2}{1-\rho^2}, \qquad E(\epsilon_t) = 0. \tag{2}$$

Stacking (2) gives $\epsilon \mid \sigma^2, \rho \sim N(0, \sigma^2\Omega_\rho)$ with $(\Omega_\rho)_{ij} = \rho^{|i-j|}/(1-\rho^2)$ (slide 23), and $y \mid X, \beta, \sigma^2, \rho \sim N(X\beta, \sigma^2\Omega_\rho)$: model (1) of L07-01-general-covariance-gls with a one-parameter Ω. The full conditionals of β and σ^2 are (7)–(8) of that note with $\Omega = \Omega_\rho$ (slide 23).



Left: (2) for four ρ*. Right: the large-sample ratios (10) and (11), exact expressions; solid = OLS/GLS variance, dashed = true/naive OLS variance.*

The left panel is the whole Ω: geometric decay, alternating in sign for $\rho < 0$. The right panel is §3: with an iid regressor ($\phi = 0$) the naive OLS variance is right (dashed line flat at 1) yet OLS still loses to GLS; with a persistent regressor ($\phi = 0.9$) and positive ρ the naive formula understates the variance (by 9.5 times at $\rho = 0.9$) — standard errors too small by $\sqrt{9.53} = 3.1$.

2 · C — GLS by quasi-differencing (W §12.3a, pp. 382–383)

Ω_ρ^{-1} is tridiagonal. Claim:

$$\Omega_\rho^{-1} = \begin{pmatrix} 1 & -\rho & & & \\ -\rho & 1+\rho^2 & -\rho & & \\ & \ddots & \ddots & \ddots & \\ & & -\rho & 1+\rho^2 & -\rho \\ & & & -\rho & 1 \end{pmatrix}. \tag{3}$$

It factors as $\Omega_\rho^{-1} = P'P$ with the **Prais–Winsten matrix**

$$P = \begin{pmatrix} \sqrt{1-\rho^2} & 0 & \cdots & 0 \\ -\rho & 1 & & \\ & \ddots & \ddots & \\ 0 & & -\rho & 1 \end{pmatrix}. \tag{4}$$

Check (4)⇒(3): column j of P has 1 (or $\sqrt{1-\rho^2}$ for $j = 1$) on the diagonal and $-\rho$ just below. So $(P'P)_{jj} = 1 + \rho^2$ for $1 < j < n$, $(P'P)_{11} = (1-\rho^2) + \rho^2 = 1$, $(P'P)_{nn} = 1$, $(P'P)_{j,j+1} = -\rho$. ■ (Both identities asserted in `L07_python_checks.py` §3.)

P applied to the model. Row 1 of Py: $\sqrt{1-\rho^2}\, y_1$; row $t \geq 2$: $y_t - \rho y_{t-1}$. So

$$\sqrt{1-\rho^2}\, y_1 = \sqrt{1-\rho^2}\, x_1'\beta + \sqrt{1-\rho^2}\, \epsilon_1 \qquad \text{(W 12.31)}, \tag{5}$$

$$y_t - \rho y_{t-1} = (x_t - \rho x_{t-1})'\beta + u_t, \quad t \geq 2 \qquad \text{(W 12.28, 12.32)}. \tag{6}$$

The errors are now white: $u_t = \epsilon_t - \rho\epsilon_{t-1}$ for $t \geq 2$, and by (1) $\mathrm{Var}(\sqrt{1-\rho^2}\epsilon_1) = (1-\rho^2)\sigma^2/(1-\rho^2) = \sigma^2$. OLS on (5)–(6) is the BLUE (W p. 383). With an intercept, the constant column becomes $\sqrt{1-\rho^2}$ in row 1 and $1-\rho$ in rows $t \geq 2$ (W 12.33).

Feasible GLS (W §12.3b, p. 383). (i) OLS residuals $\hat{u}_t$; (ii) $\hat{\rho}$ from regressing $\hat{u}_t$ on $\hat{u}_{t-1}$ (W 12.14); (iii) OLS on (5)–(6) with $\hat{\rho}$. **Prais–Winsten** keeps (5), **Cochrane–Orcutt** drops it; both are usually iterated to convergence of $\hat{\rho}$. Asymptotically the same; in a small sample the first row can matter.

Worked: W Example 12.4 (Table 12.1, p. 384), BARIUM, $n = 131$**.** Log Chinese barium imports on log chemical production, log gasoline production, log exchange rate and three antidumping dummies. Reproduced exactly from the data (`L07_python_checks.py` §3; iterated PW converges in 11 steps):

	OLS	Prais–Winsten
log(chempi)	3.12 (0.48)	2.94 (0.63)
afdec6	−0.565 (0.286)	−0.577 (0.342)
intercept	−17.80 (21.05)	−37.08 (22.78)
$\hat{\rho}$	—	0.293

Every PW standard error is larger: OLS ignores the positive serial correlation and understates sampling variation (W p. 384), and t_{afdec6} falls to $-0.577/0.342 = -1.69$. Iterated Cochrane–Orcutt gives $\hat{\rho} = 0.293$ and 2.95 for log(chempi) — close, as expected with 131 points.

T §2.9 (pp. 90–93), the same issue in finance. Regressing the 3-year on the 1-year Treasury rate in levels gives $r_{3t} = 0.832 + 0.930 r_{1t} + e_t$ with $R^2 = 96.5\%$ (2.44) but unit-root-like residuals; in changes, $c_{3t} = 0.792 c_{1t} + e_t$ (2.45) still has serially correlated residuals, which T models jointly as MA(1) errors (2.46). "One of the most commonly misused econometric models because the serial dependence in e_t is often overlooked" (T p. 90).

3 · C/G — how much OLS loses, in closed form

Single demeaned regressor, x_t AR(1)(φ) with variance σ_x^2, independent of the errors. OLS: $\hat{\beta} - \beta = \sum_t x_t\epsilon_t / \sum_t x_t^2$, with $\sum_t x_t^2 \approx n\sigma_x^2$. The numerator's variance:

$$\mathrm{Var}\Big(\sum_t x_t\epsilon_t\Big) = \sum_s\sum_t E(x_s x_t)E(\epsilon_s\epsilon_t) = \sum_s\sum_t \sigma_x^2\phi^{|s-t|}\gamma_0\rho^{|s-t|}. \tag{7}$$

For large n, $\sum_s\sum_t(\rho\phi)^{|s-t|} \approx n\big[1 + 2\sum_{k\geq 1}(\rho\phi)^k\big] = n\frac{1+\rho\phi}{1-\rho\phi}$. Substitute into (7) and divide by $(n\sigma_x^2)^2$:

$$\mathrm{Var}(\hat{\beta}_{OLS}) \approx \frac{\gamma_0}{n\sigma_x^2}\cdot\frac{1+\rho\phi}{1-\rho\phi}. \tag{8}$$

The **naive** formula $\hat{\gamma}_0/\sum x_t^2 \approx \gamma_0/(n\sigma_x^2)$ drops the last factor. **GLS** is OLS on (6): $\tilde{x}_t = x_t - \rho x_{t-1}$ has variance $\sigma_x^2(1+\rho^2-2\rho\phi)$ and the error u_t has variance $\sigma^2 = \gamma_0(1-\rho^2)$ by (1), so

$$\mathrm{Var}(\hat{\beta}_{GLS}) \approx \frac{\gamma_0(1-\rho^2)}{n\sigma_x^2(1+\rho^2-2\rho\phi)}. \tag{9}$$

Dividing (8) by the naive formula, and (8) by (9):

$$\frac{\text{true Var}_{OLS}}{\text{naive Var}_{OLS}} = \frac{1+\rho\phi}{1-\rho\phi}, \tag{10}$$

$$\frac{\text{Var}_{OLS}}{\text{Var}_{GLS}} = \frac{1+\rho\phi}{1-\rho\phi}\cdot\frac{1+\rho^2-2\rho\phi}{1-\rho^2}. \tag{11}$$

Numbers at $\rho = 0.9$: with $\phi = 0.9$, (10) $= 1.81/0.19 = 9.53$; with $\phi = 0$, (10) $= 1$ but (11) $= (1 + 0.81)/(1 - 0.81) = 9.53$ (asserted in `make_figures.py`). **Reading:** the bias of the usual SE comes from the *product* $\rho\varphi$ (both series persistent in the same direction, the typical macro case); the efficiency loss exists even when the SE is right.

4 · C — the conditional of ρ (slides 24–26)

Exact (slide 24). With prior $\rho \sim N_{[-1,1]}(\rho_0, V_\rho)$ and $\epsilon = y - X\beta$:

$$p(\rho \mid \epsilon, \sigma^2) \propto |\Omega_\rho|^{-1/2} \exp\Big\{-\frac{1}{2\sigma^2}\epsilon'\Omega_\rho^{-1}\epsilon\Big\} \exp\Big\{-\frac{(\rho-\rho_0)^2}{2V_\rho}\Big\} 1(|\rho|<1). \tag{12}$$

Slip on slide 24: its exponent reads $-0.5\,\epsilon'\Omega_\rho^{-1}\epsilon$ without σ^{-2}; the factor is needed since $\text{Var}(\epsilon) = \sigma^2\Omega_\rho$ (and the prior term is written $\rho^2 - 2\rho\rho_0$, the same up to a constant). Because $|\Omega_\rho| = (1-\rho^2)^{-1}$ and (3) is quadratic in ρ except for its corners, (12) is not a standard density: **MH step**.

"A small sin" (slide 25). Treat ϵ_1 as given (drop its factor). Then the likelihood is that of the regression $\epsilon_t = \rho\epsilon_{t-1} + u_t$, $t = 2, \dots, n$. With $\tilde\epsilon = (\epsilon_1, \dots, \epsilon_{n-1})'$ and $\epsilon = (\epsilon_2, \dots, \epsilon_n)'$, the exponent in ρ is

$$-\frac{1}{2}\Big[\frac{(\epsilon-\rho\tilde\epsilon)'(\epsilon-\rho\tilde\epsilon)}{\sigma^2} + \frac{(\rho-\rho_0)^2}{V_\rho}\Big] = -\frac{1}{2}\Big[\rho^2\Big(\frac{\tilde\epsilon'\tilde\epsilon}{\sigma^2}+\frac{1}{V_\rho}\Big) - 2\rho\Big(\frac{\tilde\epsilon'\epsilon}{\sigma^2}+\frac{\rho_0}{V_\rho}\Big)\Big] + c,$$

a normal kernel restricted to $[-1, 1]$:

$$\rho \mid y, X, \beta, \sigma^2 \sim N_{[-1,1]}(\rho_1, \bar V_\rho), \quad \bar V_\rho = (V_\rho^{-1} + \sigma^{-2}\tilde\epsilon'\tilde\epsilon)^{-1}, \quad \rho_1 = \bar V_\rho(V_\rho^{-1}\rho_0 + \sigma^{-2}\tilde\epsilon'\epsilon). \tag{13}$$

The "sin" costs one observation's information out of n — W's Cochrane–Orcutt choice, now in a Bayesian sampler.

Drawing from $N_{[a,b]}(\mu, \sigma^2)$ (slide 26). For $y \in [a, b]$ the CDF is

$$u = P(Y \le y) = \frac{\Phi\big(\frac{y-\mu}{\sigma}\big) - A}{B - A}, \qquad A = \Phi\Big(\frac{a-\mu}{\sigma}\Big),\ B = \Phi\Big(\frac{b-\mu}{\sigma}\Big). \tag{14}$$

Solve (14) for y:

$$y = \mu + \sigma\,\Phi^{-1}(uB + (1-u)A), \qquad u \sim U(0,1). \tag{15}$$

(15) is exact (the inverse-CDF method); `L07_python_checks.py` §3 confirms it equals scipy's `truncnorm.ppf` on 999 quantiles. One practical caveat: if $[a, b]$ lies deep in the upper tail, A and B both round to 1; reflect ($y \mapsto -y$) and work in the lower tail — the `rtruncnorm` helper does this, and it is the sampler reused by the Tobit and probit notes.

5 · C — the simulated example (slides 27–30) and a re-run

Design (slide 27). $n = 200$, $\rho = 0.9$, $\sigma^2 = 1$, $\beta = (2.0, 4.0)$ on two regressors; priors $\beta_0 = 0$, $V_0 = 100 I_2$, $\nu_0 = 5$, $s_0^2 = 1$, $\rho_0 = 0.9$, $V_\rho = 100$; start at GLS $\widehat{\beta} = (1.94, 3.93)$, $\widehat{\sigma}^2 = 1.12$, $\widehat{\rho} = 0.94$; burn-in 1000, keep 2000 draws. Slides 29–30's chains for $\beta_1, \beta_2, \sigma^2$ mix quickly; ρ's posterior sits in 0.85–1.00.

Re-run (`L07_python_checks.py` §3: β and σ^2 from the exact Ω conditionals via (3), ρ from (13) by (15); 1,000 burn-in + 5,000 draws): posterior means $\beta = (1.97, 4.06)$, $\sigma^2 = 1.03$, $\rho = 0.871$. No MH needed anywhere — the small sin buys an exact Gibbs step.

Links

- Unit: L07-00-map · L07-01-general-covariance-gls (the general-Ω conditionals) · L07-05-tobit (the truncated-normal draw (15) again) · L07-07-formulas · L07-08-pitfalls · L07-09-python §3.
- Class 1: L01 §7 (AR(1): stationarity $|\phi| < 1$, the equilibrium variance (1), OLS = conditional MLE — the same "drop the first observation" choice as (13)).
- Class 8: L08 (ARMA errors and model building in their own right).
- **Practice** (statements in the books): W 12.C2 (p. 397, WAGEPRC, testing and correcting for AR(1) errors); W 12.C10 (p. 399, PHILLIPS, Prais–Winsten / Cochrane–Orcutt).

HLM slides 3–23 · Partial pooling: every unit's estimate is a precision-weighted compromise with the population

> **Claim.** With N units observed T times there are three models: **pooled** (one line for all), **individual effects** (one intercept per unit) and **random coefficients** (one line per unit). Giving the unit parameters a common prior whose mean and variance are themselves estimated — the **hierarchical prior** — makes each unit's posterior mean a weighted average of its own estimate and the population mean, with weight $TV_\alpha/(TV_\alpha+\sigma^2)$ on its own data. Units with little or noisy data borrow strength from the others; the data decide how much through the between-unit variance (V_α, or τ in the eight schools). Every conditional is normal, inverse gamma or Wishart: **a pure Gibbs sampler**. Prev: L07-03-ar1-errors · Next: L07-05-tobit · Map: L07-00-map

Notation (slide 3).

- $i=1,\dots,N$ units, $t=1,\dots,T$ occasions; y_{it} response, x_{it} $(k\times 1)$ regressors.
- $y_i=(y_{i1},\dots,y_{iT})'$, X_i $(T\times k)$ with rows x_{it}'; $X_i=(1_T,\tilde{X}_i)$ splits the constant from the other regressors $\tilde{X}_i$; y, X stack the units.
- $\beta=(\alpha,\tilde{\beta})$ common coefficients; α_i unit intercepts; $\beta_i=(\alpha_i,\tilde{\beta}_i)$ unit coefficient vectors.
- First-level prior parameters μ_α, V_α (intercepts) or β_0, V_0 (coefficient vectors); second-level (hyper)priors $\mu_\alpha\sim N(\mu_0,V_\mu)$, $V_\alpha\sim IG(\eta_0/2,\eta_0\sigma_0^2/2)$, $\beta_0\sim N(\beta_{00},V_\beta)$, $V_0^{-1}\sim W(\nu_0,V_{00}^{-1})$.
- Eight schools: y_i estimated effect, σ_i its standard error, θ_i true effect, μ,τ population mean and sd (slide 23).

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1 · U — pool, separate, or compromise

Pooling all 427 weight-gain measurements of 68 pregnant women into one regression ignores that some women gain systematically more; fitting 68 separate lines uses 5 to 7 points per line and produces wild intercepts (from −14.4 to 5.2). The hierarchical model says the 68 lines are draws from a population of lines: each woman's line is estimated from her data **and** from what the other 67 say about the population. G §4.6 (p. 52) gives the formal shape: data $f(y\mid\theta)$, prior $\pi(\theta\mid\theta_0)$, hyperprior $\pi(\theta_0\mid\theta_{00})$ — and notes that θ_0 "is not identified" by itself (the likelihood does not depend on it) but is introduced "to facilitate computations or modeling".

2 · C — three models (slides 4–6)

$$\text{Pooled:}\qquad y_i\sim N(X_i\beta,\sigma^2 I_T),\quad \beta=(\alpha,\tilde{\beta}). \tag{1}$$

$$\text{Individual effects:}\qquad y_i\sim N(1_T\alpha_i+\tilde{X}_i\tilde{\beta},\sigma^2 I_T). \tag{2}$$

$$\text{Random coefficients:}\qquad y_i\sim N(X_i\beta_i,\sigma^2 I_T),\quad \beta_i=(\alpha_i,\tilde{\beta}_i). \tag{3}$$

(1) has k mean parameters, (2) has $N+k-1$, (3) has Nk. Slides 5–6 draw them as graphs: in the hierarchical versions an arrow runs from a population node (α^*, or (β,V_β)) into every unit's coefficient.

Pooled conditionals (slides 7–8). Stacking, $y\sim N(X\beta,\sigma^2 I_{TN})$ and the conditionals are note 01's (7)–(8) with $\Omega=I$ and $n=TN$: $V_1^{-1}=V_0^{-1}+\sigma^{-2}X'X$, $V_1^{-1}\beta_1=V_0^{-1}\beta_0+\sigma^{-2}X'y$, $\nu_1=\nu_0+TN$, $\nu_1 s_1^2=\nu_0 s_0^2+(y-X\beta)'(y-X\beta)$.

Individual effects, non-hierarchical prior (slides 9–10). Stack (2) with $\beta^* = (\alpha_1, \dots, \alpha_N, \tilde\beta')'$ and the dummy design

$$X^* = \begin{pmatrix} 1_T & 0 & \cdots & 0 & \tilde X_1 \\ 0 & 1_T & \cdots & 0 & \tilde X_2 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1_T & \tilde X_N \end{pmatrix}; \tag{4}$$

then it is again the pooled algebra with X^*, $\beta^* \sim N(\beta_0^*, V_0^*)$. **Slip on slide 9:** the last block row of X^* ends in $\tilde X_1$; it is $\tilde X_N$. With a vague V_0^* this is the frequentist **fixed-effects / dummy-variable regression** (W §14.1a, p. 438): each α_i is learned only from unit i.

3 · C — individual effects with a hierarchical prior (slides 11–13)

First level: $\alpha_i \sim N(\mu_\alpha, V_\alpha)$ iid, $\tilde\beta \sim N(\beta_0, V_\beta)$, $\sigma^2 \sim IG(\nu_0/2, \nu_0 s_0^2/2)$; second level: $\mu_\alpha \sim N(\mu_0, V_\mu)$, $V_\alpha \sim IG(\eta_0/2, \eta_0\sigma_0^2/2)$ (slide 11 prints the V_α hyperparameters as ν_0, σ_0^2; slide 13's update uses η_0, η_1 — the letters, not the algebra, differ).

α_i given the rest. Write $r_i = y_i - \tilde X_i\tilde\beta$, so $r_i = 1_T\alpha_i + \epsilon_i$. The terms in α_i:

$$-\frac{1}{2}\Big[\frac{(r_i - 1_T\alpha_i)'(r_i - 1_T\alpha_i)}{\sigma^2} + \frac{(\alpha_i - \mu_\alpha)^2}{V_\alpha}\Big] = -\frac{1}{2}\Big[\alpha_i^2\Big(\frac{T}{\sigma^2} + \frac{1}{V_\alpha}\Big) - 2\alpha_i\Big(\frac{1_T'r_i}{\sigma^2} + \frac{\mu_\alpha}{V_\alpha}\Big)\Big] + c,$$

so (slide 12)

$$\alpha_i \mid \tilde\beta, \sigma^2, \mu_\alpha, V_\alpha \sim N(\tilde\mu_{i\alpha}, \tilde V_\alpha), \quad \tilde V_\alpha^{-1} = V_\alpha^{-1} + \sigma^{-2}T, \quad \tilde V_\alpha^{-1}\tilde\mu_{i\alpha} = V_\alpha^{-1}\mu_\alpha + \sigma^{-2}1_T'(y_i - \tilde X_i\tilde\beta). \tag{5}$$

The partial-pooling weight. Divide the mean in (5) by the precision, with $\bar r_i = 1_T'r_i/T$:

$$\tilde\mu_{i\alpha} = \frac{(T/\sigma^2)\,\bar r_i + (1/V_\alpha)\,\mu_\alpha}{T/\sigma^2 + 1/V_\alpha} = w\,\bar r_i + (1-w)\,\mu_\alpha, \qquad w = \frac{TV_\alpha}{TV_\alpha + \sigma^2}. \tag{6}$$

> **(6) is the whole idea.** $w \to 1$ (no pooling, the fixed-effects estimate $\bar r_i$) when $T \to \infty$ or $V_\alpha \to \infty$; $w \to 0$ (complete pooling, everyone gets μ_α) when $V_\alpha \to 0$ or σ^2 is huge. Because V_α is *estimated*, the data choose the degree of pooling.

$\tilde\beta$ given the intercepts (slide 12). $y_i - 1_T\alpha_i = \tilde X_i\tilde\beta + \epsilon_i$ for all i, a pooled regression: $\tilde V_\beta^{-1} = V_\beta^{-1} + \sigma^{-2}\sum_i \tilde X_i'\tilde X_i$, $\tilde V_\beta^{-1}\beta_1 = V_\beta^{-1}\beta_0 + \sigma^{-2}\sum_i \tilde X_i'(y_i - 1_T\alpha_i)$. **$\sigma^2$** (slide 13): $\nu_1 = \nu_0 + TN$, $\nu_1 s_1^2 = \nu_0 s_0^2 + \sum_i (y_i - 1_T\alpha_i - \tilde X_i\tilde\beta)'(\cdots)$.

μ_α given the intercepts. The α_i are N iid draws from $N(\mu_\alpha, V_\alpha)$ — a normal-mean problem:

$$-\frac{1}{2}\Big[\frac{\sum_i(\alpha_i - \mu_\alpha)^2}{V_\alpha} + \frac{(\mu_\alpha - \mu_0)^2}{V_\mu}\Big] \;\Rightarrow\; \tilde V_\mu^{-1} = V_\mu^{-1} + V_\alpha^{-1}N, \quad \tilde V_\mu^{-1}\mu_1 = V_\mu^{-1}\mu_0 + V_\alpha^{-1}\sum_{i=1}^N \alpha_i. \tag{7}$$

Slip on slide 13: it prints $\sigma^{-2}N$ and $\sigma^{-2}\sum_i \alpha_i$. The α_i scatter around μ_α with variance V_α, not σ^2 — the precision is V_α^{-1}, as (7) and the site's own code (`V = diag(1/diag(iA0+I*iSig))`, `iSig` = the inverse prior variances of the unit coefficients) both say.

V_α (slide 13): $V_\alpha \mid \alpha, \mu_\alpha \sim IG(\eta_1/2, \eta_1\sigma_1^2/2)$, $\eta_1 = \eta_0 + N$, $\eta_1\sigma_1^2 = \eta_0\sigma_0^2 + \sum_i(\alpha_i - \mu_\alpha)^2$ — again the conjugate variance update, with the N intercepts as the "data".

4 · C — random coefficients (slides 14–16)

First level $\beta_i \sim N(\beta_0, V_0)$ iid, $\sigma^2 \sim IG(\eta_0/2, \eta_0 s_0^2/2)$; second level $\beta_0 \sim N(\beta_{00}, V_\beta)$, $V_0^{-1} \sim W(\nu_0, V_{00}^{-1})$. Completing squares exactly as in (5) and (7), now with vectors:

$$\beta_i \mid \sigma^2, \beta_0, V_0 \sim N(\beta_{1i}, V_{1i}), \quad V_{1i}^{-1} = V_0^{-1} + \sigma^{-2}X_i'X_i, \quad V_{1i}^{-1}\beta_{1i} = V_0^{-1}\beta_0 + \sigma^{-2}X_i'y_i; \tag{8}$$

$$\beta_0 \mid \beta_{1:N}, V_0 \sim N(\beta_{01}, \tilde{V}_\beta), \quad \tilde{V}_\beta^{-1} = V_\beta^{-1} + NV_0^{-1}, \quad \tilde{V}_\beta^{-1}\beta_{01} = V_\beta^{-1}\beta_{00} + V_0^{-1}\sum_i \beta_i; \tag{9}$$

$$V_0^{-1} \mid \beta_0, \beta_{1:N} \sim W(\nu_0 + N, V_{11}^{-1}), \quad V_{11}^{-1} \text{ with } V_{11} = V_{00} + \sum_i(\beta_i - \beta_0)(\beta_i - \beta_0)'. \tag{10}$$

(σ^2: $\eta_1 = \eta_0 + TN$, $\eta_1 s_1^2 = \eta_0 s_0^2 + \sum_i(y_i - X_i\beta_i)'(y_i - X_i\beta_i)$.) **Two slips:** slide 15 prints $V_0\beta_0^{-1}$ for $V_0^{-1}\beta_0$ in (8), and slide 16 prints NV_0 for NV_0^{-1} in (9). (10) is the matrix version of the variance update: the scatter of the unit vectors around β_0 updates the Wishart scale. (8) is (6) in matrix form: β_{1i} is a matrix-weighted average of unit i's OLS $(X_i'X_i)^{-1}X_i'y_i$ and β_0.

G §9.3 writes the same model as $y_i = X_i\beta + W_ib_i + u_i$, $b_i \mid D \sim N(0, D)$ (9.11, p. 146): "fixed effects" β and "random effects" b_i in the **statistics** sense (p. 145). His sampler (Algorithm 9.3, p. 148) draws β with the b_i integrated out, $\mathrm{Cov}(y_i) = W_iDW_i' + h_u^{-1}I_T$ (9.15), because β and the b_i are correlated a posteriori — one-block sampling mixes better.

5 · C — eight schools, computed exactly (slide 23)

Model. $y_i \mid \theta_i \sim N(\theta_i, \sigma_i^2)$, $\theta_i \sim N(\mu_\theta, \sigma_\theta^2)$, $\mu_\theta \sim N(0, 10^6)$, $\sigma_\theta \sim U(0, 100)$, $i = 1, \ldots, 8$, with the σ_i known. Below $\mu = \mu_\theta$, $\tau = \sigma_\theta$.

School	A	B	C	D	E	F	G	H
y_i	28.39	7.94	−2.75	6.82	−0.64	0.63	18.01	12.16
σ_i	14.9	10.2	16.3	11.0	9.4	11.4	10.4	17.6

The slides fit it in WinBUGS (slides 24–26). It needs no MCMC: given τ everything is normal.

Step 1, integrate the θ's. $y_i = \theta_i + e_i$ with $\theta_i \sim N(\mu, \tau^2)$ independent of $e_i \sim N(0, \sigma_i^2)$:

$$y_i \mid \mu, \tau \sim N(\mu, \sigma_i^2 + \tau^2), \text{ independent over } i. \tag{11}$$

Step 2, μ given τ. (11) is a normal-mean problem with weights $w_i = 1/(\sigma_i^2 + \tau^2)$; with the $N(0, v_0)$ prior,

$$\mu \mid y, \tau \sim N(\hat{\mu}, V_\mu), \quad V_\mu^{-1} = v_0^{-1} + \sum_i w_i, \quad \hat{\mu} = V_\mu \sum_i w_i y_i. \tag{12}$$

Step 3, θ_i given μ, τ. Normal prior $N(\mu, \tau^2)$, one normal observation y_i with variance σ_i^2:

$$E(\theta_i \mid y, \mu, \tau) = \frac{y_i/\sigma_i^2 + \mu/\tau^2}{1/\sigma_i^2 + 1/\tau^2} = (1 - B_i)y_i + B_i\mu, \quad B_i = \frac{\sigma_i^2}{\sigma_i^2 + \tau^2}, \quad \text{Var} = (1 - B_i)\sigma_i^2. \tag{13}$$

Averaging (13) over (12):

$$E(\theta_i \mid y, \tau) = (1 - B_i)y_i + B_i\hat{\mu}, \qquad \text{Var}(\theta_i \mid y, \tau) = (1 - B_i)\sigma_i^2 + B_i^2 V_\mu. \tag{14}$$

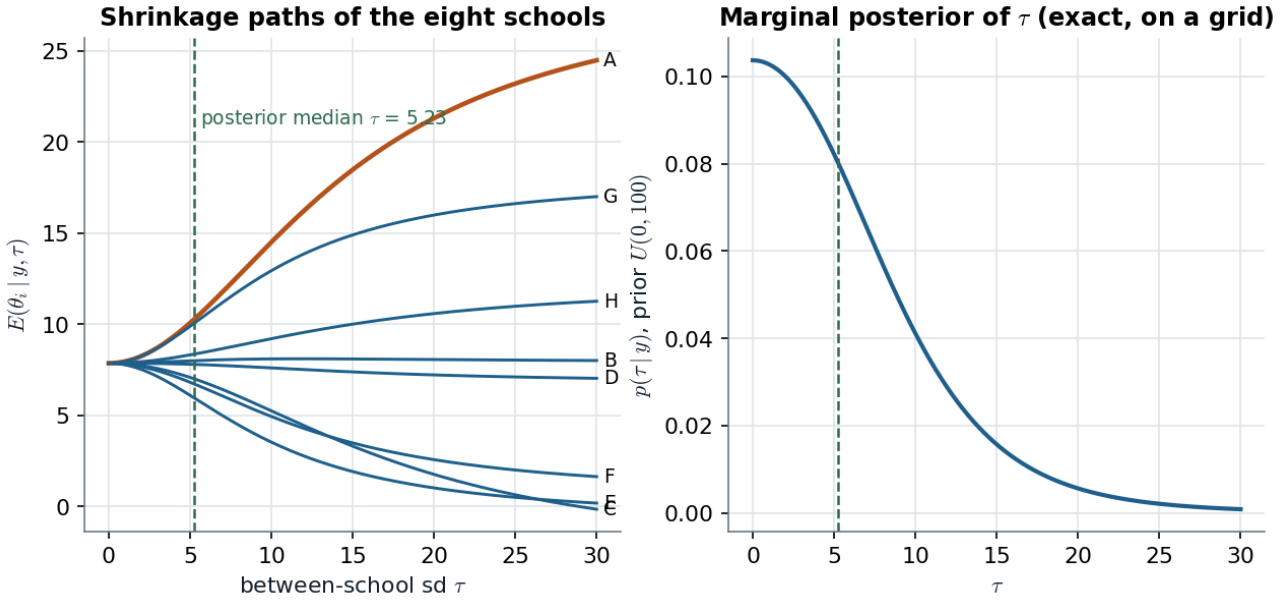
Step 4, τ. Integrating μ from (11) gives $y \mid \tau \sim N_8(0, \text{diag}(\sigma_i^2 + \tau^2) + v_0 1_8 1_8')$, so

$$p(\tau \mid y) \propto 1\{0 < \tau < 100\}\, N_8(y \mid 0,\ \text{diag}(\sigma_i^2 + \tau^2) + v_0 11'), \tag{15}$$

a one-dimensional density, normalized on a grid.

Numbers (`L07_python_checks.py` §4; `make_figures.py`).

- Complete pooling ($\tau = 0$): every $E(\theta_i \mid y)$ equals $\hat{\mu} = \sum y_i/\sigma_i^2 \big/ \sum 1/\sigma_i^2$ = **7.87**, sd $(\sum 1/\sigma_i^2)^{-1/2}$ = **4.17**.
- $p(\tau \mid y)$: mode at **0**, median **5.23**, mean **6.59**, 95% interval [**0.24, 20.87**].
- School A at $\tau = 5.2$ (the companion's default): $\hat{\mu} = 8.027$, $B_A = 14.9^2/(14.9^2 + 5.2^2) = 222.01/249.05 = 0.8914$, $E(\theta_A \mid y, \tau) = 0.1086 \times 28.39 + 0.8914 \times 8.027 = 3.083 + 7.155$ = **10.24**.
- Fully Bayes, τ integrated out: $E(\theta_A \mid y)$ = **11.64** (from 28.39); school C (−2.75) goes **up** to 6.35.



(14) for all eight schools against τ*, and (15); exact, the grid is the only approximation.*

At the posterior median the paths are still bunched within about 6 to 10: the data say the schools differ little relative to their standard errors (all $\sigma_i \geq 9.4$ against a spread of y_i of about 31). A's raw 28.39 survives only if τ is large, and the right panel puts little mass there. Shrinkage is toward $\hat{\mu}$, not toward zero — C and E move **up**.

> **Interactive companion.** The Partial-Pooling Dial (`companions/companion-partial-pooling.html`) — move τ, the prior bound A, a data-precision multiplier κ and the highlighted school; watch (14) along all eight paths, B_i, $\hat{\mu}$, the fully Bayes mean and $p(\tau \mid y)$ (10.24 for school A at the default).

6 · C — pregnant women: pooled, individual, hierarchical (site code)

Data (`hierarchicalmodel.html`): weight gains y_{ij} (kg) of $I = 68$ women at 5–7 visits, visit times x_{ij} in weeks; $n = 427$.

Pooled ($y_{ij} = \alpha + \beta x_{ij} + \epsilon_{ij}$; the site's `lm` printout, reproduced to every digit):

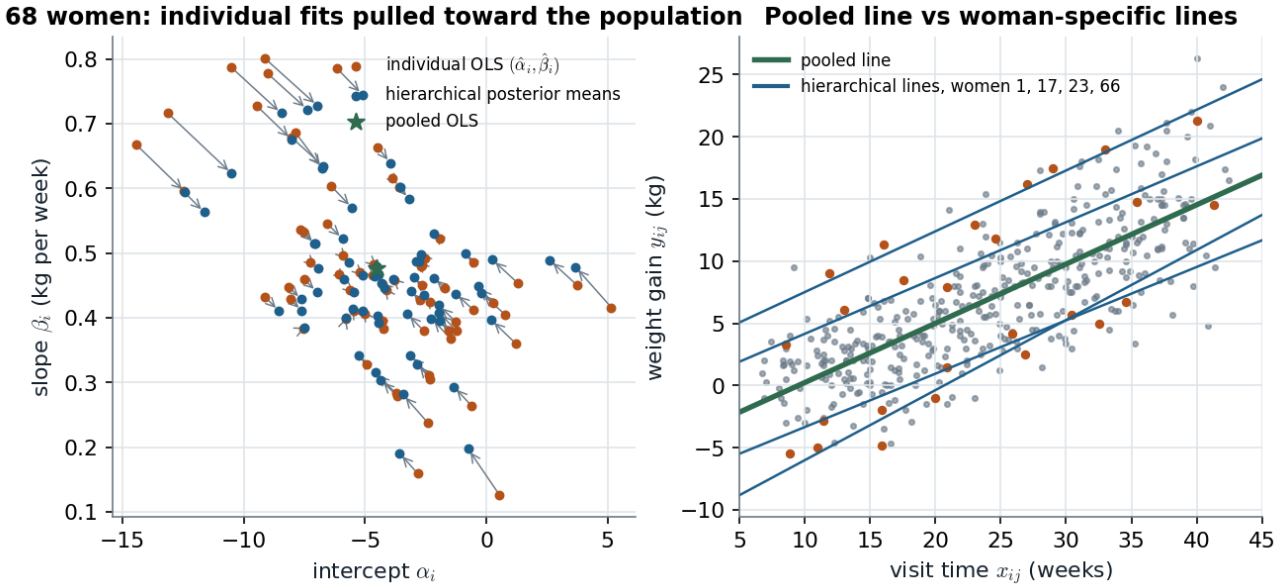
$$\hat{y} = -4.52639 + 0.47633\,x, \qquad \text{SE } (0.48680), (0.01897), \quad \hat{\sigma} = 3.612, \quad R^2 = 0.5974.$$

Individual OLS (one line per woman): intercepts from −14.432 to 5.157 (mean −4.412, median −4.260), slopes from 0.126 to 0.801 (mean 0.470, median 0.450); residual sd mean 1.013 against the pooled 3.612 — **most of the pooled noise is between-woman heterogeneity**, not visit-to-visit noise.

Hierarchical (site section 4): $y_{ij} \mid \alpha_i, \beta_i \sim N(\alpha_i + \beta_i x_{ij}, \sigma_y^2)$, $(\alpha_i, \beta_i)' \sim N((\alpha, \beta)', \text{diag}(\sigma_\alpha^2, \sigma_\beta^2))$, $(\alpha, \beta) \sim N(0_2, 1000 I_2)$, all three variances $IG(0.001, 0.001)$: model (3) with a diagonal V_0, so (10) becomes two scalar IG updates. The sampler, ported line by line (`make_figures.gibbs_hlm`, 1,000 burn-in + 2,000 draws, seeded), gives posterior means

α	β	σ_y	σ_α	σ_β
−4.363	0.4687	1.119	3.210	0.1253

(the site prints no table — only plots — so these are this run's numbers).



Left: each arrow is one woman's partial pooling (8). Right: the pooled line vs four hierarchical lines (women 1, 17, 23, 66).

The arrows point inward, longest for the most extreme individual fits — those rest on 5–7 points and are pulled most. The cloud keeps its strong negative tilt: intercept and slope are correlated because visits start at week 7–15, far from $x = 0$, so the intercept is an extrapolation. On the right, the hierarchical lines are far apart (the $\sigma_\alpha = 3.2$ kg spread is real) but each is anchored on its own points.

The site's "$p(y_{new} \mid x_{new}, \text{data})$" is the posterior of the mean line. Its code plots quantiles of `draws2[,i,1]+draws2[,i,2]*X[i,7]`, i.e. of $\alpha_i + \beta_i x$, without adding $N(0, \sigma_y^2)$ noise. For woman 1 at

her unobserved 7th visit ($x = 40.143$) this run gives median 17.68, 95% band [15.68, 19.86]; a predictive band for a new measurement would be wider by roughly $\pm 1.96 \times 1.12$.

The weight in (6) with these numbers (random-intercept reading, $T = 6$): $w = 6\sigma_\alpha^2/(6\sigma_\alpha^2 + \sigma_y^2) = 6(10.30)/(61.83 + 1.25) = 0.980$ — with this much between-woman spread, each woman's own data dominate her intercept.

7 · U — the airline intercepts (slides 20–21)

Slide 20 reports separate regressions for the 6 airlines (dependent variable output; covariates total cost, fuel price, load factor) and the pooled one: intercepts 8.56, 9.54, 8.00, 8.57, 10.65, 10.91 (SEs 0.28–0.73) vs pooled 8.08 (0.33). Slide 21 compares, firm by firm, the individual intercept, the individual-effects posterior with a non-hierarchical prior and with the hierarchical prior: the two Bayesian versions sit close together around 8.6–9.0 for every firm, while the individual OLS intercepts spread from 8.0 to 10.9 with wide intervals — with $T = 15$ and only $N = 6$ firms, the estimated between-firm variance is small and the hierarchical prior pulls hard.

8 · C/G — the frequentist neighbours

Random-intercept model, MS §6.4 (pp. 215–218). Calves weighed 8 times (weeks 12–26): $y_{ij} = \alpha + \beta(x_j - 12) + e_{ij}$ (6.27) vs $y_{ij} = \alpha + \beta(x_j - 12) + a_i + e_{ij}$ (6.28), a_i with mean 0 and variance σ_a^2, independent of e_{ij}. Under (6.28)

$$\mathrm{Var}(y_{ij}) = \sigma_a^2 + \sigma^2, \qquad \mathrm{Cov}(y_{ij}, y_{ik}) = \sigma_a^2 \ (j \neq k), \qquad \mathrm{Corr} = \frac{\sigma_a^2}{\sigma_a^2 + \sigma^2}, \tag{16}$$

the equicorrelation that MS's draftsman's plot (correlations 0.84–0.99) shows. Point estimates are the same under both ($\hat{\alpha} = 69.9$, $\hat{\beta} = 4.7$) but the standard errors differ: 5.6 vs 7.8 for α, 0.1 vs 0.4 for β (smaller under the mixed model). Model (2) with the hierarchical prior *is* (6.28) with $a_i = \alpha_i - \mu_\alpha$.

Random effects in W §14.2 (pp. 441–442). The RE estimator is OLS on quasi-demeaned data $y_{it} - \theta\bar{y}_i$ with

$$\theta = 1 - \Big[\frac{\sigma_u^2}{\sigma_u^2 + T\sigma_a^2}\Big]^{1/2} \qquad \text{(W 14.10)}. \tag{17}$$

$\theta = 0$ is pooled OLS, $\theta = 1$ fixed effects — the same pooled ↔ separate axis as (6). With the pregnancy numbers ($\sigma_u = \sigma_y = 1.119$, $\sigma_a = \sigma_\alpha = 3.210$, $T = 6$), $\theta = 0.859$ against $w = 0.980$: related, not equal (θ is a square-root version, applied to data rather than to estimates). Consistency of RE requires the unit effect to be uncorrelated with the regressors (W 14.8) — **the same assumption the exchangeable prior makes**; G p. 148 relaxes it by letting $E(b_i)$ depend on covariates. FE vs RE and the Hausman test are not on this site's schedule (`Map/class-plan.md`, "Syllabus items not on the site schedule").

G's panel example (Table 9.5, p. 149). Vella–Verbeek wages, 8 years, intercept and experience as random effects: union coefficient 0.090 (95% 0.006–0.179), smaller than the 0.129 of the single-year regression; mean experience effect 0.058 (0.038–0.078) — about 6% per year.

Named only: the dugong nonlinear growth curve and the WinBUGS/R2WinBUGS code (slides 22–33) — a software demonstration, outside the course's Python toolchain.

Links

- Unit: L07-00-map · L07-01-general-covariance-gls (the conjugate conditionals reused at every level) · L07-06-probit-ordered-multinomial (G 9.2: the binary panel version) · L07-07-formulas · L07-08-pitfalls · L07-09-python §4.
- Class 1: L01 §4 (shrinkage of a posterior mean toward the prior mean — (13) is the same formula).
- Classes 5+6: L05 (DiD and panel thinking; the FE side of §8).
- Rules pack: rules/03 (panel data, hierarchical models, FE/RE).
- **Practice** (statements in the books): G 4.6 (p. 59); G 9.2 (p. 151, binary panel data); G 9.3 (p. 151, Student-t errors in (9.11)); W 14.C5 (p. 454, RE vs FE in a wage equation).

LDV slides 4–10 · The Tobit: a regression on a latent y*, seen only above zero — fill in the hidden part and it is OLS again

> **Claim.** A **corner solution** or **censored** outcome is $y = \max(0, y^*)$ with $y^* = x'\beta + u$, $u \sim N(0, \sigma^2)$. The likelihood mixes densities (for $y > 0$) and probabilities $\Phi(-x'\beta/\sigma)$ (for $y = 0$) and is awkward for Bayes; **data augmentation** (Chib 1992) draws the hidden y_i^* of the zeros from a normal **truncated to** $(-\infty, 0]$, after which $\boldsymbol{\beta}$ and $\boldsymbol{\sigma}^2$ have the plain regression conditionals. The coefficient $\boldsymbol{\beta}$ is the effect on the *latent* mean; the effect on what is observed is $\partial E(y \mid x)/\partial x_j = \beta_j \Phi(x'\beta/\sigma)$ — and OLS on the censored data estimates roughly that, not $\boldsymbol{\beta}$: with normal regressors exactly $\beta \cdot P(y > 0)$. Prev: L07-04-hierarchical-linear-models · Next: L07-06-probit-ordered-multinomial · Map: L07-00-map

Notation.

- $y_i^* = x_i'\beta + u_i$, $u_i \sim N(0, \sigma^2)$: latent outcome; $y_i = y_i^* 1(y_i^* > 0) = \max(0, y_i^*)$ observed (slide 5; G p. 118; W 17.18–17.19).
- $C = \{i : y_i = 0\}$, $k = |C|$ the number of zeros (slide 6).
- $z_i = x_i'\beta/\sigma$; ϕ, Φ standard normal pdf and cdf; $\lambda(c) = \phi(c)/\Phi(c)$ inverse Mills ratio (W p. 538).
- $TN_{(a,b]}(\mu, \sigma^2)$: normal truncated to $(a, b]$ (slide 8 writes $N_{[-\infty,0]}$).

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1 · U — corner solutions, censoring, truncation

Slide 4 lists Tobin's (1958) durable-goods purchases, hours worked by married women, extramarital affairs, arrests, vacation spending: many exact zeros, a continuous spread above zero. Two stories produce the same data (G p. 118): **top/bottom coding** — the value exists but is reported as the limit (true censoring) — and **corner solutions** — the zero is a genuine choice. The Tobit algebra is the same; the interpretation differs (W p. 539: for corner solutions y^* often has no meaning and $\boldsymbol{\sigma}$ is not "ancillary"). **Truncation** is different again: observations with $y^* \le 0$ are not in the sample at all, x included (G p. 118; W §17.4b). The slides call the censoring rule "truncation" (slide 5) and $P(y = 0)$ "probability of truncation" (slides 9–10); read **censoring** throughout.

2 · C — the likelihood (slide 6; W 17.20–17.22)

For $y_i > 0$, $y_i = y_i^*$ and the density is the normal one:

$$f(y_i \mid x_i, \beta, \sigma) = \frac{1}{\sigma}\phi\Big(\frac{y_i - x_i'\beta}{\sigma}\Big), \quad y_i > 0. \tag{1}$$

For $y_i = 0$, a probability mass:

$$P(y_i = 0 \mid x_i) = P(u_i \le -x_i'\beta) = \Phi(-x_i'\beta/\sigma) = 1 - \Phi(z_i). \tag{2}$$

Multiplying over independent observations (slide 6 writes $\prod_{i:y_i>0} f(y_i \mid x_i, \theta)\{F(0 \mid x_i, \theta)\}^k$, the second factor as if all zeros had the same x; it is $\prod_{i\in C} F(0 \mid x_i, \theta)$):

$$L(\beta, \sigma) = \prod_{i\in C} \Phi\Big(-\frac{x_i'\beta}{\sigma}\Big) \prod_{i\notin C} \frac{1}{\sigma}\phi\Big(\frac{y_i - x_i'\beta}{\sigma}\Big) \qquad \text{(G p. 118)}, \tag{3}$$

$$\ell_i(\beta,\sigma) = 1(y_i = 0)\log[1-\Phi(z_i)] + 1(y_i > 0)\log\{\sigma^{-1}\phi[(y_i - x_i'\beta)/\sigma]\} \qquad \text{(W 17.22)}. \tag{4}$$

With a normal prior on β and inverse gamma on σ^2, (3) $\times$ prior is not a standard family (the Φ factors): "very complex" (G p. 118).

3 · C — data augmentation (slide 7; G pp. 118–120)

Add the latent y_i^* for $i \in C$ to the unknowns. Slide 7's "complete likelihood" step:

$$p(y_i \mid x_i,\theta) = \int p(y_i \mid y_i^*) f(y_i^* \mid x_i,\theta)\, dy_i^* = p(y_i \mid y_i^* > 0)[1 - F(0 \mid x_i,\theta)] + p(y_i \mid y_i^* \le 0) F(0 \mid x_i,\theta),$$

with $p(y_i \mid y_i^*) = 1(y_i = 0)1(y_i^* \le 0) + 1(y_i > 0)1(y_i = y_i^*)$: given y^*, y is known. Hence the joint posterior of (β, σ^2, y_C^*) (G eq. 8.6, p. 119):

$$\pi(\beta,\sigma^2, y_C^* \mid y) \propto \prod_{i=1}^{n} \left[1(y_i = 0)1(y_i^* \le 0) + 1(y_i > 0)\right] N(y^* \mid X\beta, \sigma^2 I)\, \pi(\beta)\pi(\sigma^2), \tag{5}$$

where y^* has $y_i^* = y_i$ for $i \notin C$. Read off the conditionals:

- **β, σ^2 given y^*:** (5) is the normal regression likelihood of y^* times the priors — the conditionals (8.2)–(8.3) of G (note 01's (7)–(8) with $\Omega = \mathrm{I}$) **with y^* in place of y.** Given y^*, β and σ^2 do not depend on y (G p. 119).
- **y_i^*, $i \in C$, given β, σ^2:** the factor in y_i^* is $1(y_i^* \le 0)N(y_i^* \mid x_i'\beta, \sigma^2)$:

$$y_i^* \mid y_i = 0, \beta, \sigma^2 \sim TN_{(-\infty,0]}(x_i'\beta, \sigma^2), \qquad y_i^* = y_i \text{ if } y_i > 0 \qquad \text{(slide 7)}. \tag{6}$$

(6) is drawn by the inverse-CDF formula of L07-03-ar1-errors (15) with $a = -\infty$, $b = 0$. This is G's Algorithm 8.2 (p. 120). The marginal posterior of (β, σ^2) is obtained by "ignoring the draws" of y_C^*.

4 · C — the linear Tobit conditionals (slide 8)

Prior $\beta \sim N(\beta_0, V_0)$, $\sigma^2 \sim IG(\nu_0/2, \nu_0 s_0^2/2)$. From (5):

$$\beta \mid \sigma^2, y^* \sim N(V_\beta(V_0^{-1}\beta_0 + \sigma^{-2}X'y^*), V_\beta), \qquad V_\beta^{-1} = V_0^{-1} + \sigma^{-2}X'X, \tag{7}$$

$$\sigma^2 \mid \beta, y^* \sim IG\Big(\frac{\nu_0 + n}{2}, \frac{\nu_0 s_0^2 + (y^* - X\beta)'(y^* - X\beta)}{2}\Big). \tag{8}$$

Two slips on slide 8: it prints $V_\beta^{-1} = V_0^{-1} + \sigma^{-2}(X'X)^{-1}$ — the data precision is $\sigma^{-2}X'X$, not its inverse — and the mean $V_\beta(V_0^{-1}\beta_0 + X'y^*)$ without the σ^{-2} on $X'y^*$. (7) is G's step 8.1 (p. 120).

5 · C — what β means: three conditional means (W §17.2a, pp. 538–539)

Positive part. For $z \sim N(0,1)$, $E(z \mid z > c) = \phi(c)/[1-\Phi(c)]$. Then

$$E(y \mid y>0, x) = x'\beta + E(u \mid u > -x'\beta) = x'\beta + \sigma\frac{\phi(-z)}{1-\Phi(-z)} = x'\beta + \sigma\lambda(z) \qquad \text{(W 17.24)}, \quad (9)$$

using $\phi(-z)=\phi(z)$, $1-\Phi(-z)=\Phi(z)$. **Unconditional.** $E(y \mid x) = P(y>0 \mid x)E(y \mid y>0, x)$ (W 17.23); substitute (9) and $\Phi\lambda = \phi$:

$$E(y \mid x) = \Phi(z)[x'\beta + \sigma\lambda(z)] = \Phi(z)\,x'\beta + \sigma\phi(z) \qquad \text{(W 17.25)}. \quad (10)$$

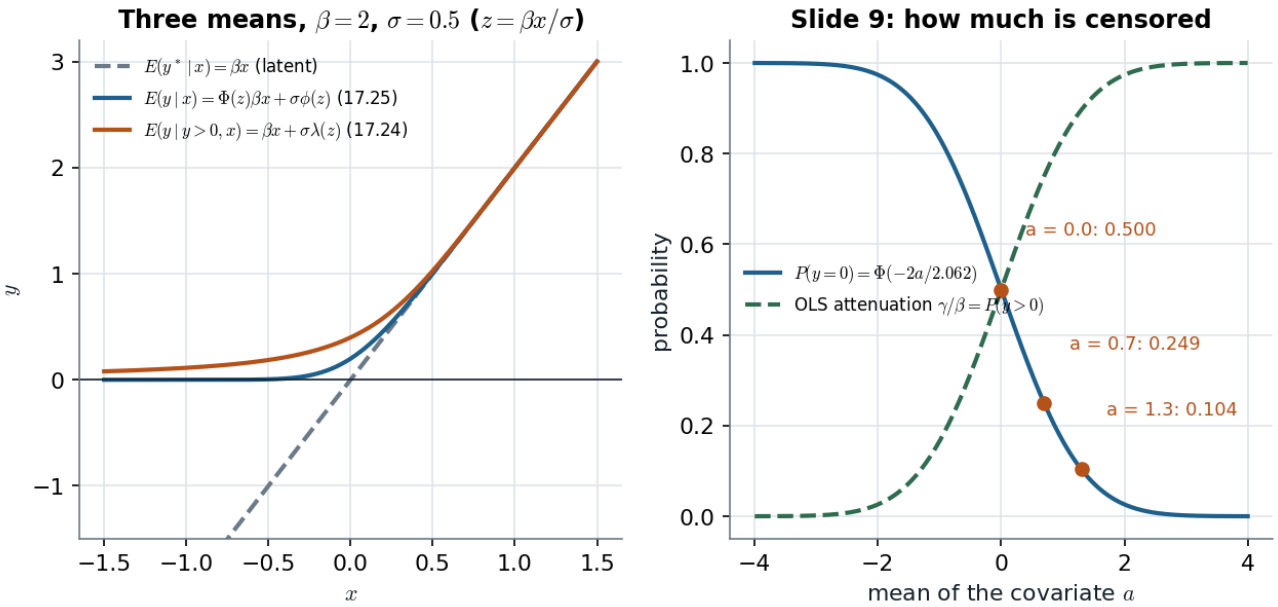
Partial effects. With $d\lambda/dc = -\lambda(c)[c+\lambda(c)]$ and $\partial z/\partial x_j = \beta_j/\sigma$, differentiate (9):

$$\frac{\partial E(y \mid y>0, x)}{\partial x_j} = \beta_j + \sigma\frac{\beta_j}{\sigma}\cdot\{-\lambda(z)[z+\lambda(z)]\} = \beta_j\{1-\lambda(z)[z+\lambda(z)]\} \qquad \text{(W 17.26)}. \quad (11)$$

Differentiate (10), with $d\phi/dz = -z\phi$:

$$\frac{\partial E(y \mid x)}{\partial x_j} = \frac{\beta_j}{\sigma}\phi(z)x'\beta + \Phi(z)\beta_j + \sigma\frac{\beta_j}{\sigma}(-z\phi(z)) = \frac{\beta_j}{\sigma}\phi(z)\,\sigma z + \Phi(z)\beta_j - \beta_j z\phi(z) = \beta_j\Phi(z) \qquad \text{(W 17.30)}. \quad (12)$$

(Both derivatives are checked numerically against (9)–(10) in `L07_python_checks.py` §5.) The bracket in (11) lies strictly between 0 and 1 (W p. 539), and $\Phi(z) \in (0,1)$: **every observable effect is β shrunk**, by a factor that depends on x and on σ.



Left: (9), (10) and the latent line, $\beta = 2$, $\sigma = 0.5$. Right: (13) and 1−(13), slide 9's example.

Far to the right all three means coincide (almost nobody is censored). Far to the left $E(y \mid x) \to 0$ while $E(y \mid y>0, x)$ stays positive — the few who are above zero are those with large u, the selection that makes OLS on the positive subsample biased (§7).

6 · C — slide 9's example and what OLS on censored data estimates

Slide 9. $y_i^* \mid x_i \sim N(\beta x_i, \sigma^2)$, $x_i \sim N(a, 1)$, so unconditionally $y^* \sim N(\beta a, \beta^2 + \sigma^2)$ and

$$P(y = 0 \mid a, \beta = 2, \sigma^2 = 0.25) = \Phi\Big(\frac{-2a}{\sqrt{4.25}}\Big) = \Phi(-2a/2.062). \tag{13}$$

At $a = 0$: $\Phi(0) = 0.500$. At $a = 0.7$: $\Phi(-1.4/2.062) = \Phi(-0.679) = 0.249$. At $a = 1.3$: $\Phi(-2.6/2.062) = \Phi(-1.261) = 0.104$ — the slide's "Prob. trunc. = 0.5, 0.25, 0.1" (slide 10).

Slide 10 compares, under a non-informative prior ($\nu_0 = 0$, $V_0^{-1} = 0$), the posterior of β that ignores the censoring (red: the zeros treated as data) with the Tobit posterior (black, centred on the true 2). The red curve is centred well below 2 at $a = 0$ and approaches 2 as censoring falls. How far below:

Claim. With normal x, the population OLS slope of y on x is $\beta\, P(y > 0)$. *Proof.* (x, y^*) is bivariate normal. Stein's lemma: for jointly normal (x, w) and smooth g, $\mathrm{Cov}(x, g(w)) = \mathrm{Cov}(x, w)\, E[g'(w)]$. Take $w = y^*$, $g(w) = \max(0, w)$, $g'(w) = 1(w > 0)$:

$$\mathrm{Cov}(x, y) = \mathrm{Cov}(x, y^*)\, P(y^* > 0) = \beta\, \mathrm{Var}(x)\, P(y > 0) \;\Rightarrow\; \frac{\mathrm{Cov}(x, y)}{\mathrm{Var}(x)} = \beta P(y > 0). \tag{14}$$

Slide 9's values: $2 \times 0.500 = 1.00$, $2 \times 0.751 = 1.50$, $2 \times 0.896 = 1.79$ (a Monte Carlo with 10^6 draws at $a = 0.7$ gives 1.5033 vs 1.5029, `L07_python_checks.py` §5). (14) is the population version of W's remark that OLS slopes estimate $\partial E(y \mid x)/\partial x_j = \beta_j \Phi$ and must be compared with $\hat{\beta}_j \times$ a scale factor (W p. 539).

> **Interactive companion.** The Censoring Dial (`companions/companion-tobit-censoring.html`) — move the covariate mean a, β, σ and an evaluation point x_0; read $P(y = 0)$ (0.249 at slide 9's $a = 0.7$), the OLS slope $\beta P(y > 0)$ and every term of (9)–(12) at x_0.

7 · C — W Example 17.2: married women's hours (Table 17.3, p. 541)

MROZ: 753 women, 325 with zero hours. Reproduced exactly here by MLE of (4) (`L07_python_checks.py` §5):

	OLS	Tobit MLE	Tobit Gibbs (flat prior) mean (sd)
nwifeinc	−3.45	−8.81 (4.46)	
educ	28.76	80.65 (21.58)	81.84 (21.76)
exper	65.67	131.56 (17.28)	
exper^2	−0.700	−1.86 (0.54)	
age	−30.51	−54.41 (7.42)	
kidslt6	−442.09	−894.02 (111.88)	−902.63 (112.86)
kidsge6	−32.78	−16.22 (38.64)	
constant	1,330.48	965.31 (446.44)	
$\hat{\sigma}$	750.18	1,122.02	1,138.8
log-likelihood	—	−3,819.09	

The Gibbs column is Algorithm 8.2 with $V_0^{-1} = 0$, $\nu_0 = 0$ (1,000 burn-in + 5,000 draws): posterior means and sds within a fraction of an SE of the MLE, as a flat-prior posterior should be.

Comparing with OLS. The Tobit educ coefficient is almost three times OLS's, but they measure different things. The APE scale factor $n^{-1}\sum_i \Phi(x_i'\hat{\beta}/\hat{\sigma}) = \mathbf{0.589}$, so the Tobit average partial effect of educ is $0.589 \times 80.65 = \mathbf{47.5}$ hours vs OLS 28.8; at the averages (with $(\overline{exper})^2$) the factor is **0.645** (W pp. 540–541; both reproduced).

G's version (Table 8.1, p. 121). Same data, covariates children under 6, children 6+, age, husband's wage, informative priors: posterior means −556.4 (young children) and −191.7 (older children). From the predictive distribution, P(zero hours) is 0.431 with young children vs 0.328 with older children; mean hours about 214 vs 587; given positive hours about 1,284 vs 1,105 (G pp. 121–122).

8 · C/G — censored and truncated regression (W §17.4, pp. 547–553)

Censored regression (W 17.36–17.39): $w_i = \min(y_i^*, c_i)$ with a known, possibly observation-specific censoring point (durations, top-coding): density $\sigma^{-1}\phi((w - x_i'\beta)/\sigma)$ below c_i, mass $1 - \Phi((c_i - x_i'\beta)/\sigma)$ at c_i — (3) with the inequality reversed. With true censoring, y^* is the variable of interest and β its regression coefficient, unlike a corner solution.

Truncated regression (W 17.40–17.41): (x_i, y_i) seen only if $y_i \le c_i$; the density of the observed y is $f(y \mid x_i'\beta, \sigma^2)/F(c_i \mid x_i'\beta, \sigma^2)$. No point masses, and no information about the excluded units. OLS on a truncated sample is biased: with slide 9's design ($a = 0.7$, $\beta = 2$), OLS on the units with $y^* > 0$ only gives a slope of **1.902** (simulated, `L07_python_checks.py` §7) — the inverse Mills ratio in (9) is an omitted regressor correlated with x (W p. 538).

Links

- Unit: L07-00-map · L07-03-ar1-errors (the truncated-normal draw) · L07-06-probit-ordered-multinomial (the same augmentation with $\sigma = 1$) · L07-02-student-t-errors (t errors: G 8.6) · L07-07-formulas · L07-08-pitfalls · L07-09-python §5.
- Class 2: L02 §4 (binary outcomes, partial effects shrunk by a density factor — the logit analogue of (12)).
- Rules pack: rules/06 (probit/Tobit listed for Class 7).
- **Practice** (statements in the books): G 8.6 (p. 132, Tobit with Student-t errors); W 17.3 (p. 559, partial effects of a log regressor in the Tobit); W 17.C3 (p. 561, FRINGE pensions); W 17.C9 (p. 562, APPLE ecolabeled apples).

LDV slides 11–15 · Probit, ordered probit, multinomial probit: one latent normal regression, three ways of cutting it

> **Claim.** Each model is a normal regression on a latent y^* observed only through a **cut**: its sign (binary probit), the interval between cut-points it falls in (ordered probit), or which latent utility is largest (multinomial probit). Only the cut is seen, so the **scale is not identified** ($\sigma = 1$, or $\Sigma_{11} = 1$). The Albert–Chib trick makes all three Gibbs samplers: draw y^* from a **truncated normal** consistent with the observed category, then β by a normal regression on y^*; cut-points get **uniform** conditionals; the multinomial version is a SUR in utility differences with a truncated multivariate normal. Coefficients are not effects: the effect on $P(y = 1)$ is $\phi(x'\beta)\beta_j \leq 0.4\beta_j$. Prev: L07-05-tobit · Next: L07-07-formulas · Map: L07-00-map

Notation.

- $y_i^* = x_i'\beta + u_i$, $u_i \sim N(0,1)$: latent index; binary $y_i = 1(y_i^* > 0)$ (G 8.8–8.9; W 17.6).
- Ordered: $y_i \in \{1, \dots, J\}$, $y_i = j \iff \gamma_{j-1} < y_i^* \leq \gamma_j$, cut-points $-\infty = \gamma_0 < \gamma_1 < \cdots < \gamma_J = \infty$ (slide 11).
- Multinomial: alternatives $0, 1, \dots, J$; U_{ij} utility; $y_{ij}^* = U_{ij} - U_{i0} = x_{ij}'\beta_j + \epsilon_{ij}$, $y_i^* = (y_{i1}^*, \dots, y_{iJ}^*)'$, $\epsilon_i \sim N_J(0, \Omega)$ (slides 13–14).
- $\Lambda(z) = e^z/(1 + e^z)$ logistic cdf.

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1 · C — the binary probit and why $\sigma = 1$ (G §8.2.2, pp. 122–124)

From the latent form,

$$P(y_i = 1) = P(x_i'\beta + u_i > 0) = P(u_i > -x_i'\beta) = 1 - \Phi(-x_i'\beta) = \Phi(x_i'\beta) \qquad \text{(G 8.7; W 17.2, 17.4).} \tag{1}$$

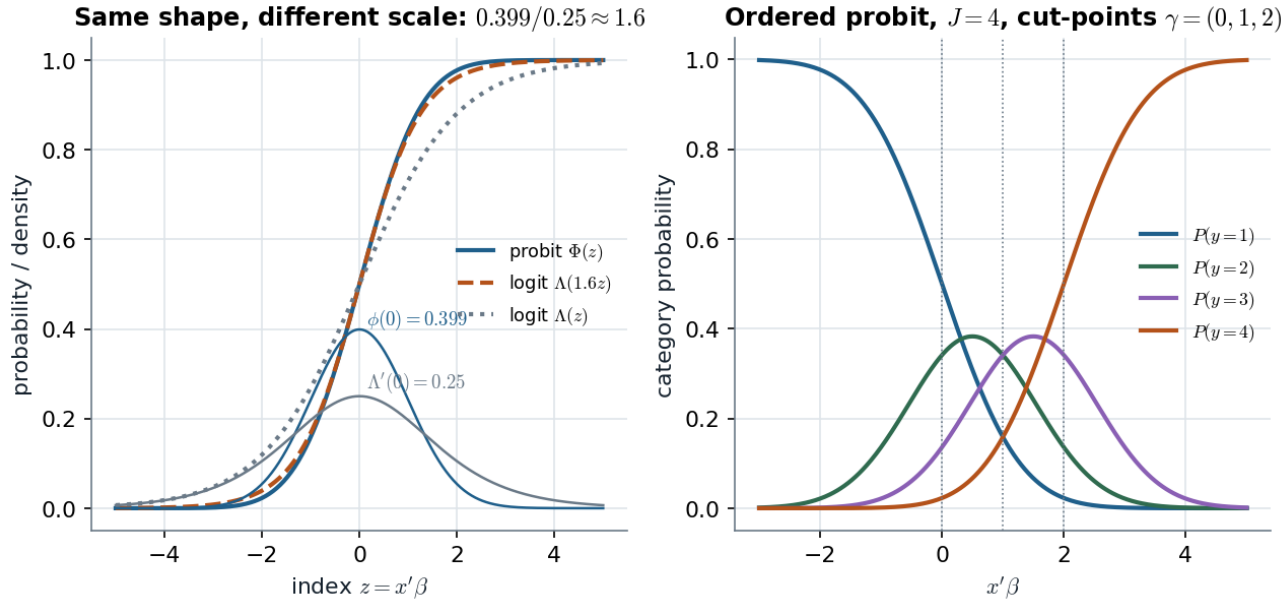
If instead $u_i \sim N(0, \sigma^2)$, $P(y_i = 1) = \Phi(x_i'\beta/\sigma)$: only β/σ enters, so (β, σ) and $(c\beta, c\sigma)$ give the same likelihood for every $c > 0$ — "only the sign of y_i^* determines the observed y_i" (G p. 124). Set $\sigma = 1$. The probit is a GLM with link Φ (G p. 123), a cdf because it must map $\mathbb{R}$ into $(0, 1)$.

Partial effects (W 17.7; G p. 124):

$$\frac{\partial P(y_i = 1 \mid x_i)}{\partial x_{ik}} = \phi(x_i'\beta)\,\beta_k, \qquad \max = \phi(0)\beta_k = \frac{\beta_k}{\sqrt{2\pi}} = 0.3989\,\beta_k. \tag{2}$$

For the logit, $\Lambda' = \Lambda(1 - \Lambda)$ peaks at $1/4$ (G p. 127). Equating the maximum effects, $0.4\beta_k^p = \beta_k^l/4$, so

$$\beta^{logit} \approx 1.6\,\beta^{probit} \qquad \text{(G p. 128).} \tag{3}$$



Left: (1) against the logit, and the rule (3). Right: the ordered probit (8) with $\gamma = (0, 1, 2)$.

$\Phi(z)$ and $\Lambda(1.6z)$ are almost indistinguishable over the whole range; $\Lambda(z)$ is visibly flatter — the coefficients differ by a scale, the fitted probabilities do not. On the right, as $x'\beta$ grows the mass moves from category 1 to 4; the middle categories rise then fall, so **a positive β need not raise** $P(y = 2)$.

2 · C — Albert–Chib data augmentation (G Algorithm 8.3, p. 125)

Likelihood contribution given the latent value: $p(y_i \mid y_i^*) = 1(y_i = 0)1(y_i^* \le 0) + 1(y_i = 1)1(y_i^* > 0)$. With $\beta \sim N(\beta_0, B_0)$,

$$\pi(\beta, y^* \mid y) \propto \prod_i \left[1(y_i = 0)1(y_i^* \le 0) + 1(y_i = 1)1(y_i^* > 0)\right] N_n(y^* \mid X\beta, I)\, N_K(\beta \mid \beta_0, B_0). \tag{4}$$

Conditionals from (4):

$$y_i^* \mid \beta, y_i \sim \begin{cases} TN_{(-\infty,0]}(x_i'\beta, 1) & y_i = 0 \\ TN_{(0,\infty)}(x_i'\beta, 1) & y_i = 1 \end{cases}, \qquad \beta \mid y^* \sim N_K(\bar\beta, B_1),\ B_1 = (X'X + B_0^{-1})^{-1},\ \bar\beta = B_1(X'y^* + B_0^{-1}\beta_0). \tag{5}$$

B_1 never changes because $\sigma^2 = 1$ (G p. 125). (5) is the Tobit sampler of L07-05-tobit with every observation latent and σ fixed.

Prior elicitation in probabilities (G pp. 125–126, computer ownership, $n = 500$**).** "A household with zero wage and no college degree owns a computer with probability 0.10": $\beta_{01} = \Phi^{-1}(0.10) = -1.2816$; "very likely below 0.3": $\Phi^{-1}(0.3) = -0.5244$, so $\sqrt{B_{0,11}} = (-0.5244 + 1.2816)/1.96 = 0.3863$, $B_{0,11} = 0.1492$. Wage \$2,000/week with probability 0.8: $-1.2816 + 2\beta_{02} = \Phi^{-1}(0.8) = 0.8416 \Rightarrow \beta_{02} = 1.0616$; at least 0.6: $\Phi^{-1}(0.6) = 0.2533 \Rightarrow \beta_2 = 0.767$, $B_{0,22} = [(1.0616 - 0.767)/1.96]^2 = 0.0225$. College raises 0.1 to 0.4: $\beta_{03} = \Phi^{-1}(0.4) + 1.2816 = 1.0283$. Posterior (Table 8.2, p. 126): $\beta_1 = -0.705$ (0.074), wage 0.879 (0.106), college 0.524 (0.154). The logit version (pp. 128–129) needs MH — a normal prior is not conditionally conjugate for (1) with Λ — and gives college 0.786 with $P(\beta_3 > 0 \mid y) = 0.88$ only.

Worked: W Example 17.1 (Table 17.1, p. 533), labour-force participation, MROZ ($n = 753$). Reproduced exactly (`L07_python_checks.py` §6):

	nwifeinc	educ	exper	exper^2	age	kidslt6	kidsge6	const	log-lik
probit	−0.012	0.131	0.123	−0.0019	−0.053	−0.868	0.036	0.270	−401.30
logit	−0.021	0.221	0.206	−0.0032	−0.088	−1.443	0.060	0.425	−401.77

Scale factors at the means: $\phi(\bar{x}'\hat{\beta}) = 0.391$ (probit), $\Lambda'(\bar{x}'\hat{\beta}) = 0.243$ (logit), ratio **1.61** — rule (3) (W p. 534). Averaged over the sample: 0.301 and 0.179, so the probit APE of educ is $0.301 \times 0.131 = 0.039$, the LPM's 0.038. 73.4% correctly predicted, pseudo-R^2 0.221. The Albert–Chib sampler (5) with a flat prior returns posterior means within 0.3 SE of every MLE (educ 0.132 vs 0.131).

3 · C — the ordered probit (slides 11–12)

Probabilities. From the cut rule,

$$P(y_i = j \mid x_i) = P(\gamma_{j-1} < x_i'\beta + u_i \le \gamma_j) = \Phi(\gamma_j - x_i'\beta) - \Phi(\gamma_{j-1} - x_i'\beta), \quad j = 1, \dots, J. \tag{6}$$

Identification. Adding c to the intercept and to every γ_j leaves (6) unchanged: fix $\gamma_1 = 0$ when x has an intercept (slide 11: "in probit models $y_i \in \{1, 2\}$ and $\gamma_1 = 0$" — the binary probit is $J = 2$). The scale is fixed by $\mathrm{Var}(u) = 1$.

Conditionals (slide 12), flat priors on β and on the ordered γ's:

$$y_i^* \mid \beta, \gamma, y_i = j \sim TN_{(\gamma_{j-1}, \gamma_j]}(x_i'\beta, 1), \qquad \beta \mid y^* \sim N((X'X)^{-1}X'y^*, (X'X)^{-1}). \tag{7}$$

Cut-point γ_j given everything else: the joint density of (y^*, γ) contains γ_j only through the indicators $1(\gamma_{j-1} < y_i^* \le \gamma_j)$ for $y_i = j$ and $1(\gamma_j < y_i^* \le \gamma_{j+1})$ for $y_i = j + 1$, and the ordering. They hold iff $\gamma_j \ge \max\{y_i^* : y_i = j\}$, $\gamma_j < \min\{y_i^* : y_i = j + 1\}$, $\gamma_{j-1} < \gamma_j < \gamma_{j+1}$. A flat density restricted to an interval is uniform:

$$\gamma_j \mid \gamma_{(-j)}, \beta, y^* \sim U(\underline{\gamma}_j, \overline{\gamma}_j), \quad \underline{\gamma}_j = \max\{\max\{y_i^* : y_i = j\}, \gamma_{j-1}\}, \quad \overline{\gamma}_j = \min\{\min\{y_i^* : y_i = j{+}1\}, \gamma_{j+1}\}. \tag{8}$$

(The slide writes the bounds as $\bar{\gamma}_{j-1}, \bar{\gamma}_{j+1}$.)

Re-run (`L07_python_checks.py` §6; $n = 500$, $x'\beta = 0.5 + 1.0x$, $\gamma = (0, 1, 2)$, $J = 4$; 2,000 burn-in + 8,000 draws): posterior means $\beta = (0.50, 0.89)$, $\gamma_2 = 1.10$, $\gamma_3 = 2.08$. **The cut-point chain has lag-1 autocorrelation 0.99.** With many observations the interval in (8) is squeezed between neighbouring latent values, so each γ moves very little per sweep (Cowles 1996 proposed an MH fix). This is the price of the elegant uniform draw: check the trace of γ, not just of β.

4 · C — the multinomial probit as a SUR (slides 13–15)

Choice rule (slide 13). Individual i picks the alternative with the largest utility. Differencing from alternative 0, $y_{ij}^* = U_{ij} - U_{i0} = x_{ij}'\beta_j + \epsilon_{ij}$, and the econometrician sees

$$y_i = 0 \iff \max_j y_{ij}^* < 0; \qquad y_i = j \iff y_{ij}^* = \max(y_i^*) \geq 0. \tag{9}$$

Stacking (slide 14). $y_i^* = X_i\beta + \epsilon_i$ with $X_i = \text{blockdiag}(x_{i1}', \ldots, x_{iJ}')$, $\beta = (\beta_1', \ldots, \beta_J')'$, $\epsilon_i \sim N_J(0, \Omega)$; over individuals $y^* = X\beta + \epsilon$ with $\text{Var}(\epsilon) = I_N \otimes \Omega$: a seemingly unrelated regression (GLM3 slides 31–32), latent.

Conditionals (slide 15). β normal (the SUR GLS form: $V_1^{-1} = V_0^{-1} + \sum_i X_i'\Omega^{-1}X_i$, $V_1^{-1}\beta_1 = V_0^{-1}\beta_0 + \sum_i X_i'\Omega^{-1}y_i^*$ — GLM3 slide 32); Ω inverse Wishart (scale updated by $\sum_i (y_i^* - X_i\beta)(y_i^* - X_i\beta)'$); and

$$y_i^* \sim \begin{cases} N(X_i\beta, \Omega)\, 1(\max(y_i^*) < 0) & y_i = 0 \\ N(X_i\beta, \Omega)\, 1(\max(y_i^*) = y_{ij}^* \geq 0) & y_i = j \end{cases}, \tag{10}$$

a **truncated multivariate normal**, drawn one coordinate at a time from its univariate conditional (normal, truncated: below 0 if $y_i = 0$; above $\max(0, \text{others})$ for the chosen j; below the chosen coordinate otherwise) — the same device G uses for the multivariate probit (9.9).

Identification. Utilities are defined up to location (hence the differencing) and **scale**: (β, Ω) and $(c\beta, c^2\Omega)$ imply the same choices, so one variance (say $\Omega_{11} = 1$) must be fixed (Keane 1992; McCulloch & Rossi 1994 sample the unidentified model and normalize afterwards; McCulloch, Polson & Rossi 2000 impose $\Omega_{11} = 1$ in the prior — slide 16's references).

Re-run (`L07_python_checks.py` §6; 3 alternatives, $n = 800$, alternative-specific intercept and slope, Ω known with $\Omega_{11} = 1$): posterior means $(0.45, 1.10, -0.11, -0.71)$ for true $(0.3, 1.0, -0.2, -0.8)$. The Ω step is omitted (marked `ponytail:` in the code, upgrade path stated).

5 · C/G — the multivariate probit (G §9.2, pp. 139–144)

J binary decisions by the same individual, $y_{ij} = 1(y_{ij}^* > 0)$, $u_i \sim N_J(0, \Sigma)$: a SUR whose responses are all binary. Only signs are seen, so all J variances are fixed at 1 and Σ is a **correlation matrix** (p. 141); its free correlations get a normal prior truncated to positive-definite matrices and a tailored MH step (Algorithm 9.2), β and the y_i^* are Gibbs draws. Example (Table 9.4, p. 144): Rubinfeld's 95 voters, public school (1) and voting for a school tax (2): income raises the probability of voting yes ($\beta_{22} = 1.113$, 95% 0.282–1.997), property taxes lower it ($\beta_{23} = -1.437$), correlation $\sigma_{12} = 0.085$ (-0.302 to 0.455).

6 · U — the logit coefficient as an odds ratio (MS §6.5, pp. 218–220)

MS fit $\log\{P(Y = 1 \mid x)/P(Y = 0 \mid x)\} = \alpha + \beta x$ (6.29) to 32 IVF patients (good responder vs poor, x = inhibin B rise): $\hat{\alpha} = -2.310$ (0.947), $\hat{\beta} = 0.026$ (0.009), residual deviance 24.758 on 30 df. $\hat{\beta}$ is the log odds ratio per unit of x: $e^{0.026} = 1.026$, +2.6% odds per unit, $e^{100 \times 0.026} = e^{2.6} = 13.5$ per 100 units. The probit counterpart of this reading does not exist — probit coefficients have no odds-ratio meaning, only the effect (2).

Links

- Unit: L07-00-map · L07-05-tobit (the same augmentation) · L07-03-ar1-errors (the truncated-normal draw) · L07-04-hierarchical-linear-models (G 9.2 binary panel) · L07-07-formulas · L07-08-pitfalls · L07-09-python §6.
- Class 2: L02 §4 (logit likelihood, IRLS, odds ratios — the frequentist side of §2).

- Rules pack: rules/06.
- **Practice** (statements in the books): G 8.7 (p. 132, probit with Student-t errors); G 9.2 (p. 151); W 17.C2 (p. 561, LOANAPP probit and logit); MS 6.18 (p. 241) and 6.21 (p. 242), logistic regressions read through odds ratios.

Class 7 · Formulas — the exam specification mapped, every result, every number

Prev: L07-06-probit-ordered-multinomial · Next: L07-08-pitfalls · Map: L07-00-map

Equation numbers are **the section notes' own**: "03 (10)" is equation (10) of L07-03-ar1-errors.

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0 · The exam specification, mapped

Sources: the class goals in Map/class-plan, the outlines of the three decks, and the topic packs rules/03 and rules/06. The exam has no code: "derive it, compute it by hand, read it".

#	Item (source)	Note, section	Be able to
1	GLM with general covariance (GLM3 outline)	L07-01-general-covariance-gls §2	show $P\Omega P' = I$ gives (3); derive GLS (5); derive the conditionals (7)–(8)
2	Heteroskedastic errors, WLS/GLS (plan)	01 §3–§4; W §8.4	write (9)–(10); read W Table 8.1; write (11)–(12) and say why α needs MH; FGLS steps
3	Student-t errors, scale mixtures (plan; GLM3 Case II)	L07-02-student-t-errors §2–§4	derive (3) from (1); write (4), (5), (6); compute prior probabilities of ν; state Geweke's conditions
4	Autoregressive errors (plan; GLM3 "parsimonious Ω")	L07-03-ar1-errors §1–§4	derive (1)–(2); verify (3)–(4); quasi-difference (5)–(6); PW vs CO; derive (13); use (15)
5	Hierarchical / multilevel linear models, partial pooling (plan; HLM)	L07-04-hierarchical-linear-models §2–§6	write (1)–(3); derive (5)–(6), (7), (8)–(10); compute (12)–(14) for a school; read the pregnancy outputs
6	Nutrition of pregnant women example (plan)	04 §6	read the pooled printout; explain the shrinkage figure; criticise the "predictive" band
7	Tobit (plan; LDV outline)	L07-05-tobit §1–§7	likelihood (3)–(4); augmentation (5)–(6); conditionals (7)–(8); derive (9)–(12); slide 9's (13); (14)

#	Item (source)	Note, section	Be able to
8	Probit (plan; W §17.1; G §8.2.2)	L07-06-probit-ordered-multinomial §1–§2	(1); identification; (2)–(3); Albert–Chib (4)–(5); read W Table 17.1 with scale factors
9	Ordered probit (plan; LDV slides 11–12)	06 §3	(6); identification; (7); derive (8)
10	Multinomial probit (plan; LDV slides 13–15)	06 §4	choice rule (9); SUR stacking; conditionals and (10); scale identification
11	Frequentist neighbours (plan: panel FE/RE note)	04 §8	(16) equicorrelation; (17) RE θ; FE = individual effects with a flat prior

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1 · Every definition and result

1.1 · General covariance → L07-01-general-covariance-gls

Eq.	Statement	Source
(1)	$y = X\beta + \epsilon$, $\epsilon \sim N(0, \sigma^2\Omega)$	GLM3 slide 3
(2)–(3)	$P\Omega P' = I \Rightarrow \mathrm{Var}(P\epsilon) = \sigma^2 I$: standard regression of Py on PX	slide 4
(5)	$\hat{\beta}_{GLS} = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}y$, $\mathrm{Var} = \sigma^2(X'\Omega^{-1}X)^{-1}$; $\Omega^{-1} = P'P$	derived
(7)	$\sigma^2 \mid \beta \sim IG(\nu_1/2, \nu_1 s_1^2/2)$, $\nu_1 = \nu_0 + n$, $\nu_1 s_1^2 = \nu_0 s_0^2 + (y - X\beta)'\Omega^{-1}(y - X\beta)$	slide 5
(8)	$\beta \mid \sigma^2 \sim N(\beta_1, V_1)$, $V_1^{-1} = V_0^{-1} + \sigma^{-2}X'\Omega^{-1}X$, $V_1^{-1}\beta_1 = V_0^{-1}\beta_0 + \sigma^{-2}X'\Omega^{-1}y$	slide 5; G (8.2)
(9)–(10)	WLS: divide every column (constant too) by $\sqrt{h_i}$; minimise $\sum(y_i - x_i'\beta)^2/h_i$	W (8.25)–(8.27)
(11)–(12)	$p(\alpha \mid \cdot) \propto p(\alpha)\prod h_i^{-1/2} e^{-e_i^2/(2\sigma^2 h_i)}$; non-standard → MH	slide 7
—	FGLS: $\log \hat{u}^2$ on x, $\hat{h} = \exp(\hat{g})$, WLS with $1/\hat{h}$	W (8.30)–(8.33)

1.2 · Student-t errors → L07-02-student-t-errors

Eq.	Statement	Source
(1)–(3)	$N(0,\sigma^2\omega)\times IG(\nu/2,\nu/2)$ integrated over $\omega = t_\nu(0,\sigma^2)$	slide 16; G p. 52
(4)	$\omega_i \mid \cdot \sim IG(\frac{\nu+1}{2}, \frac{\nu+\epsilon_i^2/\sigma^2}{2})$	slide 18; G p. 115
(5)	$E(\omega_i^{-1} \mid \cdot) = (\nu+1)/(\nu+\epsilon_i^2/\sigma^2)$	derived
—	β, σ^2 conditionals: 01 (7)–(8) with $\Omega = \mathrm{diag}(\omega_i)$	G p. 115
(6)	$p(\nu \mid \omega) \propto [(\nu/2)^{\nu/2}/\Gamma(\nu/2)]^n \exp\{-[1/\nu^0 + \frac{1}{2}\sum(\ln\omega_i + \omega_i^{-1})]\nu\}$ → MH	slide 18
—	$\nu \sim$ Exp(mean ν^0); flat β prior: posterior mean needs $P(\nu \le 2) = 0$, variance $P(\nu \le 4) = 0$	slide 17

1.3 · AR(1) errors → L07-03-ar1-errors

Eq.	Statement	Source
(1)–(2)	$\gamma_0 = \sigma^2/(1-\rho^2)$; $\mathrm{Cov}(\epsilon_t, \epsilon_{t+k}) = \rho^k\gamma_0$	slide 22; W (12.27)
(3)–(4)	Ω_ρ^{-1} tridiagonal $(1, 1+\rho^2, \dots, 1; -\rho) = P'P$, Prais–Winsten P	derived
(5)–(6)	$\sqrt{1-\rho^2}y_1$; $y_t - \rho y_{t-1} = (x_t - \rho x_{t-1})'\beta + u_t$	W (12.28)–(12.32)
(7)–(9)	large-n OLS variance with ϕ, naive variance, GLS variance	derived (G)
(10)–(11)	true/naive $= (1+\rho\phi)/(1-\rho\phi)$; OLS/GLS = (10) $\times(1+\rho^2-2\rho\phi)/(1-\rho^2)$	derived (G)
(12)	exact $p(\rho \mid \epsilon, \sigma^2)$, MH	slide 24 (corrected)
(13)	small sin: $\rho \sim N_{[-1,1]}(\rho_1, \bar{V}_\rho)$, $\bar{V}_\rho = (V_\rho^{-1} + \sigma^{-2}\tilde{\epsilon}'\tilde{\epsilon})^{-1}$, $\rho_1 = \bar{V}_\rho(V_\rho^{-1}\rho_0 + \sigma^{-2}\tilde{\epsilon}'\epsilon)$	slide 25
(14)–(15)	$y = \mu + \sigma\Phi^{-1}(uB + (1-u)A)$ draws $N_{[a,b]}(\mu, \sigma^2)$	slide 26

1.4 · Hierarchical models → L07-04-hierarchical-linear-models

Eq.	Statement	Source
(1)–(3)	pooled $N(X_i\beta, \sigma^2 I)$; individual effects $N(1_T\alpha_i + \tilde{X}_i\tilde{\beta}, \sigma^2 I)$; random coefficients $N(X_i\beta_i, \sigma^2 I)$	HLM slides 4–6

Eq.	Statement	Source
(4)	dummy design X^* (individual effects, non-hierarchical prior)	slide 9 (corrected)
(5)	$\tilde{V}_\alpha^{-1} = V_\alpha^{-1} + \sigma^{-2}T$, $\tilde{V}_\alpha^{-1}\tilde{\mu}_{i\alpha} = V_\alpha^{-1}\mu_\alpha + \sigma^{-2}1_T'(y_i - \tilde{X}_i\tilde{\beta})$	slide 12
(6)	$\tilde{\mu}_{i\alpha} = w\bar{r}_i + (1-w)\mu_\alpha$, $w = TV_\alpha/(TV_\alpha + \sigma^2)$ — **partial pooling**	derived
(7)	$\tilde{V}_\mu^{-1} = V_\mu^{-1} + NV_\alpha^{-1}$, $\tilde{V}_\mu^{-1}\mu_1 = V_\mu^{-1}\mu_0 + V_\alpha^{-1}\sum\alpha_i$; V_α: IG with $\eta_0 + N$, $\eta_0\sigma_0^2 + \sum(\alpha_i - \mu_\alpha)^2$	slide 13 (corrected)
(8)–(10)	β_i, β_0, V_0^{-1} (Wishart) conditionals	slides 15–16 (corrected)
(11)–(15)	eight schools: $y_i \mid \mu, \tau \sim N(\mu, \sigma_i^2 + \tau^2)$; $\hat{\mu}, V_\mu$; $B_i = \sigma_i^2/(\sigma_i^2 + \tau^2)$; $E(\theta_i \mid y, \tau) = (1-B_i)y_i + B_i\hat{\mu}$; $p(\tau \mid y)$	slide 23, derived
(16)	random intercept: Var $= \sigma_a^2 + \sigma^2$, Cov $= \sigma_a^2$	MS (6.28)
(17)	RE: $\theta = 1 - [\sigma_u^2/(\sigma_u^2 + T\sigma_a^2)]^{1/2}$	W (14.10)

1.5 · Tobit → L07-05-tobit

Eq.	Statement	Source
(1)–(4)	$\sigma^{-1}\phi((y - x'\beta)/\sigma)$ for $y > 0$; $\Phi(-x'\beta/\sigma)$ for $y = 0$; log-likelihood	slide 6; G p. 118; W (17.20)–(17.22)
(5)–(6)	augmented posterior; $y_i^* \sim TN_{(-\infty,0]}(x_i'\beta, \sigma^2)$ for zeros	slide 7; G (8.6), Alg. 8.2
(7)–(8)	$\beta \mid \cdot \sim N(V_\beta(V_0^{-1}\beta_0 + \sigma^{-2}X'y^*), V_\beta)$, $V_\beta^{-1} = V_0^{-1} + \sigma^{-2}X'X$; σ^2 IG on y^*	slide 8 (corrected)
(9)	$E(y \mid y > 0, x) = x'\beta + \sigma\lambda(x'\beta/\sigma)$	W (17.24)
(10)	$E(y \mid x) = \Phi(z)x'\beta + \sigma\phi(z)$	W (17.25)
(11)	$\partial E(y \mid y > 0, x)/\partial x_j = \beta_j\{1 - \lambda(z)[z + \lambda(z)]\}$	W (17.26)
(12)	$\partial E(y \mid x)/\partial x_j = \beta_j\Phi(z)$	W (17.30)
(13)	slide 9: $P(y = 0) = \Phi(-2a/2.062)$	slide 9
(14)	normal x: OLS slope $= \beta P(y > 0)$	derived (Stein, G)

1.6 · Probit family → L07-06-probit-ordered-multinomial

Eq.	Statement	Source
(1)	$P(y=1)=\Phi(x'\beta)$; σ not identified → 1	G (8.7)–(8.9); W (17.2), (17.6)
(2)	$\partial P/\partial x_k=\phi(x'\beta)\beta_k\le 0.3989\beta_k$	W (17.7); G p. 124
(3)	$\beta^{logit}\approx 1.6\beta^{probit}$	G p. 128
(4)–(5)	Albert–Chib: y_i^* truncated at 0 by y_i; $\beta\sim N(B_1(X'y^*+B_0^{-1}\beta_0),B_1)$, $B_1=(X'X+B_0^{-1})^{-1}$	G Alg. 8.3
(6)	ordered: $P(y=j)=\Phi(\gamma_j-x'\beta)-\Phi(\gamma_{j-1}-x'\beta)$; $\gamma_1=0$	slide 11
(7)–(8)	$y_i^*\sim TN_{(\gamma_{j-1},\gamma_j]}$; $\gamma_j\sim U(\underline{\gamma}_j,\overline{\gamma}_j)$	slide 12
(9)–(10)	MNP: $y_{ij}^*=x_{ij}'\beta_j+\epsilon_{ij}$; choice = argmax (0 if all <0); β normal, Ω inverse Wishart, y_i^* truncated MVN	slides 13–15

---

2 · Every number and worked case

Case	Inputs	Result	Note
W Ex. 8.6	nettfa on inc, $n=2017$, weights $1/inc$	OLS 0.821 (0.104), −10.57; WLS 0.787 (0.063), −9.58 (1.65); R^2 0.0827 / 0.0709	01 §3
GLS cost	$h=e^x$, normal design	Var OLS/GLS slope 4.38	01 §4.1
Case I re-run	$n=100$, α = (1, 1), β = (2, 4)	β = (2.20, 3.97), $\bar{\alpha}$ = (0.93, 1.12), acceptance 0.55	01 §4
Airline	slide 14	β_2 1.117, β_3 −0.504, β_4 1.895, α_1 −3.968, α_2 7.085 (3.293); $e^{-3.968}=0.0189$; $e^{0.7085}=2.03$	01 §5
ν prior	Exp(25)	0.3297, 0.2019	02 §4
t re-run	slide 19 design + outlier	β (0.43, 0.87, 1.93) vs OLS (0.54, 1.07, 2.03); $\bar{\nu}$ 3.8; outlier weight 0.018	02 §5
Vella–Verbeek	G pp. 112–116	β_U 0.129 (0.036) / 0.128 (0.033); σ^2 0.165 / 0.105; $\log_{10}$ BF 4.83	02 §6

Case	Inputs	Result	Note
Barium	W Table 12.1	OLS 3.12 (0.48); PW 2.94 (0.63), −37.08, $\hat{\rho}$ 0.293, t −1.69; CO 2.95	03 §2
AR ratios	$\rho = 0.9$	$\varphi = 0.9$: true/naive 9.53; $\varphi = 0$: OLS/GLS 9.53	03 §3
AR re-run	slide 27 design	β (1.97, 4.06), $\bar{\sigma}^2$ 1.03, $\bar{\rho}$ 0.871	03 §5
T §2.9	interest rates	$r_3 = 0.832 + 0.930 r_1$ (R^2 96.5%); $c_3 = 0.792 c_1$	03 §2
Eight schools	slide 23	pooled 7.87 (4.17); τ: mode 0, median 5.23, mean 6.59, 95% [0.24, 20.87]; A at $\tau = 5.2$: $\hat{\mu}$ 8.027, B 0.8914, 10.24; Bayes A 11.64, C 6.35	04 §5
Pregnancy	site Rmd	pooled −4.52639 (0.48680) + 0.47633 (0.01897)x, $\hat{\sigma}$ 3.612, R^2 0.5974; individual means −4.412, 0.470; Gibbs α −4.363, β 0.4687, σ_y 1.119, σ_α 3.210, σ_β 0.1253; woman 1 at 40.143: 17.68 [15.68, 19.86]	04 §6
Weights	T = 6	w = 0.980; RE θ = 0.859	04 §6, §8
MS calves	(6.27)/(6.28)	$\hat{\alpha}$ 69.9, β 4.7; SE 7.8 → 5.6, 0.4 → 0.1	04 §8
G panel	Table 9.5	β_U 0.090 (0.006, 0.179); experience 0.058	04 §8
Slide 9	$\beta = 2$, $\sigma^2 = 0.25$	P(y = 0) 0.500, 0.249, 0.104; OLS slopes 1.00, 1.50, 1.79	05 §6
Mroz Tobit	W Table 17.3	educ 80.65 (21.58), kidslt6 −894.02 (111.88), $\hat{\sigma}$ 1122.02, ℓ −3819.09; APE 0.589 → 47.5; PEA 0.645; OLS 28.76	05 §7
G Mroz	Table 8.1	P(0 hours) 0.431 vs 0.328; mean 214 vs 587 h	05 §7
Truncated OLS	slide 9 design	slope 1.902	05 §8

Case	Inputs	Result	Note
Mroz probit/logit	W Table 17.1	educ 0.131 / 0.221; ℓ -401.30 / -401.77; scale 0.391 / 0.243 = 1.61; APE 0.301	06 §2
G computer	Table 8.2	-0.705, 0.879, 0.524; priors -1.2816, 0.1492, 1.0616, 0.0225, 1.0283	06 §2
Ordered re-run	$\gamma = (0, 1, 2)$	β (0.50, 0.89), $\bar{\gamma}_2$ 1.10, $\bar{\gamma}_3$ 2.08; lag-1 corr 0.99	06 §3
MNP re-run	Ω known	(0.45, 1.10, -0.11, -0.71) vs (0.3, 1.0, -0.2, -0.8)	06 §4
Rubinfeld MVP	G Table 9.4	β_{22} 1.113, β_{23} -1.437, σ_{12} 0.085	06 §5
MS inhibin	(6.29)	-2.310 (0.947), 0.026 (0.009); OR 1.026 per unit	06 §6

Class 7 · Pitfalls — fourteen confusions, and a 30-item True/False self-test

Prev: L07-07-formulas · Next: L07-09-python · Map: L07-00-map

Every counterexample uses a number derived in a section note; the new ones are checked in `L07_python_checks.py` §7 (L07-09-python §7).

Part I — the confusions

1 · "Heteroskedasticity biases OLS." No: OLS stays unbiased; its *variance formula* is wrong and it is inefficient. Counterexample: W Example 8.6, OLS 0.821 vs WLS 0.787 — both estimate the same slope; what differs is the SE (0.104 vs 0.063) (L07-01-general-covariance-gls §3).

2 · "Robust standard errors make OLS as good as GLS." They make OLS *inference* valid, not OLS *efficient*. At $\alpha_1 = 1$ the true OLS slope variance is 4.38 times GLS's whatever SE you report (01 §4.1).

3 · "WLS weights are $1/\sqrt{h_i}$." The *variables* are divided by $\sqrt{h_i}$; the *squared residuals* are weighted by $1/h_i$ (W p. 256). And the constant column is divided too — the transformed regression has no column of ones (01 (9)).

4 · "σ² in a t model is the error variance." It is the scale; the variance is $\sigma^2\nu/(\nu-2)$. G's Vella–Verbeek σ^2 falls from 0.165 to 0.105 under t_5, yet $0.105 \times 5/3 = 0.175$: the spread hardly changed (L07-02-student-t-errors §6).

5 · "The ω_i are random effects to be estimated and reported." They are a computational device: the model is the marginal t; the ω_i are integrated out by ignoring their draws (G p. 114). Their posterior means are still useful diagnostics — an outlier's weight 0.018 (02 §5).

6 · "A large ν posterior is evidence of normality." ν is weakly identified; the slide-21 chain wanders over 5–60 with ACF 0.5 at lag 30, and one outlier drags the posterior mean to 3.8 (02 §5). Look at the prior too: Exp(25) already puts 0.20 on $\nu \geq 40$.

7 · "With AR(1) errors OLS standard errors are always too small." Only if the regressor is persistent in the same direction: true/naive variance $= (1+\rho\phi)/(1-\rho\phi)$, equal to 1 when $\varphi = 0$ and **below** 1 when $\rho\varphi < 0$. The efficiency loss of OLS remains even when the SE is right (9.53 at $\rho = 0.9$, $\varphi = 0$) (L07-03-ar1-errors §3).

8 · "Cochrane–Orcutt and Prais–Winsten are different estimators of different things." Same model, same ρ iteration; PW keeps the first observation rescaled by $\sqrt{1-\rho^2}$. Barium: 2.94 vs 2.95 (03 §2).

9 · "Shrinkage pulls estimates toward zero." Toward the estimated population mean: school C (-2.75) moves **up** to 7.02 at $\tau = 5.23$ (L07-04-hierarchical-linear-models §5).

10 · "Hierarchical = random effects = consistent no matter what." The exchangeable prior, like RE, assumes unit effects unrelated to the regressors (W 14.8); FE (individual effects with a flat prior) does not (04 §8).

11 · "The site's band is a predictive interval." It is the posterior of $\alpha_i + \beta_i x$; a new measurement adds $N(0, \sigma_y^2)$ (04 §6).

12 · "The Tobit coefficient is the marginal effect on hours." It is the effect on $E(y^*)$; on $E(y \mid x)$ it is $\beta_j \Phi(z)$, at the averages $0.645 \times \beta$, on average $0.589 \times \beta$ for Mroz (L07-05-tobit §5, §7).

13 · "Censored = truncated." Censored samples keep the units (with y at the limit); truncated samples lose them, x included. The slides' "probability of truncation" is a censoring probability (05 §1). OLS is biased in both, differently: full censored sample $\to$ slope $\beta P(y > 0) = 1.50$; positive part only $\to 1.902$ (05 §6, §8).

14 · "Probit coefficients are comparable to logit ones." Only after scaling: logit $\approx 1.6 \times$ probit (Mroz $0.221/0.131 = 1.69$; scale factors $0.391/0.243 = 1.61$). The probabilities agree; the coefficients do not (L07-06-probit-ordered-multinomial §1–§2).

Part II — True/False self-test (original)

Score per `rules/00_estilo_avaliacao.md`: correct − wrong/2. Tags: **[E]** essential, **[M]** medium, **[P]** probing.

Block A — general covariance and heteroskedasticity

1. **[E]** If $P\Omega P' = I$, OLS on (Py, PX) equals $(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}y$ for every such P. **True.** $\Omega^{-1} = P'P$ and only $P'P$ appears in the OLS formula on the rotated data (01 (4)–(5)).

2. **[E]** Under heteroskedasticity with known h_i, OLS is biased, which is why WLS is used. **False.** OLS remains unbiased; WLS is used for efficiency and for valid usual standard errors (01 §1, W p. 255).

3. **[M]** In the full conditional of β, setting $V_0^{-1} = 0$ gives a posterior mean equal to the GLS estimator. **True.** $V_1^{-1}\beta_1 = \sigma^{-2}X'\Omega^{-1}y$ and $V_1^{-1} = \sigma^{-2}X'\Omega^{-1}X$ (01 (8)).

4. **[M]** For $h(z, \alpha) = \exp(z'\alpha)$ with a normal prior on α, the full conditional of α is normal. **False.** α appears in $-\frac{1}{2}\sum z_i'\alpha$ and inside $e^{-z_i'\alpha}$: no conjugate form, hence MH (01 (12)).

5. **[P]** In FGLS, the intercept of the regression of $\log \hat{u}^2$ on x estimates $\log \sigma^2$ consistently. **False.** It absorbs $E \log v$ (-1.27 for normal errors); harmless, because a constant factor in h cancels from WLS (01 §4.1).

6. **[M]** A WLS standard error 40% smaller than the robust OLS one proves the variance model is correct. **False.** The WLS SE is computed *assuming* $h = inc$; if h is wrong, it is wrong too (W p. 257, 01 §4.1).

Block B — Student-t errors

7. **[E]** If $\epsilon \mid \omega \sim N(0, \sigma^2\omega)$ and $\omega \sim IG(\nu/2, \nu/2)$, then $\epsilon \sim t_\nu(0, \sigma^2)$. **True.** Integrating the $IG((\nu+1)/2, (\nu + \epsilon^2/\sigma^2)/2)$ kernel leaves $(1 + \epsilon^2/(\nu\sigma^2))^{-(\nu+1)/2}$ (02 (1)–(3)).

8. **[E]** If $\omega_i^{-1} \sim IG(\nu/2, \nu/2)$, the marginal of ϵ_i is t_ν. **False.** The reciprocal relation: $\omega_i \sim IG$, equivalently $\omega_i^{-1} \sim$ Gamma (the slide-6 slip, 02 §2).

9. **[M]** In the Gibbs sampler, an observation with $|\epsilon_i| < \sigma$ gets an expected weight above 1. **True.** $(\nu+1)/(\nu + \epsilon_i^2/\sigma^2) > 1 \iff \epsilon_i^2 < \sigma^2$ (02 (5)).

10. **[M]** The full conditional of ν is a gamma distribution when ν has an exponential prior. **False.** ν enters through $[(\nu/2)^{\nu/2}/\Gamma(\nu/2)]^n$; the conditional (02 (6)) is non-standard — MH.

11. [**P**] With a flat prior on β and $\nu \sim \text{Exp}(25)$, the posterior median of β exists but the posterior mean may not. **True.** Geweke: the mean needs $P(\nu \leq 2) = 0$, and the exponential puts mass there; quantiles always exist (02 §4).

12. [**M**] Under t_5 errors the scale σ^2 equals the error variance. **False.** Variance $= \sigma^2 \cdot 5/3$ (02 §6).

Block C — AR(1) errors

13. [**E**] With AR(1) errors, $\text{Cov}(\epsilon_t, \epsilon_{t+k}) = \rho^k \sigma^2/(1-\rho^2)$ for $|\rho| < 1$. **True.** (03 (1)–(2)).

14. [**M**] Ω_ρ is tridiagonal. **False.** Its *inverse* is tridiagonal; Ω_ρ itself is full, entries $\rho^{|i-j|}/(1-\rho^2)$ (03 (3)).

15. [**M**] Cochrane–Orcutt and Prais–Winsten differ only in the treatment of the first observation. **True.** PW keeps $\sqrt{1-\rho^2}(y_1, x_1)$; CO drops it (03 §2, W p. 383).

16. [**P**] If the regressor is serially uncorrelated, AR(1) errors leave OLS both efficient and with correct usual standard errors. **False.** The SE is (asymptotically) right — ratio 1 — but GLS is still more efficient: 9.53 at $\rho = 0.9$ (03 (10)–(11)).

17. [**M**] After conditioning on the first error, the full conditional of ρ under a truncated-normal prior is a truncated normal. **True.** The likelihood becomes that of a regression of ϵ_t on ϵ_{t-1} (03 (13)).

Block D — hierarchical models

18. [**E**] In the individual-effects model with a hierarchical prior, the posterior mean of α_i moves toward μ_α more when T is small. **True.** Weight on own data $w = TV_\alpha/(TV_\alpha + \sigma^2)$ falls with T (04 (6)).

19. [**M**] In the update of μ_α, the precision of each α_i is σ^{-2}. **False.** It is V_α^{-1} — the slide-13 slip (04 (7)).

20. [**M**] In the eight-schools model, every posterior mean $E(\theta_i \mid y, \tau)$ lies between y_i and $\hat{\mu}(\tau)$. **True.** It is the convex combination $(1 - B_i)y_i + B_i\hat{\mu}$ (04 (14)).

21. [**P**] Shrinkage always moves the extreme schools toward zero. **False.** Toward $\hat{\mu} \approx 8$: school C (-2.75) moves up (04 §5, Part I.9).

22. [**M**] The individual-effects model with a vague non-hierarchical prior reproduces the dummy-variable (fixed-effects) regression. **True.** Each α_i is learned from its own unit only (04 §2, W p. 438).

23. [**P**] The site's 95% band for woman 1 at $x = 40.143$ covers a new weight measurement with 95% probability. **False.** It omits the σ_y noise: it is a band for the mean line (04 §6).

Block E — Tobit

24. [**E**] In the Tobit, $P(y = 0 \mid x) = 1 - \Phi(x'\beta/\sigma)$. **True.** (05 (2), W 17.21).

25. [**M**] In the Tobit, $\partial E(y \mid x)/\partial x_j = \beta_j$. **False.** It is $\beta_j \Phi(x'\beta/\sigma)$ (05 (12)); β is the latent-scale effect.

26. [**M**] In Chib's data augmentation, the latent value of a censored observation is drawn from a normal truncated to $(-\infty, 0]$. **True.** (05 (6)).

27. [**P**] With $x \sim N(a, 1)$, OLS of the censored y on x estimates β times the share of uncensored observations. **True.** Stein's lemma: $\text{Cov}(x, \max(0, y^*)) = \text{Cov}(x, y^*)P(y^* > 0)$ (05 (14)).

Block F — probit family

28. **[E]** In the probit, (β, σ) and $(2\beta, 2\sigma)$ give the same likelihood, so σ is fixed at 1. **True.** Only $x'\beta/\sigma$ enters (06 (1)).

29. **[M]** In the ordered probit Gibbs sampler, each cut-point is drawn from a normal distribution. **False.** From a uniform between the largest latent value of its lower category and the smallest of its upper one (06 (8)).

30. **[P]** In the multinomial probit, fixing one variance of Ω is needed even after differencing utilities from a base alternative. **True.** Differencing removes the location; $(c\beta, c^2\Omega)$ still gives the same choices — the scale (06 §4).

Links

- Unit: L07-00-map · L07-07-formulas · L07-09-python §7 (the counterexamples' code).
- Style: rules/00.

Class 7 · Python — every sampler of the class, asserted against the sources

> Every routine the class needs, each followed by the `asserts` that tie it to a printed number. numpy, scipy, statsmodels and the `wooldridge` data only — **no PyMC, no scikit-learn**: every Gibbs and Metropolis–Hastings step is written out, as the slides write it. The lecturer's R is the reference, never the code. **The file is the source of truth:** `Notas/L07/L07_python_checks.py`; the blocks below are copied from it verbatim, section by section. Prev: L07-08-pitfalls · Map: L07-00-map · Formulas: L07-07-formulas

Run: `python Notas/L07/L07_python_checks.py` — about two minutes (the samplers); it prints one line per section and ends with `ALL L07 CHECKS PASSED` (§8). The first disagreement aborts with an `AssertionError` naming the quantity and the printed value. Data: `pip install wooldridge` (allowed by the project plan).

Reused, not rewritten.

Existing code	What it supplies	Used in
`figures/make_figures.py` (imported as `mf`)	eight-schools data, `schools_conditional`, `schools_tau_posterior`; pregnancy data and `gibbs_hlm` (the site's sampler, ported); `gls_efficiency`; `ar1_ratios`; `tobit_moments`, `p_zero_slide`; `t_weight`, `mixture_pdf`	§1, §2, §4, §5, §7
`companions/check_companions.py`	the two pages recomputed and compared with their JavaScript under `node`	cited, not imported: it checks the **pages**
`statsmodels` `OLS`, `WLS`, `Probit`, `Logit`	the frequentist fits of W's tables	§1, §3, §5, §6

What is new here, and why: no existing project code samples these models. `statsmodels` has no Bayesian samplers and no Tobit; its `GLSAR` iterates Cochrane–Orcutt, not Prais–Winsten, so PW is three lines here.

```
"""Class 7 (L07) reference implementation with self-checks against the sources'
 ↪  printed numbers.

Run: python Notas/L07/L07_python_checks.py   (numpy, scipy, statsmodels, wooldridge;
 ↪  no scikit-learn, no PyMC)
Every section asserts the numbers of one section note and prints a one-line summary;
 ↪  the first
mismatch aborts with an AssertionError naming it. Shown section by section in
 ↪  L07-09-python.md.

Reused, not rewritten: figures/make_figures.py (imported as mf) supplies the
 ↪  eight-schools data and
```

```
exact posterior in tau, the pregnant-women data and the port of the site's hierarchical Gibbs sampler,
the GLS/OLS efficiency ratio, the AR(1) variance ratios, the Tobit moments and the ordered-probit
probabilities. Importing it draws nothing (build() runs only as __main__).
Data for the Wooldridge tables come from the `wooldridge` package (pip install wooldridge).
"""
import pathlib
import sys

import numpy as np
import statsmodels.api as sm
import wooldridge
from scipy import integrate, optimize, stats
from scipy.special import gammaln

sys.path.insert(0, str(pathlib.Path(__file__).resolve().parent / "figures"))
import make_figures as mf  # noqa: E402


def near(x, printed, dec):
    """True when x rounds to the printed value at `dec` decimals (half a unit in the last place)."""
    return abs(x - printed) <= 0.5 * 10.0 ** -dec + 1e-9


def check(x, printed, dec, what):
    assert near(x, printed, dec), f"{what}: computed {x!r}, source prints {printed}"
```

0 · Set-up, and the two samplers every note reuses

`check(x, printed, dec)` passes when `x` rounds to the printed value at `dec` decimals. `rtruncnorm` is slide 26's inverse-CDF draw (note 03 (15)), with the upper-tail reflection; `draw_beta` is the general-Ω conditional (note 01 (8)) — the boxed result of the map.

```
# ============================================================================ shared samplers
def rtruncnorm(rng, mu, sd, a, b):
    """Draw N(mu, sd^2) truncated to [a, b] by the inverse CDF (GLM3 slide 26).

    y = mu + sd * Phi^-1( u Phi((b-mu)/sd) + (1-u) Phi((a-mu)/sd) ), u ~ U(0, 1).
    Vectorized over mu, a, b. Works on the upper tail by symmetry so Phi^-1 is never asked for 1 - 1e-17.
    """
    mu, a, b = np.broadcast_arrays(np.asarray(mu, float), np.asarray(a, float), np.asarray(b, float))
    u = rng.uniform(size=mu.shape)
    al, be = (a - mu) / sd, (b - mu) / sd
    flip = al > 0                                   # whole interval in the upper tail: reflect
```

```
    lo = np.where(flip, -be, al)
    hi = np.where(flip, -al, be)
    z = stats.norm.ppf(u * stats.norm.cdf(hi) + (1 - u) * stats.norm.cdf(lo))
    return mu + sd * np.where(flip, -z, z)


def draw_beta(rng, X, y, prec_w, sigma2, b0, V0inv):
    """beta | . ~ N(b1, V1), V1^-1 = V0^-1 + X' W X / sigma2, V1^-1 b1 = V0^-1 b0 +
     ↪  X' W y / sigma2.

    prec_w: vector of weights (diagonal of Omega^-1) or a full matrix Omega^-1. GLM3
 ↪  slide 5.
    """
    XtW = X.T * prec_w if np.ndim(prec_w) == 1 else X.T @ prec_w
    P = V0inv + XtW @ X / sigma2
    V1 = np.linalg.inv(P)
    b1 = V1 @ (V0inv @ b0 + XtW @ y / sigma2)
    return rng.multivariate_normal(b1, V1), b1, V1
```

1 · GLS, WLS and Case I → L07-01-general-covariance-gls

OLS on (Py, PX) = GLS; W Table 8.1 from the `401KSUBS` data; FGLS; a Gibbs sampler with a random-walk MH step for α on the slide-8 design.

```
# ============================================================================ 1 ·
 ↪  general covariance, GLS, WLS (L07-01)
def section1():
    rng = np.random.default_rng(1)
    n = 40
    X = np.column_stack([np.ones(n), rng.normal(size=n)])
    A = rng.normal(size=(n, n))
    Omega = A @ A.T / n + np.eye(n)                          # an arbitrary known
 ↪  covariance
    y = X @ np.array([1.0, 2.0]) + np.linalg.cholesky(Omega) @ rng.normal(size=n)
    L = np.linalg.cholesky(Omega)
    P = np.linalg.inv(L)                                     # P Omega P' = I (slide 4)
    assert np.allclose(P @ Omega @ P.T, np.eye(n))
    b_star = np.linalg.lstsq(P @ X, P @ y, rcond=None)[0]    # OLS on (Py, PX)
    Oi = np.linalg.inv(Omega)
    b_gls = np.linalg.solve(X.T @ Oi @ X, X.T @ Oi @ y)      # (X'O^-1X)^-1 X'O^-1 y
    assert np.allclose(b_star, b_gls)
    # full conditional with a vague prior collapses to GLS (slide 5 with V0^-1 -> 0)
    _, b1, V1 = draw_beta(rng, X, y, Oi, 2.0, np.zeros(2), np.zeros((2, 2)))
    assert np.allclose(b1, b_gls) and np.allclose(V1, 2.0 * np.linalg.inv(X.T @ Oi @
     ↪  X))

    # Wooldridge Example 8.6 (Table 8.1, p. 257): nettfa on inc, single people (fsize
     ↪  = 1), WLS weights 1/inc
    d = wooldridge.data("401ksubs")
    d = d[d.fsize == 1]
    Xd = sm.add_constant(d"inc")
```

```
ols = sm.OLS(d.nettfa, Xd).fit(cov_type="HC0")
wls = sm.WLS(d.nettfa, Xd, weights=1 / d.inc).fit()
check(len(d), 2017, 0, "Table 8.1 n")
check(ols.params["inc"], 0.821, 3, "OLS inc")
check(ols.bse["inc"], 0.104, 3, "OLS inc robust SE")
check(ols.params["const"], -10.57, 2, "OLS intercept")
check(wls.params["inc"], 0.787, 3, "WLS inc")
check(wls.bse["inc"], 0.063, 3, "WLS inc SE")
check(wls.params["const"], -9.58, 2, "WLS intercept")
check(wls.bse["const"], 1.65, 2, "WLS intercept SE")
check(ols.rsquared, 0.0827, 4, "OLS R2")
check(wls.rsquared, 0.0709, 4, "WLS R2")
# WLS is OLS on the transformed variables (8.25)-(8.26): divide every column, the
 ↪  constant too, by sqrt(h)
sh = np.sqrt(d.inc.values)
b_tr = np.linalg.lstsq(Xd.values / sh[:, None], d.nettfa.values / sh,
↪ rcond=None)[0]
assert np.allclose(b_tr, wls.params.values)

# Case I, h(x, alpha) = exp(alpha0 + alpha1 x), slide-8 design: n = 100, alpha =
 ↪  (1, 1), sigma^2 = 1
n = 100
x = rng.normal(size=n)
Xh = np.column_stack([np.ones(n), x])
beta_true, alpha_true = np.array([2.0, 4.0]), np.array([1.0, 1.0])
yh = Xh @ beta_true + np.sqrt(np.exp(Xh @ alpha_true)) * rng.normal(size=n)
# FGLS (W 8.30-8.33) = the slide's initial value for alpha: regress log(e_ols^2)
 ↪  on x
b_ols = np.linalg.lstsq(Xh, yh, rcond=None)[0]
g = np.linalg.lstsq(Xh, np.log((yh - Xh @ b_ols) ** 2), rcond=None)[0]
b_fgls = sm.WLS(yh, Xh, weights=np.exp(-Xh @ g)).fit().params
# slope of the log-variance regression is consistent for alpha1; the intercept is
 ↪  off by E log chi2_1 = -1.27
check(stats.chi2(1).expect(np.log), -1.27, 2, "E log chi2_1")
# Gibbs for beta + random-walk MH for alpha (slide 7: alpha needs MH), priors as
 ↪  slide 8
V0inv, Vainv = np.eye(2) / 100, np.eye(2) / 100

def loglik_alpha(a, b):
    e = yh - Xh @ b
    lh = Xh @ a
    return -0.5 * np.sum(lh) - 0.5 * np.sum(e ** 2 * np.exp(-lh))

a, b = g.copy(), b_ols.copy()
keep, acc = [], 0
for it in range(12000):
    b, _, _ = draw_beta(rng, Xh, yh, np.exp(-Xh @ a), 1.0, np.zeros(2), V0inv)
    prop = a + 0.15 * rng.normal(size=2)
    lr = loglik_alpha(prop, b) - loglik_alpha(a, b) - 0.5 * (prop @ Vainv @ prop
↪ - a @ Vainv @ a)
```

```
        if np.log(rng.uniform()) < lr:
            a, acc = prop, acc + 1
        if it >= 2000:
            keep.append(np.r_[b, a])
    keep = np.array(keep)
    pm = keep.mean(0)
    assert 0.15 < acc / 12000 < 0.7, acc / 12000
    assert np.all(np.abs(pm[2:] - alpha_true) < 0.4) and np.all(np.abs(pm[:2] -
     ↪ beta_true) < 0.4), pm
    # efficiency ratio used in the figure (exact, fixed normal-quantile design)
    r = mf.gls_efficiency(1.0)
    check(r[1], 4.38, 2, "Var OLS / Var GLS slope at alpha1 = 1")
    print(f"1 GLS: OLS on (Py, PX) = GLS; W Table 8.1 OLS .821 (.104) / WLS .787
     ↪ (.063) reproduced; "
          f"Case I MH acceptance {acc/12000:.2f}, posterior means beta {pm[0]:.2f},
           ↪ {pm[1]:.2f}, alpha {pm[2]:.2f}, {pm[3]:.2f}; "
          f"FGLS slope {b_fgls[1]:.2f}; OLS/GLS slope variance at alpha1 = 1:
           ↪ {r[1]:.2f}")
    return dict(acc=acc / 12000, pm=pm, b_ols=b_ols, g=g, b_fgls=b_fgls)
```

2 · Student-t errors → L07-02-student-t-errors

The scale-mixture identity by quadrature, the ν prior probabilities, and the Gibbs + MH sampler of slides 16–19 with one planted outlier.

```
# ============================================================================== 2 ·
 ↪ Student-t errors (L07-02)
def gibbs_t(rng, X, y, b0, V0inv, nu0_s, s0sq, nu_prior_mean, M0, M, nu_init=10.0,
 ↪ step=0.25):
    """Gibbs + MH sampler of GLM3 slides 16-19 for y = X beta + e, e_i | w_i ~ N(0,
     ↪ sigma2 w_i), w_i ~ IG(nu/2, nu/2).

    Full conditionals: w_i ~ IG((nu+1)/2, (nu + e_i^2/sigma2)/2); beta ~ N with
 ↪ weights 1/w_i;
    sigma2 ~ IG((nu0+n)/2, (nu0 s0^2 + sum e_i^2/w_i)/2); nu by random-walk MH on
    log p(nu|w) = n[(nu/2)log(nu/2) - lgamma(nu/2)] - [1/nu^0 + 0.5 sum(log w_i +
 ↪ 1/w_i)] nu, nu > 0 (slide 18).
    Returns draws (M x (k+2)): beta, sigma2, nu; plus the MH acceptance rate and the
 ↪ mean weights 1/w_i.
    """
    n, k = X.shape
    beta = np.linalg.lstsq(X, y, rcond=None)[0]
    sigma2 = np.sum((y - X @ beta) ** 2) / (n - k)
    nu, w = nu_init, np.ones(n)
    out, acc, wbar = np.zeros((M, k + 2)), 0, np.zeros(n)

    def lp_nu(v, w):
        if v <= 0:
            return -np.inf
        return n * (0.5 * v * np.log(0.5 * v) - gammaln(0.5 * v)) - (1 /
         ↪ nu_prior_mean + 0.5 * np.sum(np.log(w) + 1 / w)) * v
```

```
    for it in range(M0 + M):
        e = y - X @ beta
        w = 1 / rng.gamma((nu + 1) / 2, 2 / (nu + e ** 2 / sigma2))       # 1/w ~
↪   G((nu+1)/2, rate (nu+e^2/s2)/2)
        beta, _, _ = draw_beta(rng, X, y, 1 / w, sigma2, b0, V0inv)
        e = y - X @ beta
        sigma2 = 1 / rng.gamma((nu0_s + n) / 2, 2 / (nu0_s * s0sq + np.sum(e ** 2 /
↪   w)))
        prop = nu + step * rng.normal()
        if np.log(rng.uniform()) < lp_nu(prop, w) - lp_nu(nu, w):
            nu, acc = prop, acc + 1
        if it >= M0:
            out[it - M0] = np.r_[beta, sigma2, nu]
            wbar += 1 / w / M
    return out, acc / (M0 + M), wbar


def section2():
    # the scale-mixture identity, numerically (G p. 52)
    for nu in (1, 3, 5, 10, 30):
        for e in (0.0, 0.8, 2.5, 6.0):
            assert abs(mf.mixture_pdf(e, nu) - stats.t.pdf(e, nu)) < 1e-8, (nu, e)
    # prior of nu ~ Exp(mean 25) (slide 17): mass on nu <= 10 and nu >= 40
    p10, p40 = 1 - np.exp(-10 / 25), np.exp(-40 / 25)
    check(p10, 0.3297, 4, "P(nu <= 10)")
    check(p40, 0.2019, 4, "P(nu >= 40)")
    # Var of t_nu is nu/(nu-2) sigma^2: the scale sigma^2 is not the variance
    ↪  (pitfall)
    check(stats.t(5).var(), 5 / 3, 6, "Var t_5")
    # slide-19 simulation: n = 100, nu = 10, sigma^2 = 1, beta = (0.5, 1, 2); priors
    ↪  beta0 = 0, V0 = 10 I3, nu0 = 5, s0^2 = 1, nu^0 = 25
    rng = np.random.default_rng(19)
    n = 100
    X = np.column_stack([np.ones(n), rng.normal(size=(n, 2))])
    beta_true = np.array([0.5, 1.0, 2.0])
    y = X @ beta_true + stats.t(10).rvs(size=n, random_state=rng)
    y[7] += 15.0                                                    # one gross
↪   outlier
    draws, acc, wbar = gibbs_t(rng, X, y, np.zeros(3), np.eye(3) / 10, 5.0, 1.0,
↪   25.0, M0=3000, M=20000)
    pm = draws.mean(0)
    ols = np.linalg.lstsq(X, y, rcond=None)[0]
    assert np.all(np.abs(pm[:3] - beta_true) < 0.3), pm
    assert wbar[7] < 0.2 * np.median(wbar), (wbar[7], np.median(wbar))   # the
    ↪  outlier is downweighted
    keep = np.arange(n) != 7
    ols_clean = np.linalg.lstsq(X[keep], y[keep], rcond=None)[0]
    shift_ols, shift_t = abs(ols[0] - ols_clean[0]), abs(pm[0] - ols_clean[0])
```

```
    assert shift_t < 0.5 * shift_ols, (shift_t, shift_ols)                # the
     ↪  outlier moves OLS, barely the t fit
    print(f"2 t errors: mixture identity holds; P(nu<=10) = {p10:.4f}, P(nu>=40) =
     ↪  {p40:.4f}; "
          f"Gibbs-MH (one outlier +15): beta {pm[0]:.2f}, {pm[1]:.2f}, {pm[2]:.2f} vs
           ↪  OLS {ols[0]:.2f}, {ols[1]:.2f}, {ols[2]:.2f}; "
          f"posterior mean nu {pm[4]:.1f}; MH acceptance {acc:.2f}; outlier weight
           ↪  {wbar[7]:.3f} vs median {np.median(wbar):.3f}")
    return dict(pm=pm, ols=ols, acc=acc, w_out=wbar[7], w_med=np.median(wbar))
```

3 · AR(1) errors → L07-03-ar1-errors

Ω_ρ^{-1} tridiagonal and $P'P = \Omega_\rho^{-1}$; iterated Prais–Winsten reproducing W Table 12.1; Cochrane–Orcutt; the truncated-normal draw against `truncnorm`; the slide-27 sampler with the small-sin ρ step.

```
# ============================================================================== 3 ·
 ↪  AR(1) errors (L07-03)
def omega_rho(n, rho):
    """Omega_rho of GLM3 slide 23: entries rho^|i-j| / (1 - rho^2)."""
    i = np.arange(n)
    return rho ** np.abs(i[:, None] - i[None, :]) / (1 - rho ** 2)


def omega_rho_inv(n, rho):
    """Omega_rho^-1 in closed form: tridiagonal, diagonal (1, 1+rho^2, ..., 1+rho^2,
     ↪  1), off-diagonal -rho."""
    return np.diag(np.r_[1, np.full(n - 2, 1 + rho ** 2), 1]) - rho * (np.eye(n, k=1)
     ↪  + np.eye(n, k=-1))


def prais_winsten(y, X, rho):
    """Quasi-differenced data of W (12.31)-(12.32): first row scaled by sqrt(1 -
     ↪  rho^2), then y_t - rho y_{t-1}."""
    ys = np.r_[np.sqrt(1 - rho ** 2) * y[0], y[1:] - rho * y[:-1]]
    Xs = np.vstack([np.sqrt(1 - rho ** 2) * X[0], X[1:] - rho * X[:-1]])
    return ys, Xs


def section3():
    n, rho = 6, 0.7
    Om = omega_rho(n, rho)
    Oi = np.linalg.inv(Om)
    tri = omega_rho_inv(n, rho)
    assert np.allclose(Oi, tri)                                          # tridiagonal
     ↪  inverse
    Pm = np.vstack([np.r_[np.sqrt(1 - rho ** 2), np.zeros(n - 1)],
                    np.eye(n)[1:] - rho * np.eye(n, k=-1)[1:]])
    assert np.allclose(Pm.T @ Pm, Oi) and np.allclose(Pm @ Om @ Pm.T, np.eye(n))   #
     ↪  PW is the P of slide 4
```

```
    # Wooldridge Example 12.4 (Table 12.1, p. 384): iterated Prais-Winsten on BARIUM
    b = wooldridge.data("barium")
    y = b.lchnimp.values
    X = sm.add_constant(b"lchempi", "lgas", "lrtwex", "befile6", "affile6",
↪   "afdec6").values
    ols = sm.OLS(y, X).fit()
    check(ols.params[1], 3.12, 2, "OLS log(chempi)")
    check(ols.bse[1], 0.48, 2, "OLS SE log(chempi)")
    check(ols.params[0], -17.80, 2, "OLS intercept")
    r, n_pw = 0.0, 0
    while True:
        fit = sm.OLS(*prais_winsten(y, X, r)).fit()
        u = y - X @ fit.params
        r_new = np.sum(u[1:] * u[:-1]) / np.sum(u[:-1] ** 2)          # W (12.14):
↪   regress u_t on u_{t-1}
        n_pw += 1
        if abs(r_new - r) < 1e-10:
            break
        r = r_new
    check(r, 0.293, 3, "PW rho-hat")
    check(fit.params[1], 2.94, 2, "PW log(chempi)")
    check(fit.bse[1], 0.63, 2, "PW SE log(chempi)")
    check(fit.params[0], -37.08, 2, "PW intercept")
    check(fit.params[6], -0.577, 3, "PW afdec6")
    check(fit.params[6] / fit.bse[6], -1.69, 2, "PW t afdec6")
    # Cochrane-Orcutt drops the first row: same rho iteration, different numbers in a
     ↪  sample of 131
    r_co = 0.0
    for _ in range(200):
        ys, Xs = prais_winsten(y, X, r_co)
        f_co = sm.OLS(ys[1:], Xs[1:]).fit()
        u = y - X @ f_co.params
        r_co = np.sum(u[1:] * u[:-1]) / np.sum(u[:-1] ** 2)

    # truncated-normal inverse-CDF draw (slide 26) equals truncnorm's quantile
     ↪  function, exactly
    uu = np.linspace(0.001, 0.999, 999)
    mu, sd, lo, hi = 0.9, 0.05, -1.0, 1.0
    yq = mu + sd * stats.norm.ppf(uu * stats.norm.cdf((hi - mu) / sd) + (1 - uu) *
↪   stats.norm.cdf((lo - mu) / sd))
    assert np.allclose(yq, stats.truncnorm((lo - mu) / sd, (hi - mu) / sd, loc=mu,
     ↪  scale=sd).ppf(uu), atol=1e-10)
    rng = np.random.default_rng(26)
    tail = rtruncnorm(rng, np.zeros(20000), 1.0, 8.0, np.inf)             # deep upper
↪   tail: no Phi^-1(1) blow-up
    assert np.all(tail >= 8.0) and abs(tail.mean() - stats.truncnorm(8,
     ↪  np.inf).mean()) < 0.01

    # slide-27 simulation: n = 200, rho = 0.9, sigma^2 = 1, beta = (2, 4); priors
     ↪  beta0 = 0, V0 = 100 I2,
```

```
    # nu0 = 5, s0^2 = 1, rho0 = 0.9, V_rho = 100; rho by the "small sin"
     ↪ truncated-normal conditional (slide 25)
    n = 200
    Xs = rng.normal(size=(n, 2))
    e = np.zeros(n)
    e[0] = rng.normal() / np.sqrt(1 - 0.81)
    for t in range(1, n):
        e[t] = 0.9 * e[t - 1] + rng.normal()
    ys = Xs @ np.array([2.0, 4.0]) + e
    beta, s2, rh = np.linalg.lstsq(Xs, ys, rcond=None)[0], 1.0, 0.5
    keep = []
    for it in range(6000):
        Oi = omega_rho_inv(n, rh)                                    # closed form, no
↪   200 x 200 inversion
        beta, _, _ = draw_beta(rng, Xs, ys, Oi, s2, np.zeros(2), np.eye(2) / 100)
        res = ys - Xs @ beta
        s2 = 1 / rng.gamma((5 + n) / 2, 2 / (5 * 1.0 + res @ Oi @ res))
        et, el = res[1:], res[:-1]
        Vr = 1 / (1 / 100 + el @ el / s2)
        r1 = Vr * (0.9 / 100 + el @ et / s2)
        rh = float(rtruncnorm(rng, r1, np.sqrt(Vr), -1.0, 1.0))
        if it >= 1000:
            keep.append([beta[0], beta[1], s2, rh])
    pm = np.array(keep).mean(0)
    assert abs(pm[0] - 2) < 0.2 and abs(pm[1] - 4) < 0.2 and abs(pm[3] - 0.9) < 0.07,
     ↪ pm
    print(f"3 AR(1): Omega_rho^-1 tridiagonal, PW matrix P'P = Omega^-1; W Table 12.1
     ↪ reproduced (PW 2.94 (0.63), "
          f"rho-hat {r:.3f}, t_afdec6 {fit.params[6]/fit.bse[6]:.2f}; {n_pw}
           ↪ iterations); Cochrane-Orcutt rho {r_co:.3f}, "
          f"log(chempi) {f_co.params[1]:.2f}; Gibbs (slide 27 design) beta
           ↪ {pm[0]:.2f}, {pm[1]:.2f}, sigma2 {pm[2]:.2f}, rho {pm[3]:.3f}")
    return dict(rho=r, pw=fit, co=f_co, r_co=r_co, pm=pm)
```

4 · Hierarchical models → L07-04-hierarchical-linear-models

Eight schools exactly; the site's pooled and individual regressions to every printed digit; the site's Gibbs sampler (ported in `figures/make_figures.py`, `gibbs_hlm`) and the weights w and θ.

```
# ============================================================================ 4 ·
 ↪ hierarchical models (L07-04)
def section4():
    # eight schools: exact p(tau | y) under tau ~ U(0, 100), mu ~ N(0, 1e6) (HLM
     ↪ slide 23)
    grid = np.linspace(0, 100, 4001)
    p = mf.schools_tau_posterior(grid)
    med = mf.quantile_on_grid(grid, p, 0.5)
    mean_tau = np.trapezoid(grid * p, grid)
    q025, q975 = mf.quantile_on_grid(grid, p, 0.025), mf.quantile_on_grid(grid, p,
↪   0.975)
```

```
    w = 1 / mf.S8 ** 2
    pooled, pooled_sd = np.sum(w * mf.Y8) / w.sum(), 1 / np.sqrt(w.sum())
    check(pooled, 7.87, 2, "complete-pooling mean")
    check(pooled_sd, 4.17, 2, "complete-pooling sd")
    check(med, 5.23, 2, "posterior median tau")
    # fully Bayes E(theta_i | y): integrate E(theta_i | y, tau) against p(tau | y)
    cond = np.array([mf.schools_conditional(t)["mean"] for t in grid])
    eb = np.trapezoid(cond * p[:, None], grid, axis=0)
    cA = mf.schools_conditional(med)
    assert np.all(np.diff(np.sort(eb)) >= 0) and eb.max() < mf.Y8.max() and eb.min()
     ↪  > mf.Y8.min()
    # school A at the posterior median of tau, by hand: B_A = 14.9^2/(14.9^2 + tau^2)
    BA = 14.9 ** 2 / (14.9 ** 2 + med ** 2)
    check(cA["B"][0], BA, 10, "B_A")
    check(cA["mean"][0], (1 - BA) * 28.39 + BA * cA["mu"], 10, "E(theta_A | y, tau)")

    # pregnant women (hierarchicalmodel.html): pooled, individual and hierarchical
     ↪  regressions
    ys, xs = mf.pregnancy()
    x1, y1 = np.concatenate(xs), np.concatenate(ys)
    fit = sm.OLS(y1, sm.add_constant(x1)).fit()
    check(len(y1), 427, 0, "n")
    check(fit.params[0], -4.52639, 5, "pooled intercept")
    check(fit.params[1], 0.47633, 5, "pooled slope")
    check(fit.bse[0], 0.48680, 5, "pooled SE intercept")
    check(fit.bse[1], 0.01897, 5, "pooled SE slope")
    check(np.sqrt(fit.scale), 3.612, 3, "residual standard error")
    check(fit.rsquared, 0.5974, 4, "R2")
    ind = mf.ols_by_woman(ys, xs)
    sig = np.array([np.sqrt(np.sum((y - c[0] - c[1] * x) ** 2) / (len(y) - 2)) for y,
↪   x, c in zip(ys, xs, ind)])
    for col, (mn, q1, md, mean, q3, mx) in zip(ind.T, ((-14.432, -6.744, -4.260,
     ↪  -4.412, -2.330, 5.157),
                                                       (0.126, 0.395, 0.450, 0.470,
↪   0.525, 0.801))):
        got = (col.min(), np.quantile(col, 0.25), np.median(col), col.mean(),
↪   np.quantile(col, 0.75), col.max())
        for g_, pr in zip(got, (mn, q1, md, mean, q3, mx)):
            check(g_, pr, 3, "individual-regression summary")
    check(sig.mean(), 1.0133, 4, "mean individual sigma")
    check(sig.min(), 0.3484, 4, "min individual sigma")
    d1, d2 = mf.gibbs_hlm(ys, xs)
    post = d2.mean(0)
    pm1 = d1.mean(0)
    # partial pooling: posterior means less dispersed than individual OLS, centred
     ↪  near the population means
    assert np.all(post.std(0) < ind.std(0))
    assert abs(pm1[0] - ind[:, 0].mean()) < 1.0 and abs(pm1[1] - ind[:, 1].mean()) <
     ↪  0.05
```

```
    # the conditional shrinkage weight of a woman's intercept (alpha_i given
     ↪ everything else), HLM slide 12 form:
    # precision 1/s_a^2 + T/s_y^2  =>  weight on her own data  w = T s_a^2 / (T s_a^2
     ↪ + s_y^2)
    sy, sa = pm1[2], pm1[3]
    w6 = 6 * sa ** 2 / (6 * sa ** 2 + sy ** 2)
    # random-effects quasi-demeaning, W (14.10) p. 442: theta = 1 - sqrt(s_u^2 /
     ↪ (s_u^2 + T s_a^2))
    theta_re = 1 - np.sqrt(sy ** 2 / (sy ** 2 + 6 * sa ** 2))
    assert 0 < theta_re < w6 < 1
    # predictive for woman 1 at her 7th (unobserved) visit, x = 40.143 (site section
     ↪ 4.4.3)
    pred1 = d2[:, 0, 0] + d2[:, 0, 1] * 40.143
    q1 = np.quantile(pred1, [0.025, 0.5, 0.975])
    print(f"4 hierarchical: eight schools pooled {pooled:.2f} (sd {pooled_sd:.2f}),
     ↪ tau median {med:.2f}, mean {mean_tau:.2f}, "
          f"95% [{q025:.2f}, {q975:.2f}]; E(theta_A|y) = {eb[0]:.2f}; pregnancy
           ↪ pooled -4.52639 + 0.47633 x reproduced; "
          f"Gibbs means alpha {pm1[0]:.3f}, beta {pm1[1]:.4f}, s_y {pm1[2]:.3f}, s_a
           ↪ {pm1[3]:.3f}, s_b {pm1[4]:.4f}; "
          f"weight on own data (T=6) {w6:.3f}, RE theta {theta_re:.3f}; woman 1 at
           ↪ x=40.143: median {q1[1]:.2f} [{q1[0]:.2f}, {q1[2]:.2f}]")
    return dict(med=med, mean_tau=mean_tau, q=(q025, q975), eb=eb, pooled=pooled,
     ↪ pooled_sd=pooled_sd, cA=cA,
                pm1=pm1, post=post, ind=ind, w6=w6, theta_re=theta_re, q1=q1)
```

5 · Tobit → L07-05-tobit

Slide 9's probabilities; the Stein attenuation by Monte Carlo; (17.24)–(17.30) against simulation and numerical derivatives; W Table 17.3 by MLE; Chib's data augmentation.

```
# ============================================================================ 5 ·
 ↪ Tobit (L07-05)
def tobit_mle(X, h):
    def nll(th):
        b, s = th[:-1], np.exp(th[-1])
        xb = X @ b
        return -np.sum(np.where(h > 0, stats.norm.logpdf((h - xb) / s) - np.log(s),
         ↪ stats.norm.logcdf(-xb / s)))
    ols = sm.OLS(h, X).fit()
    th = np.r_[ols.params, np.log(np.sqrt(ols.scale))]
    for meth in ("BFGS", "Nelder-Mead", "BFGS"):
        th = optimize.minimize(nll, th, method=meth, options=dict(maxiter=200000,
↪   maxfev=200000, xatol=1e-10,
                                                                  fatol=1e-10) if
↪   meth == "Nelder-Mead" else dict(gtol=1e-9)).x
    return th[:-1], np.exp(th[-1]), -nll(th)


def gibbs_tobit(rng, X, h, M0=1000, M=5000):
```

```
    """Chib (1992) data augmentation, LDV slides 7-8 / G Algorithm 8.2, flat prior
     ↪  (V0^-1 = 0, nu0 = 0).

    y*_i = y_i if y_i > 0, else y*_i ~ N(x_i b, s2) truncated to (-inf, 0]; b | y* ~
↪   N((X'X)^-1 X'y*, s2 (X'X)^-1);
    s2 | b, y* ~ IG(n/2, (y* - Xb)'(y* - Xb)/2).
    """
    n = len(h)
    cen = h <= 0
    XtXi = np.linalg.inv(X.T @ X)
    Lc = np.linalg.cholesky(XtXi)
    b = XtXi @ X.T @ h
    s2 = np.var(h)
    ystar = h.astype(float).copy()
    out = np.zeros((M, X.shape[1] + 1))
    for it in range(M0 + M):
        ystar[cen] = rtruncnorm(rng, X[cen] @ b, np.sqrt(s2), -np.inf, 0.0)
        bhat = XtXi @ X.T @ ystar
        b = bhat + np.sqrt(s2) * Lc @ rng.normal(size=len(bhat))
        r = ystar - X @ b
        s2 = 1 / rng.gamma(n / 2, 2 / (r @ r))
        if it >= M0:
            out[it - M0] = np.r_[b, np.sqrt(s2)]
    return out


def section5():
    # slide 9: y* ~ N(a beta, beta^2 + sigma^2) with beta = 2, sigma^2 = 0.25
    check(np.sqrt(4.25), 2.062, 3, "sd of y*")
    for a, pr in ((0.0, 0.5), (0.7, 0.25), (1.3, 0.1)):
        check(mf.p_zero_slide(a), pr, 2, f"Prob. trunc. at a = {a}")
    # OLS on censored data with normal x: slope = beta P(y > 0) (Stein's lemma),
     ↪  Monte Carlo check
    rng = np.random.default_rng(5)
    xs = rng.normal(0.7, 1, 1_000_000)
    yc = np.maximum(0, 2 * xs + 0.5 * rng.normal(size=xs.size))
    slope = np.polyfit(xs, yc, 1)[0]
    assert abs(slope - 2 * (1 - mf.p_zero_slide(0.7))) < 0.005, slope
    # pointwise Tobit moments (W 17.24, 17.25, 17.30) against Monte Carlo at x0 =
     ↪  0.3
    m = mf.tobit_moments(0.3, 2.0, 0.5)
    ysim = np.maximum(0, 0.6 + 0.5 * rng.normal(size=2_000_000))
    assert abs(ysim.mean() - m["e_all"]) < 2e-3 and abs(ysim[ysim > 0].mean() -
     ↪  m["e_pos"]) < 2e-3
    eps = 1e-5
    num = (mf.tobit_moments(0.3 + eps, 2.0, 0.5)["e_all"] - mf.tobit_moments(0.3 -
↪   eps, 2.0, 0.5)["e_all"]) / (2 * eps)
    assert abs(num - m["dE_all"]) < 1e-6                                  # (17.30) is
     ↪  the derivative of (17.25)
```

```
    num2 = (mf.tobit_moments(0.3 + eps, 2.0, 0.5)["e_pos"] - mf.tobit_moments(0.3 -
↪   eps, 2.0, 0.5)["e_pos"]) / (2 * eps)
    assert abs(num2 - m["dE_pos"]) < 1e-6                                  # (17.26)

    # Wooldridge Example 17.2 (Table 17.3, p. 541): hours of 753 married women, 325
     ↪  zeros
    d = wooldridge.data("mroz")
    cols = ["nwifeinc", "educ", "exper", "expersq", "age", "kidslt6", "kidsge6"]
    X = sm.add_constant(d[cols]).values
    h = d.hours.values.astype(float)
    check(int((h == 0).sum()), 325, 0, "zeros")
    ols = sm.OLS(h, X).fit()
    for j, pr in zip(range(8), (1330.48, -3.45, 28.76, 65.67, -0.700, -30.51,
     ↪  -442.09, -32.78)):
        check(ols.params[j], pr, 2 if abs(pr) > 1 else 3, f"OLS col {j}")
    check(np.sqrt(ols.scale), 750.18, 2, "OLS sigma")
    b, s, ll = tobit_mle(X, h)
    for j, pr in zip(range(8), (965.31, -8.81, 80.65, 131.56, -1.86, -54.41, -894.02,
     ↪  -16.22)):
        check(b[j], pr, 2, f"Tobit col {j}")
    check(s, 1122.02, 2, "Tobit sigma")
    check(ll, -3819.09, 2, "Tobit log-likelihood")
    ape = stats.norm.cdf(X @ b / s).mean()
    xm = X.mean(0)
    xm[4] = xm[3] ** 2                                                     # (average
↪   exper)^2, as W says
    pea = stats.norm.cdf(xm @ b / s)
    check(ape, 0.589, 3, "APE scale")
    check(pea, 0.645, 3, "PEA scale")
    check(ape * b[2], 47.50, 1, "APE educ")
    # Chib data augmentation reproduces the MLE (flat prior: posterior means ~ MLE)
    dr = gibbs_tobit(np.random.default_rng(17), X, h)
    pm, psd = dr.mean(0), dr.std(0)
    se = np.array([21.58, 111.88])
    assert abs(pm[2] - b[2]) < 0.25 * se[0] and abs(pm[6] - b[6]) < 0.25 * se[1],
     ↪  (pm[2], pm[6])
    assert abs(psd[2] / se[0] - 1) < 0.15 and abs(psd[6] / se[1] - 1) < 0.15
    print(f"5 Tobit: slide 9 P(y=0) 0.500/0.249/0.104; OLS-on-censored slope
     ↪  {slope:.4f} = beta P(y>0) "
          f"{2*(1-mf.p_zero_slide(0.7)):.4f}; W Table 17.3 Tobit reproduced (educ
           ↪  {b[2]:.2f}, kidslt6 {b[6]:.2f}, sigma {s:.2f}, "
          f"loglik {ll:.2f}); APE {ape:.3f}, PEA {pea:.3f}; Gibbs posterior means
           ↪  educ {pm[2]:.2f} (sd {psd[2]:.2f}), "
          f"kidslt6 {pm[6]:.2f} (sd {psd[6]:.2f}), sigma {pm[-1]:.1f}")
    return dict(b=b, s=s, ll=ll, ape=ape, pea=pea, pm=pm, psd=psd, slope=slope)
```

6 · Probit, ordered and multinomial probit → L07-06-probit-ordered-multinomial

W Table 17.1 (probit and logit), the scale factors, Albert–Chib, Greenberg's prior arithmetic, the ordered-probit sampler with uniform cut-points, and a multinomial probit sampler with Ω known.

```
# ============================================================================ 6 · probit, ordered, multinomial (LO7-06)
def gibbs_probit(rng, X, y, M0=1000, M=5000):
    """Albert & Chib (1993), G Algorithm 8.3 with a flat prior: z_i ~ TN(x_i b, 1) on (0, inf) if y_i = 1 else (-inf, 0];
    b | z ~ N((X'X)^-1 X'z, (X'X)^-1). B1 does not change across iterations because sigma^2 = 1."""
    XtXi = np.linalg.inv(X.T @ X)
    Lc = np.linalg.cholesky(XtXi)
    b = np.zeros(X.shape[1])
    lo = np.where(y == 1, 0.0, -np.inf)
    hi = np.where(y == 1, np.inf, 0.0)
    out = np.zeros((M, X.shape[1]))
    for it in range(M0 + M):
        z = rtruncnorm(rng, X @ b, 1.0, lo, hi)
        b = XtXi @ X.T @ z + Lc @ rng.normal(size=len(b))
        if it >= M0:
            out[it - M0] = b
    return out


def gibbs_ordered(rng, X, y, J, M0=2000, M=8000):
    """Ordered probit (LDV slide 12): gamma_1 = 0 fixed, flat priors.
    z_i ~ TN(x_i b, 1) on (gamma_{y_i - 1}, gamma_{y_i}]; b | z as in the probit;
    gamma_j ~ U(max{max z_i: y_i = j, gamma_{j-1}}, min{min z_i: y_i = j+1, gamma_{j+1}}), j = 2..J-1."""
    XtXi = np.linalg.inv(X.T @ X)
    Lc = np.linalg.cholesky(XtXi)
    b = np.zeros(X.shape[1])
    gam = np.r_[-np.inf, 0.0, np.arange(1, J - 1, dtype=float), np.inf]   # gamma_0 .. gamma_J
    out = np.zeros((M, X.shape[1] + J - 2))
    for it in range(M0 + M):
        z = rtruncnorm(rng, X @ b, 1.0, gam[y - 1], gam[y])
        b = XtXi @ X.T @ z + Lc @ rng.normal(size=len(b))
        for j in range(2, J):
            lo = max(z[y == j].max(), gam[j - 1])
            hi = min(z[y == j + 1].min(), gam[j + 1])
            gam[j] = rng.uniform(lo, hi)
        if it >= M0:
            out[it - M0] = np.r_[b, gam[2:J]]
    return out


def gibbs_mnp(rng, Xlist, y, Sigma, M0=1000, M=4000):
    """Multinomial probit as a SUR in utility differences (LDV slides 13-15), Sigma known (identified: Sigma_11 = 1).

    w_i = X_i beta + e_i, e_i ~ N(0, Sigma), J differences; y_i = 0 if max(w_i) < 0, else y_i = argmax w_i + 1.
```

```
    w_ij | w_i,-j is univariate normal (A.9-type conditional) truncated by the
↪   choice; beta | w is the SUR draw
    V1^-1 = sum X_i' Sigma^-1 X_i, V1^-1 b1 = sum X_i' Sigma^-1 w_i (flat prior).
    ponytail: Sigma fixed at its true value; the McCulloch-Rossi inverse-Wishart step
↪   is the upgrade path.
    """
    n, J, k = Xlist.shape
    Si = np.linalg.inv(Sigma)
    P = np.einsum("ijk,jl,ilm->km", Xlist, Si, Xlist)
    V1 = np.linalg.inv(P)
    Lc = np.linalg.cholesky(V1)
    b = np.zeros(k)
    w = np.where(np.arange(J)[None, :] == (y - 1)[:, None], 0.5, -0.5)
    out = np.zeros((M, k))
    for it in range(M0 + M):
        mu = np.einsum("ijk,k->ij", Xlist, b)
        for j in range(J):
            o = [q for q in range(J) if q != j]
            # conditional of component j given the others: mean mu_j + S_jo S_oo^-1
             ↪  (w_o - mu_o), var S_jj - S_jo S_oo^-1 S_oj
            Soo_i = np.linalg.inv(Sigma[np.ix_(o, o)])
            cm = mu[:, j] + (w[:, o] - mu[:, o]) @ Soo_i @ Sigma[o, j]
            cs = np.sqrt(Sigma[j, j] - Sigma[j, o] @ Soo_i @ Sigma[o, j])
            mo = w[:, o].max(1)
            lo = np.where(y == j + 1, np.maximum(0, mo), -np.inf)
            # chosen j: above 0 and all others; base chosen: below 0; another k
             ↪  chosen: below w_ik
            hi = np.where(y == 0, 0.0, np.where(y == j + 1, np.inf, w[np.arange(n),
↪   np.clip(y - 1, 0, J - 1)]))
            w[:, j] = rtruncnorm(rng, cm, cs, lo, hi)
        rhs = np.einsum("ijk,jl,il->k", Xlist, Si, w)
        b = V1 @ rhs + Lc @ rng.normal(size=k)
        if it >= M0:
            out[it - M0] = b
    return out


def section6():
    # Wooldridge Example 17.1 (Table 17.1, p. 533): labor force participation, MROZ
    d = wooldridge.data("mroz")
    cols = ["nwifeinc", "educ", "exper", "expersq", "age", "kidslt6", "kidsge6"]
    X = sm.add_constant(d[cols]).values
    y = d.inlf.values.astype(int)
    pr = sm.Probit(y, X).fit(disp=0)
    lg = sm.Logit(y, X).fit(disp=0)
    for j, pv in zip(range(8), (0.270, -0.012, 0.131, 0.123, -0.0019, -0.053, -0.868,
     ↪  0.036)):
        check(pr.params[j], pv, 4 if abs(pv) < 0.01 else 3, f"probit col {j}")
    for j, pv in zip(range(8), (0.425, -0.021, 0.221, 0.206, -0.0032, -0.088, -1.443,
     ↪  0.060)):
```

```
        check(lg.params[j], pv, 4 if abs(pv) < 0.01 else 3, f"logit col {j}")
    check(pr.llf, -401.30, 2, "probit loglik")
    check(lg.llf, -401.77, 2, "logit loglik")
    check(pr.prsquared, 0.221, 3, "probit pseudo R2")
    check(100 * np.mean((pr.predict(X) > 0.5) == y), 73.4, 1, "percent correctly
↪   predicted")
    xm = X.mean(0)
    s_pr, s_lg = stats.norm.pdf(xm @ pr.params), (lambda e: e / (1 + e) **
↪   2)(np.exp(xm @ lg.params))
    check(s_pr, 0.391, 3, "probit PEA scale")
    check(s_lg, 0.243, 3, "logit PEA scale")
    check(s_pr / s_lg, 1.61, 2, "ratio of scale factors")
    check(stats.norm.pdf(X @ pr.params).mean(), 0.301, 3, "probit APE scale")
    # Albert-Chib Gibbs reproduces the probit MLE
    dr = gibbs_probit(np.random.default_rng(23), X, y)
    pm, psd = dr.mean(0), dr.std(0)
    assert np.all(np.abs(pm - pr.params) < 0.3 * pr.bse), (pm, pr.params)
    assert np.all(np.abs(psd / pr.bse - 1) < 0.2)
    # Greenberg's prior elicitation (pp. 125-126 probit, p. 128 logit)
    check(stats.norm.ppf(0.10), -1.2816, 4, "beta01 probit")
    check(((stats.norm.ppf(0.3) - stats.norm.ppf(0.1)) / 1.96) ** 2, 0.1492, 4,
↪   "B0,11 probit")
    check((stats.norm.ppf(0.8) + 1.2816) / 2, 1.0616, 4, "beta02 probit")
    check((stats.norm.ppf(0.6) + 1.2816) / 2, 0.767, 3, "lower beta2 probit (G prints
↪   0.7674, truncated)")
    check(((1.0616 - 0.7674) / 1.96) ** 2, 0.0225, 4, "B0,22 probit")
    check(stats.norm.ppf(0.4) + 1.2816, 1.0283, 4, "beta03 probit")
    check(np.log(1 / 9), -2.1972, 4, "beta01 logit")
    check((np.log(3 / 7) - np.log(1 / 9)) / 1.96, 0.6887, 4, "prior sd beta1 logit")
    check((np.log(4) + 2.1972) / 2, 1.7917, 4, "beta02 logit")
    check(stats.norm.pdf(0), 0.3989, 4, "phi(0)")

    # ordered probit (slides 11-12) on a simulated sample: x'b = 0.5 + 1.0 x, gamma =
    ↪  (0, 1, 2), J = 4
    rng = np.random.default_rng(14)
    n = 500
    Xo = np.column_stack([np.ones(n), rng.normal(size=n)])
    zs = Xo @ np.array([0.5, 1.0]) + rng.normal(size=n)
    yo = 1 + np.searchsorted(np.array([0.0, 1.0, 2.0]), zs)
    dro = gibbs_ordered(rng, Xo, yo, 4)
    pmo = dro.mean(0)
    assert abs(pmo[1] - 1.0) < 0.2 and abs(pmo[2] - 1.0) < 0.25 and abs(pmo[3] - 2.0)
    ↪  < 0.3, pmo
    lag1 = np.corrcoef(dro[1:, 2], dro[:-1, 2])[0, 1]                    # cut-point
↪   chain: slow mixing (Cowles 1996)

    # multinomial probit: 3 alternatives (0 = base), 2 utility differences,
    ↪  alternative-specific (intercept, x)
    n = 800
    x = rng.normal(size=n)
```

```
    Xl = np.zeros((n, 2, 4))
    Xl[:, 0, 0], Xl[:, 0, 1], Xl[:, 1, 2], Xl[:, 1, 3] = 1, x, 1, x
    Sig = np.array([[1.0, 0.5], [0.5, 1.5]])
    btrue = np.array([0.3, 1.0, -0.2, -0.8])
    wtrue = np.einsum("ijk,k->ij", Xl, btrue) + rng.multivariate_normal(np.zeros(2),
↪   Sig, size=n)
    ym = np.where(wtrue.max(1) < 0, 0, wtrue.argmax(1) + 1)
    drm = gibbs_mnp(rng, Xl, ym, Sig)
    pmm = drm.mean(0)
    assert np.all(np.abs(pmm - btrue) < 0.25), pmm
    print(f"6 probit: W Table 17.1 probit/logit reproduced (educ .131/.221, loglik
     ↪  -401.30/-401.77); scale factors "
          f"{s_pr:.3f}/{s_lg:.3f} = {s_pr/s_lg:.2f}; Albert-Chib means educ
           ↪  {pm[2]:.3f} (MLE {pr.params[2]:.3f}); "
          f"ordered probit Gibbs b {pmo[0]:.2f}, {pmo[1]:.2f}, gamma2 {pmo[2]:.2f},
           ↪  gamma3 {pmo[3]:.2f} (lag-1 corr of gamma2 {lag1:.2f}); "
          f"MNP Gibbs {np.round(pmm, 2)} vs true {btrue}")
    return dict(pr=pr, lg=lg, s_pr=s_pr, s_lg=s_lg, pm=pm, pmo=pmo, lag1=lag1,
     ↪  pmm=pmm)
```

7 · The counterexamples of the pitfalls note → L07-08-pitfalls

```
# ============================================================================== 7 ·
 ↪  counterexamples of the pitfalls note (L07-08)
def section7():
    # (a) random-coefficient shrinkage is not "toward zero": with tau = 5.23 school C
     ↪  (y = -2.75) moves UP
    c = mf.schools_conditional(5.23)
    assert c["mean"][2] > mf.Y8[2] and c["mean"][0] < mf.Y8[0]
    # (b) robust SEs fix inference, not efficiency: GLS variance beats OLS's true
     ↪  variance at alpha1 = 1
    assert mf.gls_efficiency(1.0)[1] > 4
    # (c) Tobit coefficient is not the effect on E(y|x): at z = 0 the slope is
     ↪  beta/2
    check(mf.tobit_moments(0.0, 2.0, 0.5)["dE_all"], 1.0, 10, "dE/dx at z = 0")
    # (d) probit sigma is not identified: scaling (beta, sigma) by c leaves P(y = 1)
     ↪  unchanged
    xb = np.linspace(-2, 2, 9)
    assert np.allclose(stats.norm.cdf(xb / 1.0), stats.norm.cdf((3 * xb) / 3.0))
    # (e) t scale vs variance: sigma^2 = 1, nu = 5 gives Var = 5/3
    check(stats.t(5, scale=1).var(), 1.6667, 4, "Var t5")
    # (f) every Gibbs weight equals 1 at |e/sigma| = 1, whatever nu
    assert all(abs(mf.t_weight(1.0, nu) - 1) < 1e-12 for nu in (1, 2, 5, 30))
    # (g) censoring vs truncation: OLS on the positive subsample is also biased
     ↪  (slope flatter than beta)
    rng = np.random.default_rng(7)
    xs = rng.normal(0.7, 1, 400_000)
    ystar = 2 * xs + 0.5 * rng.normal(size=xs.size)
    pos = ystar > 0
    s_trunc = np.polyfit(xs[pos], ystar[pos], 1)[0]
```

```
    assert s_trunc < 2.0
    # (h) AR errors with an iid regressor: OLS keeps the right naive SE (ratio 1),
     ↪  GLS still gains
    n0, e0 = mf.ar1_ratios(0.9, 0.0)
    assert abs(n0 - 1) < 1e-12 and e0 > 9
    print(f"7 pitfalls: school C shrinks up to {c['mean'][2]:.2f}; truncated-sample
     ↪  OLS slope {s_trunc:.3f} < 2; "
          f"AR(0.9) errors + iid x: naive OLS var right, GLS gain x{e0:.2f}")
    return dict(s_trunc=s_trunc, cC=c["mean"][2])


if __name__ == "__main__":
    section1()
    section2()
    section3()
    section4()
    section5()
    section6()
    section7()
    print("ALL LO7 CHECKS PASSED")
```

8 · The run

```
if __name__ == "__main__":
    section1()
    section2()
    section3()
    section4()
    section5()
    section6()
    section7()
    print("ALL LO7 CHECKS PASSED")
```

Output of the last run (verbatim):

```
1 GLS: OLS on (Py, PX) = GLS; W Table 8.1 OLS .821 (.104) / WLS .787 (.063)
 ↪  reproduced; Case I MH acceptance 0.55, posterior means beta 2.20, 3.97, alpha
 ↪  0.93, 1.12; FGLS slope 3.97; OLS/GLS slope variance at alpha1 = 1: 4.38
2 t errors: mixture identity holds; P(nu<=10) = 0.3297, P(nu>=40) = 0.2019; Gibbs-MH
 ↪  (one outlier +15): beta 0.43, 0.87, 1.93 vs OLS 0.54, 1.07, 2.03; posterior mean
 ↪  nu 3.8; MH acceptance 0.83; outlier weight 0.018 vs median 1.063
3 AR(1): Omega_rho^-1 tridiagonal, PW matrix P'P = Omega^-1; W Table 12.1 reproduced
 ↪  (PW 2.94 (0.63), rho-hat 0.293, t_afdec6 -1.69; 11 iterations); Cochrane-Orcutt
 ↪  rho 0.293, log(chempi) 2.95; Gibbs (slide 27 design) beta 1.97, 4.06, sigma2
 ↪  1.03, rho 0.871
4 hierarchical: eight schools pooled 7.87 (sd 4.17), tau median 5.23, mean 6.59, 95%
 ↪  [0.24, 20.87]; E(theta_A|y) = 11.64; pregnancy pooled -4.52639 + 0.47633 x
 ↪  reproduced; Gibbs means alpha -4.363, beta 0.4687, s_y 1.119, s_a 3.210, s_b
 ↪  0.1253; weight on own data (T=6) 0.980, RE theta 0.859; woman 1 at x=40.143:
 ↪  median 17.68 [15.68, 19.86]
```

```
5 Tobit: slide 9 P(y=0) 0.500/0.249/0.104; OLS-on-censored slope 1.5033 = beta P(y>0)
 ↪  1.5029; W Table 17.3 Tobit reproduced (educ 80.65, kidslt6 -894.02, sigma
 ↪  1122.02, loglik -3819.09); APE 0.589, PEA 0.645; Gibbs posterior means educ 81.84
 ↪  (sd 21.76), kidslt6 -902.63 (sd 112.86), sigma 1138.8
6 probit: W Table 17.1 probit/logit reproduced (educ .131/.221, loglik
 ↪  -401.30/-401.77); scale factors 0.391/0.243 = 1.61; Albert-Chib means educ 0.132
 ↪  (MLE 0.131); ordered probit Gibbs b 0.50, 0.89, gamma2 1.10, gamma3 2.08 (lag-1
 ↪  corr of gamma2 0.99); MNP Gibbs [ 0.45  1.1  -0.11 -0.71] vs true [ 0.3  1.  -0.2
 ↪  -0.8]
7 pitfalls: school C shrinks up to 7.02; truncated-sample OLS slope 1.902 < 2;
 ↪  AR(0.9) errors + iid x: naive OLS var right, GLS gain x9.53
ALL L07 CHECKS PASSED
```

9 · What is deliberately absent

- **PyMC** (not installed; the class plan lists it as optional): every sampler is hand-written, which is what the exam asks you to be able to write down.
- **The multinomial-probit Ω step** (inverse Wishart with a scale normalization, McCulloch–Rossi): Ω is held at its true value, marked `ponytail:` in `gibbs_mnp` with the upgrade path.
- **The airline data** (Greene) are not on disk: the airline posteriors of GLM3 slide 14 and HLM slides 20–21 are read, not reproduced.
- **Convergence diagnostics** beyond what the notes use (one lag-1 autocorrelation for the cut-points): the samplers are long enough for the asserted tolerances, which are a fraction of a posterior sd.

Links

- Unit: L07-00-map · L07-07-formulas · L07-08-pitfalls · figures script `figures/make_figures.py` · companion checks `companions/check_companions.py`.
- Class 2: L02 Python (IRLS logit and Poisson — the frequentist fits behind §6's logit).